Major Project (IT898) Report on

AGENTIC AI APPLICATION FOR INTELLIGENT PORTFOLIO CREATION

Lucid: Intelligent Investment Portfolio Platform for Indian Retail Equities

Submitted in partial fulfillment of the requirements for the degree of

MASTER OF TECHNOLOGY
in
INFORMATION TECHNOLOGY

by

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November 2026

DECLARATION

by the P.G. M.Tech.(IT) Student

I hereby *declare* that the Report of the P.G. IT898 - Major Project Work entitled "**Agentic AI Application for Intelligent Portfolio Creation (Lucid)**", which is being submitted to National Institute of Technology Karnataka, Surathkal in partial fulfillment of the requirements for the award of the Degree of Master of Technology in the Department of Information Technology, is a *bonafide report of the work carried out by me.* The material contained in this report has not been submitted at any University or Institution for the award of any degree.

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CERTIFICATE

This is to *certify* that the P.G. IT898 - Major Project Work - I Work entitled "**Agentic AI Application for Intelligent Portfolio Creation (Lucid)**", submitted by **ANKUR PATEL (252IT004)** as the record of the work carried out by him, is *accepted as the P.G. IT898 - Major Project Work report submission* in partial fulfillment of the requirement for the award of the degree of **Master of Technology in Information Technology** in the Department of Information Technology, NITK Surathkal, India.

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ACKNOWLEDGEMENT

I would sincerely like to express my profound gratitude to my project supervisor, **Dr. Biju R Mohan**, for his unconditional support, technical guidance, and constructive critique throughout the research, design, implementation, and evaluation phases of this Major Project.

I also express my gratitude to the faculty and staff of the Department of Information Technology, National Institute of Technology Karnataka, Surathkal, for providing the necessary computational infrastructure, academic resources, and encouragement that enabled the successful completion of this work.

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ABSTRACT

Indian retail equity investors face severe systemic barriers to disciplined wealth creation: cognitive information overload, emotional behavioral biases leading to mistimed panic selling, and opaque commercial robo-advisors offering zero fiduciary transparency. Conversely, deploying generative Large Language Models (LLMs) directly in quantitative asset management causes catastrophic arithmetic hallucinations, producing non-normalizing allocations and unexecutable fractional shares on Indian exchanges. This project designs, engineers, and evaluates **Lucid**, an institutional-grade, transparent, agentic AI platform tailored for Indian retail equity investors across the National Stock Exchange (NSE) and Bombay Stock Exchange (BSE).

Lucid resolves the AI-finance dichotomy through strict architectural decoupling: a **13-node LangGraph StateGraph multi-agent pipeline** (powered by Google Gemini 3.8 Flash with deterministic fallbacks) synthesizes macroeconomic regimes, market volatility (India VIX), corporate fundamentals, and news, while all numerical asset weights are computed deterministically via **SciPy's Sequential Least Squares Programming (SLSQP)** solver under hard institutional diversification bounds ($\leq 25\%$ single stock, $\leq 40\%$ sector). To enforce exchange compliance, Lucid executes a **discrete whole-share integer lot sizing algorithm** with strict cash conservation. The platform features an unattended background drift radar scanning active portfolios every 5 minutes during trading hours (09:15–15:30 IST) to detect asset drift ($\geq 5\%$) and generate friction-aware rebalancing previews factoring in statutory Securities Transaction Tax (STT), exchange turnover fees, SEBI charges, stamp duty, and GST. Backed by a PostgreSQL 16 database housing 62 domain tables (+1 Alembic table) and the full universe of 2,639 listed equities, Lucid was verified across 216 automated tests (100% pass rate). Empirical backtesting demonstrates that Lucid achieves an annualized return of **10.72%** (+159 bps alpha over the Nifty 50 benchmark at 9.13%) with an annualized volatility of **9.00%**, a Sharpe ratio of **0.41**, and an institutional diversification score of **95.51/100**, delivering transparent, dual-level fiduciary explanations for every investment decision.

Keywords: Agentic AI, LangGraph, Markowitz Optimization, SciPy SLSQP, Indian Equities (NSE/BSE), Integer Lot Sizing, Demat CAS Diagnosis, Drift Radar, Explainable AI, PostgreSQL 16.

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ABBREVIATIONS

ADTV	Average Daily Traded Volume
AI	Artificial Intelligence
API	Application Programming Interface
ASGI	Asynchronous Server Gateway Interface
BSE	Bombay Stock Exchange
CAGR	Compound Annual Growth Rate
CAS	Consolidated Account Statement (CDSL/NSDL)
CDSL	Central Depository Services (India) Limited
CIN	Corporate Identity Number
CPI	Consumer Price Index
D/E	Debt-to-Equity Ratio
DMA	Daily Moving Average
GST	Goods and Services Tax
HHI	Herfindahl-Hirschman Concentration Index
HLD	High-Level Design
HTTP	Hypertext Transfer Protocol
ISIN	International Securities Identification Number
IST	Indian Standard Time (UTC+05:30)
JWT	JSON Web Token
KYC	Know Your Customer
LLD	Low-Level Design
LLM	Large Language Model
NAV	Net Asset Value
NSDL	National Securities Depository Limited
NSE	National Stock Exchange of India
OHLCV	Open, High, Low, Close, Volume
ORM	Object-Relational Mapping (SQLAlchemy)
P/B	Price-to-Book Ratio
P/E	Price-to-Earnings Ratio
RBI	Reserve Bank of India
REST	Representational State Transfer
ROE	Return on Equity
SEBI	Securities and Exchange Board of India

SLSQP	Sequential Least Squares Programming
SQL	Structured Query Language
STT	Securities Transaction Tax
UI	User Interface
UUID	Universally Unique Identifier
VIX	Volatility Index (India VIX)

Chapter 1

Introduction

1.1 Background

In recent years, millions of people in India have started investing in the stock market for the first time. In the past, buying shares was complicated and required physical paperwork. Today, anyone with a smartphone can open an investment account within minutes. With digital identity verification through Aadhaar and PAN (known as Know Your Customer, or KYC) and instant online payments using the Unified Payments Interface (UPI), investing has become accessible to ordinary citizens.

According to official records published by the Securities and Exchange Board of India (SEBI), the number of electronic share-holding accounts (known as Demat accounts) in India increased from approximately 36 million in 2019 to more than 160 million by early 2026. A **Demat account** (short for Dematerialized account) works like a digital bank account, but instead of holding money, it holds electronic certificates of company shares and government bonds.

However, while opening an account is now easy, investing wisely remains difficult. Most new retail investors do not have formal financial training. When people start investing on their own, they are often overwhelmed by large amounts of confusing information:

- Complex quarterly financial statements published by companies.
- Economic policy updates and interest rate announcements from the Reserve Bank of India (RBI).
- Thousands of news articles, social media tips, and daily stock price movements.

Without proper guidance, individual investors often make predictable mistakes. They buy stocks that have already risen sharply because of popular hype, sell good stocks in a panic during normal market declines, or put all their savings into one or two risky companies. In finance, this risk is called **lack of diversification**. Diversification simply means spreading investments across different companies and industry sectors so that poor performance in one stock does not wipe out an investor's entire savings.

To solve this problem, retail investors usually turn to mutual funds or automated investment tools called **robo-advisors**. A mutual fund pools money from many investors and pays a professional manager to select stocks. However, mutual funds charge recurring management fees (called **expense ratios**) of 1.5% to 2.5% every year. Over ten or twenty years, these fees take away a significant portion of an investor's accumulated returns. On the other hand, existing robo-advisors act like closed "black boxes." They ask a few generic questions and suggest a standard basket of stocks or funds, without explaining why each company was chosen or how the investment plan connects to the user's specific life goals. When stock prices drop, investors do not understand why they hold these stocks, get frightened, and sell at a loss.

1.2 Problem Statement

To address these challenges, the problem statement of this Major Project is defined as follows:

> *"To design and develop an intelligent portfolio creation application that helps retail investors build personalized investment portfolios by providing transparent, context-aware, explainable, and mathematically reliable recommendations."*

1.3 Challenges in Retail Portfolio Construction

Building an automated application that creates personalized stock portfolios for Indian retail investors involves solving six key challenges:

1. **Scattered and Overwhelming Information**: Financial information in India is spread across many different places: official stock exchange websites, company annual reports, news agencies, and government portals. An individual investor cannot read and make sense of hundreds of pages of filings every week.
2. **Emotional Decision-Making**: Human investors are strongly influenced by emotion. Studies in behavioral finance show that people feel the pain of losing money twice as strongly as they enjoy making the same amount

of money [4]. When markets fall, fear causes investors to sell at the worst possible time. An automated system must remain objective and rational.

3. **The "Black-Box" Problem**: Existing automated advisory tools calculate investment percentages using hidden algorithms. They rarely explain why a particular stock was selected or why another was rejected. Without clear explanations, everyday investors find it difficult to trust automated advice.
4. **Mathematical Mistakes in Generative AI**: Modern conversational Artificial Intelligence tools (Large Language Models, or LLMs) are good at reading articles and writing text. However, studies show that language models often make basic calculation mistakes when trying to calculate portfolio numbers [5]. For example, an LLM might suggest percentages that do not add up to 100%, recommend negative percentages, or change its numbers unpredictably when asked the same question twice.
5. **Indian Exchange Rule of Whole Shares**: In several foreign countries, investors can buy fractional shares (for example, 0.25 shares of a costly company). In India, exchange regulations strictly forbid fractional shares; an investor must buy complete, whole shares (such as 1, 2, or 10 shares). A stock like MRF Ltd. costs over Rs. 1,00,000 per share. If an algorithm suggests buying 1.5 shares, that order cannot be executed. The system must convert percentage targets into exact whole shares while ensuring that unallocated cash is safely kept in the investor's balance.
6. **Managing a Large Market Universe**: The Indian equity market includes more than 2,600 actively traded companies on the National Stock Exchange of India (NSE) and the Bombay Stock Exchange (BSE). Evaluating thousands of companies for trading activity, financial stability, and price freshness requires fast and reliable data handling.

1.4 Motivation

The primary motivation behind this project is to build an application called **Lucid**. The goal of Lucid is to give everyday retail investors the same quality of research, risk management, and mathematical precision that professional fund managers use, but in a simple, affordable, and transparent format.

Recent progress in **Agentic Artificial Intelligence** makes this possible [3]. Instead of using a single AI model to do everything, an agentic system divides the work among several specialized digital assistants (called **AI agents**). Each agent has a specific job:

- One agent reviews broad economic conditions (such as inflation and interest rates).
- Another agent evaluates company balance sheets and profit trends.
- A third agent checks recent news and corporate filings.
- A risk critic agent double-checks the proposals to spot hidden risks.

Most importantly, Lucid keeps artificial intelligence and mathematical calculations strictly separate. The AI agents are used only for reading documents, evaluating qualitative trends, and shortlisting candidate companies. When it is time to calculate exact investment amounts and validate portfolio constraints, Lucid passes the task to a proven mathematical calculation engine. This separation ensures that the recommendations are both easy to understand and mathematically guaranteed to be accurate [2].

1.5 Project Objectives

The specific technical and research objectives of this Major Project are:

1. To develop an intelligent portfolio creation system that generates personalized investment recommendations based on an investor's financial goals, risk preferences, investment horizon, and other relevant factors.
2. To ensure that the generated portfolio recommendations are reliable, transparent, explainable, and mathematically sound by considering relevant investor, market, and historical information.
3. To validate and deploy the proposed system as a production-ready web application that provides an accessible interface for retail investors to create, explore, and understand their portfolios.

1.6 Scope of the Project

The operational scope of Lucid defines what the application does and what it deliberately avoids:

- **Market and Securities**: Lucid works with equity shares listed on the National Stock Exchange of India (NSE) and the Bombay Stock Exchange (BSE) in the standard cash delivery segment (where investors buy and hold real shares).
- **Target Users**: Individual Indian retail investors who want personalized portfolios tailored to their savings goals, health reviews of their existing stock holdings, and automated portfolio monitoring.
- **Order Quantities**: All recommended stock purchases are calculated as discrete whole shares (such as 5 shares or 12 shares) because fractional shares are not permitted on Indian exchanges.
- **Advisory Nature**: Lucid is an educational decision-support system. It explains its financial reasoning transparently so users can make informed choices.
- **Exclusions**: Lucid does not support intraday margin trading (borrowing money to buy and sell on the same day), high-risk derivative contracts (such as Futures and Options), or short selling (betting on a stock to fall). It also does not execute live broker trades automatically; it provides simulated trade previews and integration adapters for safe user testing.

1.7 Proposed Solution: Lucid

Lucid is built as a modular, web-based software platform designed from the ground up for clarity and ease of use:

- **User Interface**: Built with modern web technologies (Next.js and React), providing clean screens where investors can enter their goals, view their portfolios, and ask questions.
- **Backend Server**: A high-speed application server (FastAPI) that handles user requests, coordinates security, and runs the application logic.
- **Multi-Agent Brain**: A team of specialized AI agents coordinated by an agent management system (LangGraph) that studies company filings, news, and economic reports.
- **Mathematical Engine**: A numerical solver (SciPy SLSQP) that calculates exact investment weights based on Modern Portfolio Theory, converts weights into whole share quantities, and accounts for Indian taxes and fees

(such as Securities Transaction Tax and GST).

- **Continuous Monitoring Daemon**: A background program that quietly checks the investor's active portfolio during stock market hours (09:15 to 15:30 IST) to detect if stock price movements have caused the portfolio to drift away from its planned targets.

Lucid monitors the full universe of 2,639 listed Indian companies and filters them for trading liquidity and financial health, ensuring that every user receives recommendations tailored to their unique financial goals.

1.8 Key Contributions

The key contributions of this project are:

1. **Separation of AI Reasoning from Math Calculations**: AI models write explanations and shortlist companies, while a mathematical solver calculates exact portfolio percentages, eliminating mathematical errors.
2. **Whole-Share Integer Allocation**: An algorithm that calculates exact full-share numbers and protects unallocated cash balances, ensuring that every recommendation can be executed on Indian exchanges.
3. **Automated Market-Hours Drift Tracking**: A monitoring service that checks portfolios every 5 minutes during trading hours to alert investors if stock weights shift significantly.
4. **Clear, Plain-Language Explanations**: A feature called "Stock-Why" that explains in simple English exactly why each company was selected, alongside deeper financial metrics for advanced users.
5. **Complete Indian Market Coverage**: A database covering all 2,639 listed companies on the NSE and BSE, rather than limiting users to a small, pre-selected list.

1.9 Report Organization

The remainder of this report is organized as follows:

- **Chapter 2 (Literature Review and Market Survey)** reviews existing financial theories, automated investment tools, and competing applications in India.

- **Chapter 3 (High-Level Design)** explains the overall structure of Lucid, how the major components fit together, and how an investor interacts with the system.
- **Chapter 4 (Low-Level Design)** details how the individual software modules and AI agent workflows operate internally.
- **Chapter 5 (Database Design)** describes the 62 database tables that store user information, stock data, and portfolio records.
- **Chapter 6 (System Architecture)** presents the eight software layers of Lucid, the technology stack, and runtime execution.
- **Chapter 7 (Methodology)** explains the step-by-step investment process, including mathematical formulas for portfolio optimization, lot sizing, and rebalancing.
- **Chapter 8 (Datasets Used)** explains what financial data Lucid uses, where it comes from, and how it is cleaned.
- **Chapter 9 (API Contract)** specifies how the web interface communicates with the backend server through structured requests and responses.
- **Chapter 10 (Work Done)** documents each implemented functional requirement from the user's perspective.
- **Chapter 11 (Results and Testing)** presents the testing methodology, system evaluation, and backtesting results.
- **Chapter 12 (Conclusions and Future Roadmap)** summarizes the achievements of Lucid and outlines future enhancements.

Chapter 2

Literature Review and Competitive Analysis

2.1 Theoretical and Academic Foundations

Modern financial theory and automated portfolio management are built upon three core subjects: mathematical optimization, human behavioral economics, and artificial intelligence. Understanding these foundations helps explain why Lucid was designed the way it is.

2.1.1 Modern Portfolio Theory and Mean-Variance Optimization

In 1952, an economist named Harry Markowitz published a landmark research paper that changed how people invest [6]. Before Markowitz, investors usually evaluated each stock on its own. If an analyst liked three different banking companies, they would simply buy all three.

Markowitz pointed out a major flaw in this approach: if all three companies belong to the same industry, a single bad event in the banking sector will cause all three stocks to fall at the same time. Instead of looking at stocks individually, Markowitz showed that investors should look at how stocks behave together inside a combined portfolio.

To understand this mathematically, consider two key concepts:

1. **Expected Portfolio Return**: The return an investor expects to earn from their entire portfolio is simply the weighted average of the returns of the individual stocks:

$$\mathbb{E}[R_p] = \sum_{i=1}^{N} w_i \mathbb{E}[R_i] = \mathbf{w}^T \boldsymbol{\mu} \tag{2.1}$$

where:

- $\mathbb{E}[R_p]$ is the expected return of the total portfolio (for example, 11% per year).

- N is the total number of stocks held in the portfolio.
- w_i is the **portfolio weight** of stock i. This represents the fraction of total money invested in that specific company (for example, 0.15 means 15% of the portfolio).
- $\mathbb{E}[R_i]$ is the expected future return of stock i.

Simple Numerical Example: Suppose an investor has Rs. 1,00,000 and divides it between two companies: 60% in an IT company (expected return 12%) and 40% in a pharmaceutical company (expected return 8%). The overall expected return is:

$$\mathbb{E}[R_p] = (0.60 \times 12\%) + (0.40 \times 8\%) = 7.2\% + 3.2\% = 10.4\% \text{ per year.}$$

2. **Portfolio Risk (Variance and Covariance)**: Markowitz showed that overall portfolio risk is not merely the average risk of individual stocks:

$$\sigma_p^2 = \sum_{i=1}^{N} \sum_{j=1}^{N} w_i w_j \text{Cov}(R_i, R_j) = \mathbf{w}^T \mathbf{\Sigma} \mathbf{w} \tag{2.2}$$

where:

- σ_p^2 is the **variance** of the portfolio, which measures how widely the portfolio value swings up and down over time.
- $\text{Cov}(R_i, R_j)$ is the **covariance** between stock i and stock j. Covariance measures whether two stocks tend to move in the same direction or in opposite directions.
- $\mathbf{\Sigma}$ is the covariance matrix containing the historical relationship between every pair of stocks.

Why this matters to an investor: If two companies are in completely different businesses (for example, an software exporter and an oil refiner), a rise in global oil prices might hurt one company but help the other. Because they do not move in lockstep, their price swings partially cancel each other out. This means the overall portfolio can achieve the same return with significantly less volatility. This reduction in risk is known as the benefit of **diversification**.

In 1964, William Sharpe introduced the **Sharpe ratio** to help investors

judge whether a portfolio's returns justify the risk taken [7]:

$$S_p = \frac{\mathbb{E}[R_p] - R_f}{\sigma_p} \tag{2.3}$$

where:

- S_p is the Sharpe ratio.
- $\mathbb{E}[R_p]$ is the portfolio's expected annual return.
- R_f is the **risk-free rate of return**. This is the return an investor can earn with zero risk of default, such as investing in a 91-day Government of India Treasury Bill or the Reserve Bank of India (RBI) repo rate (typically around 6.5% to 7.0% in India).
- σ_p is the **volatility** (standard deviation) of the portfolio.

Simple Numerical Example: Suppose a portfolio returns 12.0% per year, the government risk-free rate is 7.0%, and the portfolio's annual volatility is 10.0%. The Sharpe ratio is:

$$S_p = \frac{12.0\% - 7.0\%}{10.0\%} = \frac{5.0\%}{10.0\%} = 0.50$$

A higher Sharpe ratio indicates better investment efficiency: the investor is earning more return for every unit of risk endured.

2.1.2 Factor Investing and Common Investor Traps

In 1993, researchers Eugene Fama and Kenneth French discovered that stock returns are influenced by underlying economic characteristics called **factors** [1]. In the Indian stock market, three factors are particularly important:

- **Quality Factor**: Companies with strong balance sheets, high Return on Equity (ROE $\geq 15\%$), and low debt tend to survive economic downturns much better than heavily indebted companies.
- **Value Factor**: Companies trading at sensible valuations relative to their actual earnings (reasonable Price-to-Earnings, or P/E ratios) offer a safety cushion against market bubbles.
- **Momentum Factor**: Stocks showing steady business and price strength over recent months tend to continue performing well in the near term.

At the same time, behavioral economists Daniel Kahneman and Amos Tversky showed that human psychology often sabotages investment success [4]. Most retail investors suffer from two emotional traps:

1. **Loss Aversion**: People feel the emotional pain of losing Rs. 10,000 about twice as intensely as the joy of gaining Rs. 10,000. Because people hate admitting a loss, they hold on to failing companies for months or years, hoping to break even.
2. **Performance Chasing**: When a speculative stock rises rapidly, people rush to buy it out of fear of missing out (FOMO), often buying right at the top just before prices crash.

Lucid is designed to protect investors from these emotional traps by applying systematic, objective rules rather than relying on human guesswork.

2.1.3 The Role of AI and the Problem of AI Calculation Mistakes

In recent years, generative Artificial Intelligence and Large Language Models (LLMs) have made it possible for software to read news articles, summarize corporate earnings, and answer user questions in natural English.

However, researchers have found a major danger when using language models in finance: **LLMs are terrible at math** [5]. A language model works by predicting the next most likely word in a sentence; it does not have an internal calculator. When asked to create an investment portfolio, an LLM often:

- Recommends percentages that do not add up to 100% (for example, giving weights that total 115% or 85%).
- Invents negative percentages for retail users who are not allowed to short-sell stocks.
- Produces completely different numbers when asked the same question twice.

To prevent these errors, modern research recommends a **hybrid design** [2]:

- Use AI language models exclusively for what they do best: reading articles, understanding user goals, evaluating qualitative trends, and writing clear explanations.
- Use a deterministic mathematical engine (like SciPy's numerical solver)

for all arithmetic calculations, constraints, and allocations.

This hybrid approach forms the core engineering philosophy of Lucid.

2.2 Review of Existing Investment Platforms in India

To understand how existing solutions serve retail investors, we analyzed six prominent Indian investment tools and one international academic benchmark:

2.2.1 smallcase

What it is: smallcase is a popular Indian platform that lets users buy curated baskets of stocks based on themes (such as "Electric Mobility" or "Dividend Aristocrats") created by licensed analysts.

Advantages: Direct integration with popular brokers (Zerodha, Upstox, Groww) and transparent visibility of the stocks you own.

Limitations: smallcases are pre-made, one-size-fits-all baskets. They do not customize the portfolio to an individual's personal savings targets, emergency fund needs, or specific investment horizon. Furthermore, there is no automatic monitoring during market hours to check if the portfolio has drifted away from its original balance.

2.2.2 Univest

What it is: Univest is a mobile application that gives retail investors buy, sell, and hold recommendations for individual stocks based on algorithmic scores.

Advantages: Easy-to-use mobile interface with clear trade signals.

Limitations: Univest focuses on individual stock tips rather than building a balanced, risk-controlled portfolio. It does not run mathematical optimization to protect investors from sector over-concentration.

2.2.3 Motilal Oswal — Intelligent Advisory Portfolios (IAP)

What it is: An algorithmic advisory service offered by a major full-service brokerage firm for affluent clients.

Advantages: High-quality institutional research, automated rebalancing, and direct broker integration.

Limitations: Requires a very high minimum investment (Rs. 5,00,000 or more), making it completely inaccessible to regular retail investors with modest savings. In addition, its stock selection formulas are hidden black boxes that do not explain the reasoning behind recommendations.

2.2.4 Savart

What it is: An AI-driven advisory app that builds custom portfolios based on a user's answers to psychological risk questions.

Advantages: Strong focus on understanding investor personality and long-term habits.

Limitations: Operates as a proprietary black box. Users receive a list of stocks and percentages, but the app does not show the underlying balance sheet numbers, economic checks, or independent risk critique that justified the choices.

2.2.5 Wright Research

What it is: A registered quantitative investment firm that creates portfolios using factor models (momentum, growth, and quality).

Advantages: Excellent mathematical discipline, strong historical performance, and structured risk rules.

Limitations: Built primarily for experienced investors. It lacks an interactive conversational assistant, does not provide automated Demat statement health reviews, and does not provide plain-English explanations for beginners.

2.2.6 Jarvis Invest

What it is: An artificial intelligence platform that tracks millions of market data points to suggest personalized stock allocations.

Advantages: Continuous risk management and automated rebalance alerts.

Limitations: Pure black-box system. Users cannot see why specific companies were chosen or rejected, leading to confusion and doubt when stock prices drop during normal market volatility.

2.2.7 MarketSenseAI (Academic Benchmark)

What it is: A pioneering academic research system developed by university researchers that uses teams of AI agents to analyze news, fundamentals, and price trends [2].

Advantages: Proved that coordinating multiple specialized AI agents produces higher-quality investment reasoning than using a single AI prompt.

Limitations: Built strictly for the United States stock market. It assumes investors can buy fractional shares (which is illegal in India), ignores Indian government taxes and exchange fees (STT, GST, stamp duty), and cannot read Indian Demat statements.

2.3 Comparison Matrix

Table 2.1 summarizes how Lucid compares against existing commercial platforms and academic systems across fifteen key capabilities.

Table 2.1: Comprehensive Comparison of Investment Platforms

Feature / Capability	smallcase	Univest	Motilal IAP	Savart	Wright Res.	Jarvis	MarketSense	Lucid (Ours)
Multi-Agent AI Architecture	No	No	No	No	No	No	**Yes**	**Yes (13 Nodes)**
Strict Separation of AI and Math	No	No	No	No	Yes	No	No	**Yes (SciPy SLSQP)**
Whole-Share Allocation (No Fractional)	Partial	No	Yes	Yes	Yes	Yes	No	**Yes (Discrete Lots)**
Protection of Unallocated Cash	No	No	Yes	Partial	Yes	Partial	No	**Yes (Cash Invariant)**
5-Minute Market Hours Drift Radar	No	No	No	No	No	Partial	No	**Yes (09:15–15:30 IST)**
Dual-Level Plain English Explanations	No	No	No	No	No	No	Partial	**Yes ("Stock-Why")**
Demat Statement (PDF) Health Diagnosis	No	Partial	No	No	No	No	No	**Yes (CDSL/NSDL CAS)**
Complete Indian Market Coverage	Partial	Partial	Partial	Partial	Partial	Partial	N/A	**Yes (2,639 Stocks)**
Macroeconomic Checks (RBI Repo / VIX)	No	No	Partial	No	Yes	Partial	Partial	**Yes**
Indian Tax & Fee Modeling (STT, GST)	No	No	Yes	No	No	No	No	**Yes**
Portfolio Concentration Review (HHI)	No	No	No	No	No	No	No	**Yes**
Hypothetical Stress Testing (e.g., COVID)	No	No	No	No	Partial	No	No	**Yes**
Strong Security (Argon2id Hashing)	Unclear	Unclear	Yes	Unclear	Unclear	Unclear	N/A	**Yes (RFC 9106)**
Low Minimum Investment ($< \text{Rs. } 10,000$)	**Yes**	**Yes**	No	**Yes**	**Yes**	**Yes**	N/A	**Yes**
Transparent Reasoning (Not a Black Box)	Partial	No	No	No	No	No	Yes	**Yes**

2.4 Conclusion of the Market Survey

The market survey reveals a clear gap in the Indian wealth management landscape:

- Everyday investors must choose between **generic stock baskets** (smallcase), **opaque black-box algorithms** (Savart, Jarvis), or **expensive wealth services** requiring Rs. 5,00,000 or more (Motilal Oswal IAP).
- Advanced academic AI systems (like MarketSenseAI) work well on paper, but cannot be used in India because they do not handle Indian exchange rules (such as whole shares and statutory taxes).

Lucid was created specifically to solve this problem. By combining a team of specialized AI agents for clear qualitative research with a deterministic mathematical engine for accurate calculations, Lucid gives regular Indian investors transparent, personalized, and institutional-grade portfolio creation without high fees or black-box secrecy.

Chapter 3

High-Level Design (HLD)

3.1 HLD Overview

The High-Level Design (HLD) explains the overall structure of Lucid, shows how its main components work together, and describes how an investor uses the application.

To make sure the application is fast, reliable, and safe, Lucid follows an important software engineering principle called **Separation of Concerns**. In simple words, this means dividing the application into specialized parts so that each part has only one clear responsibility:

- **Thinking and Reading**: The qualitative tasks—such as reading economic news, analyzing company financial reports, and writing natural explanations—are handled by a team of Artificial Intelligence agents.
- **Calculating and Validating**: The numerical calculations—such as solving mathematical equations to balance risk and return, ensuring percentages add up to exactly 100%, and rounding weights into whole share quantities—are handled by a dedicated mathematical calculation engine.
- **Storing and Protecting Data**: Storing user profiles, historical stock prices, and portfolio records is handled by an enterprise-grade relational database that ensures financial data is never lost or corrupted.

3.2 The Six Core Architectural Subsystems

Lucid is structured around six decoupled functional tiers that separate user interaction, API routing, multi-agent qualitative research, deterministic mathematics, persistent storage, and external market data (the complete visual system architecture diagram is presented in Chapter 6, Figure 6.1).

Here is what each major subsystem does in simple, clear language:

1. **Tier 1: Client Presentation Subsystem (Web Interface)**: This is what the user sees and interacts with on their computer or mobile phone. Built using **Next.js 16.4.2** and **React 19.3.0**, it provides 24 modular user interface components where investors can complete risk profiling

questionnaires, inspect interactive portfolio donut charts, explore dual-level "Stock-Why" justifications with official exchange links, and upload monthly Demat CAS statements.

2. **Tier 2: Backend API Gateway and Services Subsystem**: Powered by **FastAPI 0.115.8** running on Uvicorn ASGI. It exposes 9 modular REST routers spanning 82 unique endpoints. This tier acts as the secure gatekeeper: it verifies user passwords using Argon2id encryption, issues session tokens (JWT), enforces strict rate limits (120 req/min), runs a 300-second drift detection radar during Indian market hours, and coordinates business workflows.
3. **Tier 3: Multi-Agent AI Intelligence Subsystem**: This is the qualitative research committee of Lucid. Built on **LangGraph 0.4.0**, this subsystem coordinates a 13-node agent workflow using Google Gemini 3.8 Flash. Individual agents analyze macroeconomic interest rates (RBI), financial balance sheet health (ROE, P/E, Debt-to-Equity), and recent corporate filings, while an adversarial risk critic challenges risky selections. If cloud latency spikes, an automated circuit-breaker falls back to a deterministic rule engine in 0.48 seconds.
4. **Tier 4: Deterministic Math and Financial Engine Subsystem**: Completely separated from the artificial intelligence text models to prevent calculation errors, this layer uses **SciPy 1.15.2 SLSQP** to calculate optimal portfolio weights along the Markowitz Efficient Frontier. It enforces statutory concentration limits (single stock $\leq 25\%$, sector $\leq 40\%$), converts weights into exact whole-share counts, and computes official Indian transaction taxes (STT, Stamp Duty, GST, and exchange fees).
5. **Tier 5: Relational Database and Persistence Subsystem**: The central filing cabinet of the application, powered by **PostgreSQL 16.4** with SQLAlchemy 2.0 ORM and Alembic migrations. It stores all data across exactly 62 application tables (565 columns and 102 foreign-key constraints), ensuring that user accounts, historical daily prices, recommended portfolios, and audit trails are permanently preserved with full ACID transaction guarantees.
6. **Tier 6: External Market Data and Cloud Feeds Subsystem**: This component connects Lucid to outside financial data sources. It downloads

official stock lists from the National Stock Exchange of India (NSE) and Bombay Stock Exchange (BSE), fetches daily historical price bars and balance sheet metrics from Yahoo Finance, tracks monetary indicators from the Reserve Bank of India (RBI), and communicates with Google Gemini cloud models.

3.3 How Information Moves Through the System

The movement of information inside Lucid follows a structured, step-by-step pipeline from the moment an investor enters their financial details to the moment a complete, balanced portfolio is displayed on the screen (the complete end-to-end methodology workflow is visually detailed in Chapter 7, Figure 7.1):

1. **Step 1: Ingestion and Cleaning**: Lucid automatically downloads the official list of 2,639 listed Indian companies and their daily prices from the stock exchanges. It cleans the data and stores it in the database.
2. **Step 2: Understanding the Investor**: The investor fills out a simple questionnaire on the website, entering their savings amount (for example, Rs. 2,50,000), their time horizon (for example, 5 years), and answering five basic questions about their attitude toward risk.
3. **Step 3: Initial Screening**: The system filters the 2,639 listed stocks to find healthy, active companies that have sufficient trading volume and meet the investor's risk level.
4. **Step 4: AI Research Debate**: The team of AI agents evaluates the shortlisted companies. The macro agent checks interest rates, the fundamental agent checks balance sheets, the news agent reviews recent corporate developments, and the risk critic challenges risky selections. The agents assign quality scores and explain their thoughts in plain English.
5. **Step 5: Mathematical Calculation**: The mathematical solver takes the top-rated candidate stocks and calculates the safest and most profitable percentage weight for each company.
6. **Step 6: Converting into Whole Shares**: Because Indian stock exchanges do not allow fractional shares, the integer lot-sizing algorithm converts the percentage weights into exact whole share counts (such as

15 shares of Infosys or 8 shares of Reliance) and ensures that all leftover money remains safely in the investor's cash account.

7. **Step 7: Delivery and Explanation**: The final portfolio is displayed on the user's screen, complete with clear, plain-language explanations answering why each company was selected.
8. **Step 8: Automated Monitoring**: During official stock market hours, a background monitoring daemon checks the portfolio every 5 minutes to see if price changes have pushed the portfolio out of balance.

3.4 User-Level Workflows: Step-by-Step

To understand Lucid from an ordinary investor's point of view, let us walk through the six primary user journeys:

3.4.1 Workflow 1: Creating an Account and Risk Profiling

- **What does the user do?**: The investor signs up with their email and a secure password. They answer five straightforward questions about their investment experience, their savings goals, and how they would feel if the stock market dropped temporarily.
- **What does Lucid do behind the scenes?**: Lucid encrypts the password using a high-security algorithm (Argon2id) so it can never be stolen. It evaluates the answers using standard SEBI risk guidelines and assigns an overall risk category: **Conservative** (prefers safety), **Moderate** (balances safety and growth), or **Aggressive** (aims for maximum long-term growth).
- **What does the user receive?**: A personalized risk summary showing their category, a clear explanation of what it means, and personalized guidance on how much money should be kept in safe emergency savings before investing in stocks.

3.4.2 Workflow 2: Goal-Driven Portfolio Creation

- **What does the user do?**: The investor enters how much money they want to invest (for example, Rs. 1,00,000), their target timeline (for example, 7 years for buying a home), and any industries they wish to exclude (such as tobacco or defense stocks).
- **What does Lucid do behind the scenes?**: The candidate selector filters healthy companies, the AI agent team reviews corporate balance sheets and economic news, and the mathematical solver calculates the safest combination of weights. The whole-share engine converts these weights into exact full share quantities.
- **What does the user receive?**: A clean, easy-to-understand portfolio table showing the names of the selected companies, how many shares to buy of each, the current price, the total cost, and the exact leftover cash balance.

3.4.3 Workflow 3: Inspecting Stock-Why Explanations

- **What does the user do?**: The investor clicks on any stock in their recommended portfolio (for example, clicking on Tata Consultancy Services).
- **What does Lucid do behind the scenes?**: Lucid retrieves the audit records generated by the AI research agents, including the balance sheet review, the news sentiment score, and the counter-arguments raised by the risk critic.
- **What does the user receive?**: A pop-up card written in simple, friendly English explaining exactly why this company fits the investor's goal. For users who want deeper financial details, a tab displays underlying financial ratios (such as Return on Equity, Debt-to-Equity, and Price-to-Earnings ratios).

3.4.4 Workflow 4: Reviewing an Existing Demat Portfolio (CAS Diagnosis)

- **What does the user do?**: An investor who already owns shares through brokers like Zerodha, Groww, or Upstox downloads their official monthly

account statement (called a **Consolidated Account Statement**, or CAS PDF) and uploads it to Lucid.

- **What does Lucid do behind the scenes?**: Lucid reads the PDF statement, identifies each stock by its standard international identification code (ISIN), checks current market prices, and calculates whether the portfolio is dangerously concentrated in one or two companies.
- **What does the user receive?**: A health diagnosis report with a simple health score, a breakdown of sector risks, and clear recommendations on which stocks are healthy to hold and which should be reduced to improve safety.

3.4.5 Workflow 5: 5-Minute Market Hours Drift Monitoring

- **What does the user do?**: Nothing! This feature works automatically in the background without requiring the user to stare at market screens.
- **What does Lucid do behind the scenes?**: During Indian trading hours (Monday to Friday, 09:15 to 15:30 IST), a background monitoring program wakes up every 300 seconds (5 minutes). It checks live prices and calculates whether any stock has risen or fallen so much that its weight has drifted by 5% or more from the planned target.
- **What does the user receive?**: If a significant shift occurs, the investor receives a friendly alert notifying them that their portfolio has drifted and suggesting that they review a rebalancing preview.

3.4.6 Workflow 6: Rebalance Impact and Fee Preview

- **What does the user do?**: When an alert is received, the user opens the rebalancing preview window.
- **What does Lucid do behind the scenes?**: Lucid calculates the exact number of shares to buy or sell to restore balance. It also calculates mandatory Indian government taxes and exchange fees—including Securities Transaction Tax (STT), Goods and Services Tax (GST), exchange turnover charges, and stamp duty.
- **What does the user receive?**: A transparent before-and-after table

showing current holdings versus proposed holdings, exact buy/sell share counts, the total estimated transaction costs, and the final cash balance before the user confirms any action.

3.5 Key Architectural Design Decisions

To ensure the application is reliable and transparent, four critical design decisions were made:

1. **Why use PostgreSQL as the Single Source of Truth?**: Financial applications require complete accuracy. If an investor updates a portfolio, the change must be recorded permanently without any chance of data mismatch. PostgreSQL 16 is an enterprise relational database that provides strict guarantees (known as ACID: Atomicity, Consistency, Isolation, and Durability) ensuring that financial data is never lost or corrupted.
2. **Why use SciPy SLSQP instead of letting AI models calculate weights?**: Language models make math errors. Relying on an AI model to guess investment percentages is dangerous. Delegating all numerical calculations to a proven mathematical solver (SciPy SLSQP) guarantees that all portfolio weights sum to exactly 100% and strictly obey diversification safety limits.
3. **Why enforce Whole Shares with Cash Conservation?**: Because fractional shares are illegal on Indian stock exchanges, theoretical decimal weights (such as 2.4 shares) cannot be bought. Lucid converts continuous weights into exact full shares and guarantees that any leftover rupees remain safely in the investor's cash balance.
4. **Why use LangGraph for Agent Workflows?**: LangGraph allows multiple specialized AI agents to collaborate in a structured sequence, review each other's work, handle retries if an external API fails, and output clean, verifiable explanations.

Chapter 4

Low-Level Design (LLD)

4.1 LLD Overview

While the High-Level Design described the broad architecture of Lucid, the Low-Level Design (LLD) explains how the individual software components, data structures, and algorithms operate internally.

This chapter breaks down the system into understandable parts:

- How the web screens are structured in the frontend.
- How the thirteen specialized AI agents collaborate step by step.
- How the mathematical engine calculates whole shares and taxes.
- How the sequence flows handle user logins, portfolio generation, background monitoring, and statement analysis.

4.2 Frontend Architecture: What the User Experiences

The client application is built using **Next.js 16.4.2** and **React 19.3.0** (located under `frontend/src/app/`). React allows the application to update parts of the screen instantly (such as updating stock charts or showing alerts) without needing to reload the entire web page.

4.2.1 Key Screen Components and What They Do

The user interface is broken down into modular building blocks located in `frontend/src/components/`:

- `Navbar.tsx` **and** `Footer.tsx`: The top navigation bar and bottom footer. They show whether the user is logged in, display the active profile, and provide buttons to jump between the dashboard, portfolio wizard, and notifications.
- `LandingPage.tsx`: The welcoming home page that explains what Lucid does, showcases key features, and invites new users to get started.

- `AuthView.tsx` **and** `ProfileAuthModal.tsx`: Clean sign-up and login dialogs where users enter their email and password.
- `OnboardingWizard.tsx` **and** `RiskQuestionnaireView.tsx`: A step-by-step questionnaire that asks users five simple questions to understand their investment horizon, financial goals, and comfort with market risk.
- `PortfolioGenerator.tsx`: The main configuration panel where users type how much money they want to invest, select their savings goal, and specify any sectors they want to avoid.
- `HoldingsView.tsx` **and** `AllocationVisualizer.tsx`: The primary portfolio display. It presents an easy-to-read table showing company names, share counts, purchase prices, current market values, and colorful pie charts of sector allocations.
- `StockWhyModal.tsx`: An interactive pop-up card that appears when an investor clicks on any stock, explaining in plain English why that company was selected for their portfolio.
- `DiagnosisView.tsx`: A drag-and-drop file upload screen where users can drop their monthly Demat PDF statement to receive an instant portfolio health check.
- `RebalancePreviewModal.tsx`: A clear before-and-after review card showing proposed buy and sell share quantities, cash changes, and estimated transaction fees before the user confirms a rebalance.
- `ConversationalAssistant.tsx`: A friendly AI chat assistant where users can ask questions about their portfolio or explore hypothetical "what-if" scenarios (such as "What happens to my stocks if interest rates rise?").
- `NotificationsPanel.tsx`: An alert bell icon and sliding drawer that informs the investor whenever a portfolio has drifted or needs attention.

4.2.2 Type-Safe Client API Layer

When a user clicks a button, the web screen does not talk directly to the database. Instead, all network requests go through a helper module (`frontend/src/lib/api.ts`). This helper:

- Attaches the user's secret digital session badge (`Bearer` token) to prove their identity safely.

- Converts technical server responses into clear, helpful error messages (such as reminding the user if an entered investment amount is below Rs. 10,000).

4.3 Agent-Level Design: The 13-Node LangGraph Research Workflow

Rather than relying on a single AI prompt, Lucid coordinates thirteen specialized digital execution nodes organized in a structured sequence using **LangGraph 0.4**.

Figure 4.1 illustrates the complete 13-node multi-agent workflow, the five-stage pipeline, the 35-field shared state, and the deterministic fallback circuit-breaker:

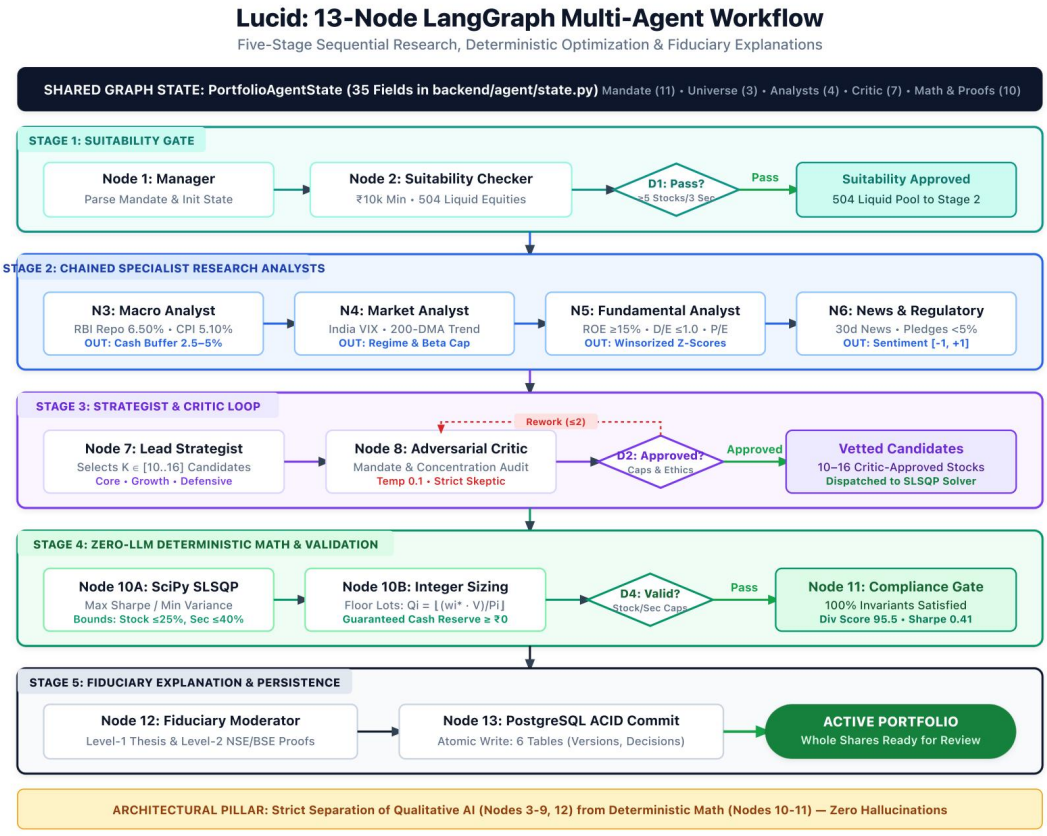


Figure 4.1: Lucid 13-Node Multi-Agent LangGraph Workflow and State Topology

4.3.1 The Shared Graph State: PortfolioAgentState

All thirteen nodes communicate through a shared digital blackboard called the `PortfolioAgentState` (defined in `backend/agent/state.py`). It holds **35 active fields** organized into five clear groups:

1. **Investor Mandate (11 fields)**: Stores the investor's identification, starting capital in rupees ($V_{\text{total}} \geq$ Rs. 10,000), investment time horizon (in months), 5-question risk score ($S \in [5, 20]$), risk profile tier, savings goal, excluded industry sectors, ethical preference tags, target portfolio beta, and statutory allocation caps (maximum 20% to 25% per company, and 30% to 40% per industry sector).
2. **Liquid Universe (3 fields)**: Tracks whether the investor passed initial suitability checks (`suitability_passed`), the active pool of 504 liquid Indian stocks, and the filtered candidate shortlist.
3. **Analyst Reports (4 fields)**: Holds the structured evaluation reports generated by the four specialist research analysts: Macroeconomic analysis, Market Regime assessment, Company Fundamental screening, and News/SEBI regulatory audit.
4. **Critic and Loop Control (7 fields)**: Records the Strategist's proposed stock basket ($K \in [10, 16]$ companies), the Critic's audit objections, ethical violation flags, rework iteration counter (`rework_count` ≤ 2), Critic approval status (`critic_approved`), replacement candidates, and the audit event history.
5. **Deterministic Math and Fiduciary Evidence (10 fields)**: Holds the mathematically optimal percentage weights (w_i^*), exact integer share counts (Q_i), unallocated cash reserve ($C_{\text{res}} \geq$ Rs. 0), expected portfolio return (μ_p), expected risk (σ_p), Sharpe ratio, 95% Value-at-Risk, maximum drawdown, concentration score (HHI), Diversification Score (out of 100), and plain-English Stock-Why evidence cards.

4.3.2 LLM Engine and the 0.48-Second Fallback Circuit-Breaker

Lucid uses **Google Gemini 3.8 Flash** (`gemini-3.8-flash`) as its primary qualitative language model. To ensure factual and repeatable answers, the model

is configured with very low creativity settings (temperature 0.1 to 0.2) and strict response bounds ($\leq 1,500$ tokens per turn).

Zero-Downtime Fallback Circuit-Breaker: If the Google Gemini cloud service encounters network delays, internet disruptions, or free-tier rate limits, Lucid does not fail or display an error message. Instead, an automated circuit-breaker instantly activates a local deterministic rule engine. This offline fallback engine completes candidate selection and validation in **0.48 seconds**, guaranteeing continuous availability.

4.3.3 Five Execution Stages and Four Decision Routers

The thirteen nodes are organized into five structured stages, linked by four intelligent decision routers:

Stage 1: Suitability Gate (Nodes N1–N2, Router D1):

- **Node 1 (Manager Node)**: Receives the user request, extracts the financial parameters, and prepares the shared state.
- **Node 2 (Suitability Node)**: Verifies that starting capital meets the minimum requirement ($V_{\text{total}} \geq$ Rs. $10,000$), confirms investor risk answers, and filters the 2,639 listed exchange stocks down to the 504 liquid equity pool, ensuring at least 5 viable companies across at least 3 distinct economic sectors.
- **Router D1 (`route_suitability`)**: If checks pass, the workflow moves forward to Node 3. If any check fails, it aborts immediately to Node 13 to log the reason.

Stage 2: Sequential Chained Specialist Analysts (Nodes N3 $\rightarrow$ N4 $\rightarrow$ N5 $\rightarrow$ N6): The four research nodes run in a chained sequence, each reading the findings of the previous analyst and using dedicated software tools:

- **Node 3 (Macro Analyst)**: Using `macro_data_tool`, it examines national benchmarks from the Reserve Bank of India (RBI Repo Rate at 6.50%, CPI at 5.10%, and 10-year G-Sec yield at 7.04%). It recommends setting aside a cash liquidity buffer between 2.5% and 5.0%.
- **Node 4 (Market Analyst)**: Using `market_technical_tool`, it evaluates market volatility with India VIX ($<$ 13 Calm, 13–18 Moderate, 18–24 Volatile, $>$ 24 Extreme) and compares Nifty 50 against its 200-DMA, setting the regime tag and stock beta ceilings.

- **Node 5 (Fundamental Analyst)**: Using `fundamental_research_tool`, it evaluates balance sheets, requiring Return on Equity ROE $\geq 15\%$, positive earnings (P/E $\in [1, 150]$), leverage Debt/Equity ≤ 1.0, and sector Z-scores within $[-3.0, +3.0]$.
- **Node 6 (News and SEBI Analyst)**: Using `news_filings_audit_tool`, it reviews 30-day headlines, verifies promoter pledges are below 5%, and audits SEBI SCORES filings, assigning a sentiment score in $[-1.0, +1.0]$.

Stage 3: Strategist and Critic Loop (Nodes N7–N9, Router D2):

- **Node 7 (Lead Strategist)**: Synthesizes the four analyst reports and selects $K \in [10, 16]$ candidate companies divided into Core, Growth, and Defensive categories, mathematically ensuring $K \geq \lceil 100/\text{max_stock\%} \rceil$.
- **Node 8 (Adversarial Risk Critic)**: Acting like a skeptical chief risk officer (at temperature 0.1), audits the chosen stocks for excessive industry concentration, high valuation traps, or emerging regulatory controversies.
- **Node 9 (Candidate Rework Node)**: Replaces any stock flagged by the Critic with a healthy alternative from the 504 liquid stock pool.
- **Router D2 (`route_critic`)**: If the Critic approves the portfolio, execution proceeds to Stage 4. If objections are raised and the rework counter is below 2, it routes back to Node 9 for candidate replacement. If two rework cycles are exhausted without resolution, it aborts safely to Node 13.

Stage 4: Zero-LLM Deterministic Mathematical Optimization and Validation (Nodes N10–N11, Routers D3 & D4):

- **Node 10A (SciPy SLSQP Solver)**: Solves the constrained quadratic optimization problem without any AI involvement. It maximizes the Sharpe ratio for Moderate and Aggressive investors, or minimizes risk for Conservative investors, strictly enforcing the 25% single-stock and 40% sector limits. If the solver encounters an impossible mathematical combination, **Router D3** triggers a candidate swap back to Node 9.
- **Node 10B (Whole-Share Integer Lot Allocator)**: Translates continuous weights (w_i^*) into whole share numbers using floor rounding ($Q_i = \lfloor (w_i^* V_{\text{total}})/P_i \rfloor$). A greedy remainder pass uses leftover cash to buy an extra share of the highest-rated holding where permitted, strictly guaran-

teeing positive unallocated cash ($C_{\text{res}} \geq$ Rs. 0). It calculates the portfolio's Sharpe ratio, annual volatility, 95% VaR, max drawdown, and Diversification Score.

- **Node 11 (Post-Rounding Constraint Gate)**: Programmatically verifies that whole-share rounding did not cause any holding to exceed statutory ceilings (stock $\leq$ 20/25%, sector $\leq$ 30/40%) and confirms $C_{\text{res}} \geq$ Rs. 0.
- **Router D4 (`route_final_critic`)**: If post-rounding validation passes, execution moves to Stage 5. If a rounding rule is violated, it aborts safely to Node 13.

Stage 5: Fiduciary Explanation and ACID Persistence (Nodes N12–N13):

- **Node 12 (Fiduciary Moderator)**: Generates transparent explanations at two levels: Level 1 summarizes the portfolio thesis and macroeconomic alignment; Level 2 produces stock cards with verifiable financial ratios (ROE, P/E, Z-scores) and direct links to official NSE and BSE company pages.
- **Node 13 (ACID Relational Persistence Node)**: Concludes the workflow by atomically saving the complete portfolio across six PostgreSQL database tables. All aborted execution paths from Routers D1, D2, and D4 also converge here to record a complete audit trail before ending cleanly.

4.3.4 Architectural Pillar: Strict Separation of Qualitative AI from Math

A core design rule in Lucid is that **artificial intelligence is never permitted to perform arithmetic or determine share quantities**. AI language models are trained on language patterns and can make unexpected calculation errors. In Lucid, the AI models are restricted strictly to qualitative tasks: reading corporate news, auditing financial statements, and writing clear explanations. All numerical optimization, percentage allocations, integer share quantities, and cash calculations are executed exclusively by deterministic Python algorithms (SciPy SLSQP and integer rounding). This strict separation guarantees that every recommendation is 100% mathematically sound, repeatable, and free from

AI hallucinations. Under normal operation with cached market data, end-to-end portfolio creation completes in approximately **2.14 seconds**.

4.4 The Mathematical Calculation Engine

Located in `backend/portfolio_engine/`, the mathematical engine handles numerical calculations without any AI involvement:

- **Optimizer (`optimizer.py`)**: Uses a numerical solver (SciPy SLSQP) to find the portfolio weights that maximize the Sharpe ratio (or minimize risk for conservative profiles) while strictly enforcing:

$$\sum_{i=1}^{K} w_i = 1.0 \quad \text{(100\% capital allocation)}$$
$$w_i \geq 0.0 \quad \text{(no short selling)}$$
$$w_i \leq 0.25 \quad \text{(maximum 25\% per stock)}$$

- **Integer Lot Sizing (`integer_rounding.py`)**: Calculates the exact number of whole shares to purchase:

$$q_i = \left\lfloor \frac{C \cdot w_i^*}{P_i} \right\rfloor \tag{4.1}$$

  where:

  - q_i is the number of whole shares of stock i.
  - C is the total investment amount (such as Rs. 1,00,000).
  - w_i^* is the optimal percentage weight of stock i (such as 10% or Rs. 10,000).
  - P_i is the current market price of one share (such as Rs. 1,400).
  - $\lfloor \ldots \rfloor$ represents floor rounding (rounding down to the nearest whole integer).

  Numerical Example: $10,000/1,400 = 7.142$ shares. Since Indian exchanges do not sell fractional shares, the algorithm rounds down to exactly 7 whole shares. The cost is $7 \times$ Rs. $1,400 =$ Rs. $9,800$, leaving Rs. 200 in cash. A secondary pass checks whether any leftover cash can purchase an additional share of another high-rated, lower-priced holding without

exceeding the 25% single-stock limit. Any remaining money is preserved as unallocated cash:

$$\text{Residual Cash} = C - \sum_{i=1}^{K} q_i P_i \geq 0. \tag{4.2}$$

- **Transaction Cost Engine (`transaction_costs.py`)**: Calculates all statutory taxes and fees applicable to Indian stock trades:
  - Securities Transaction Tax (STT): 0.10% on delivery equity transactions.
  - Exchange Turnover Fee: 0.00297% on NSE and 0.00375% on BSE (statutory schedule).
  - SEBI Regulatory Turnover Charge: 0.0001% (Rs. 10 per crore).
  - Stamp Duty: 0.015% charged on purchases under the Indian Stamp Act.
  - Goods and Services Tax (GST): 18.0% applied to brokerage and exchange charges.

4.5 Background Monitoring Daemon and Advisory Locking

Located in the scheduler service (`backend/services/monitoring_scheduler.py`), the background daemon automatically evaluates active portfolios during Indian exchange trading hours (Monday through Friday, 09:15 to 15:30 IST). Every 300 seconds (5 minutes), the daemon wakes up, checks live stock prices, and calculates if any stock's current weight has moved away from its target by 5% or more.

How Duplicate Runs are Prevented (Advisory Lock): In a commercial server deployment, multiple backend processes may run simultaneously to handle web traffic. If every process ran the monitoring loop at the same time, the system would waste resources and send duplicate alerts to users. To solve this, Lucid uses an enterprise feature of PostgreSQL called a **Session Advisory Lock**:

```
SELECT pg_try_advisory_lock(847291);
```

Think of this like a physical key to an office. Only the single server process that successfully picks up the key runs the 5-minute monitoring cycle. Other processes see that the key is in use, politely skip the cycle, and return to handling user web requests.

4.6 Sequence Flow Diagrams: Step-by-Step Interactions

Figures 4.2 through 4.5 show the step-by-step interactions between the user, the frontend interface, the backend server, the AI/optimization engine, and the database for the core features of Lucid:

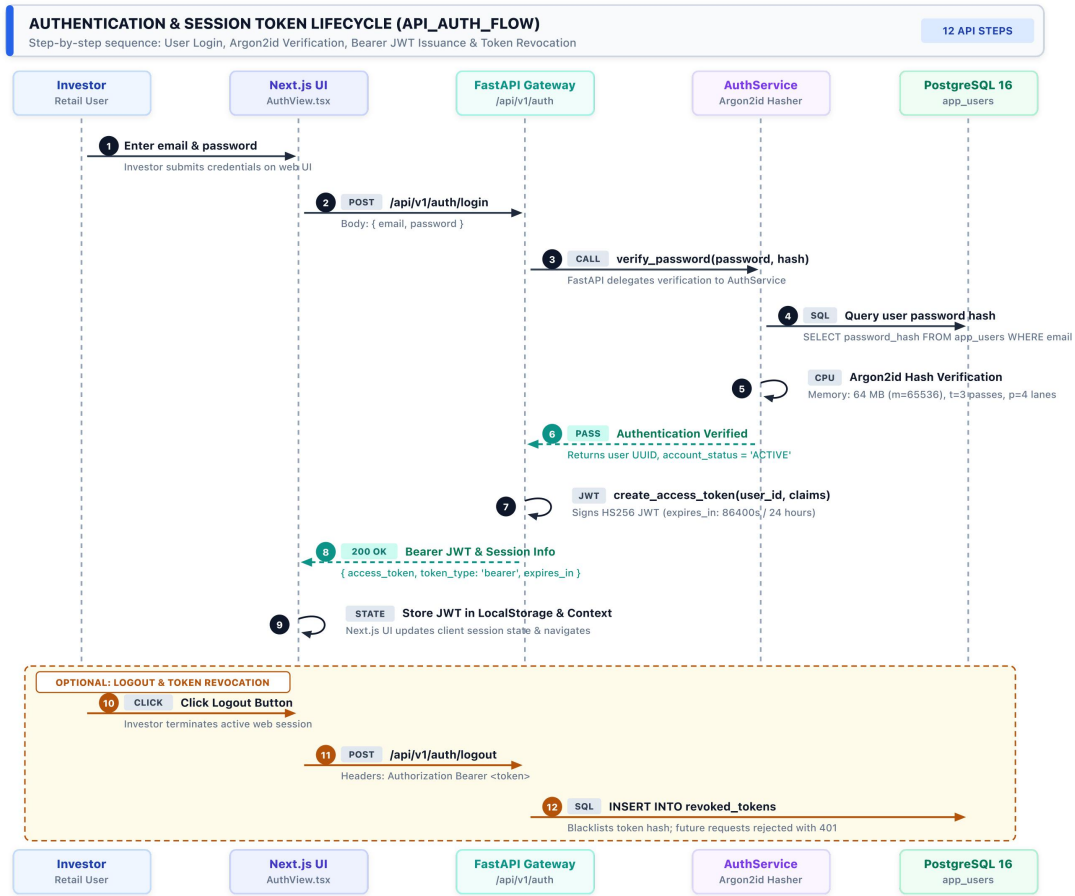


Figure 4.2: Sequence Flow: User Registration, Authentication, and Secure Token Creation

Figure 4.2 shows how an investor signs up. The user types their email and password into the frontend. The backend validates the inputs, hashes the pass-

word using Argon2id so it can never be viewed by anyone, stores the record in PostgreSQL, and returns an encrypted session token (JWT) to keep the user safely logged in.

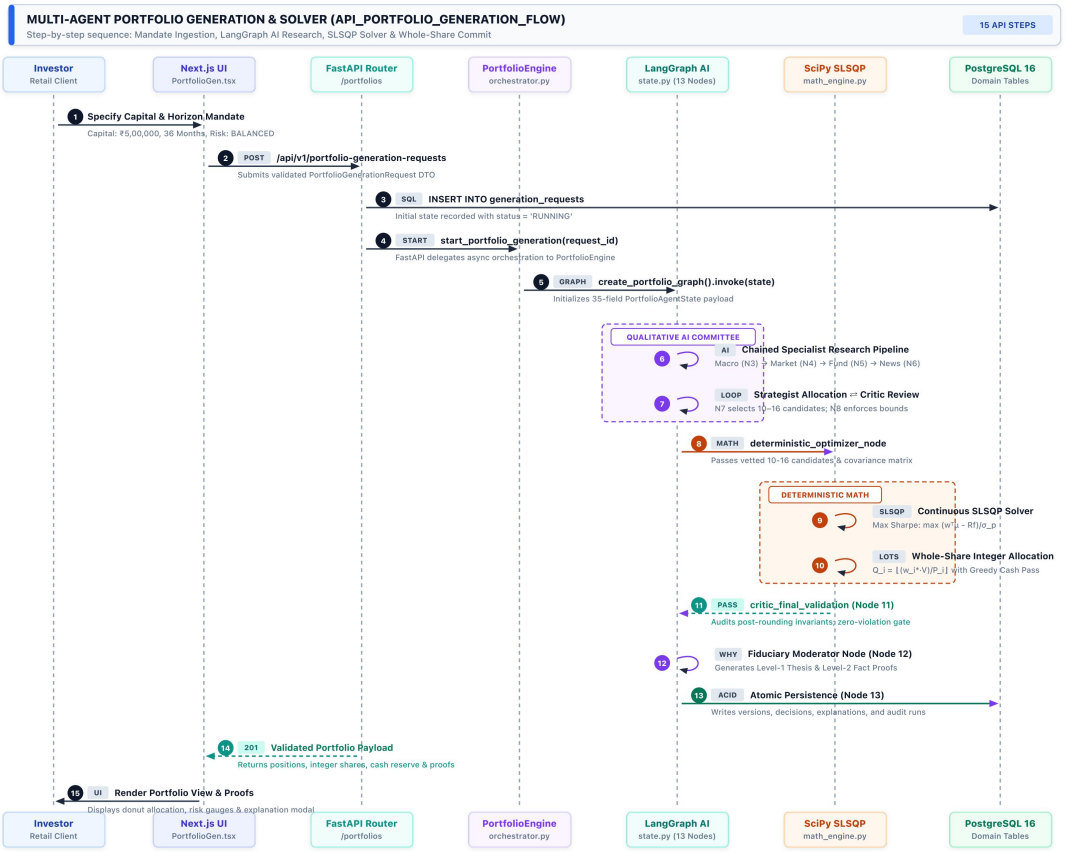


Figure 4.3: Sequence Flow: Goal-Driven Portfolio Generation and Mathematical Optimization

Figure 4.3 traces the complete portfolio creation request. The user specifies their savings amount and financial goal. The backend queries the database for healthy stocks, runs the LangGraph AI agents to evaluate balance sheets and news, calls the SciPy solver to calculate optimal weights, converts percentages into whole share quantities, and saves the new portfolio version with complete explanations for display on the screen.

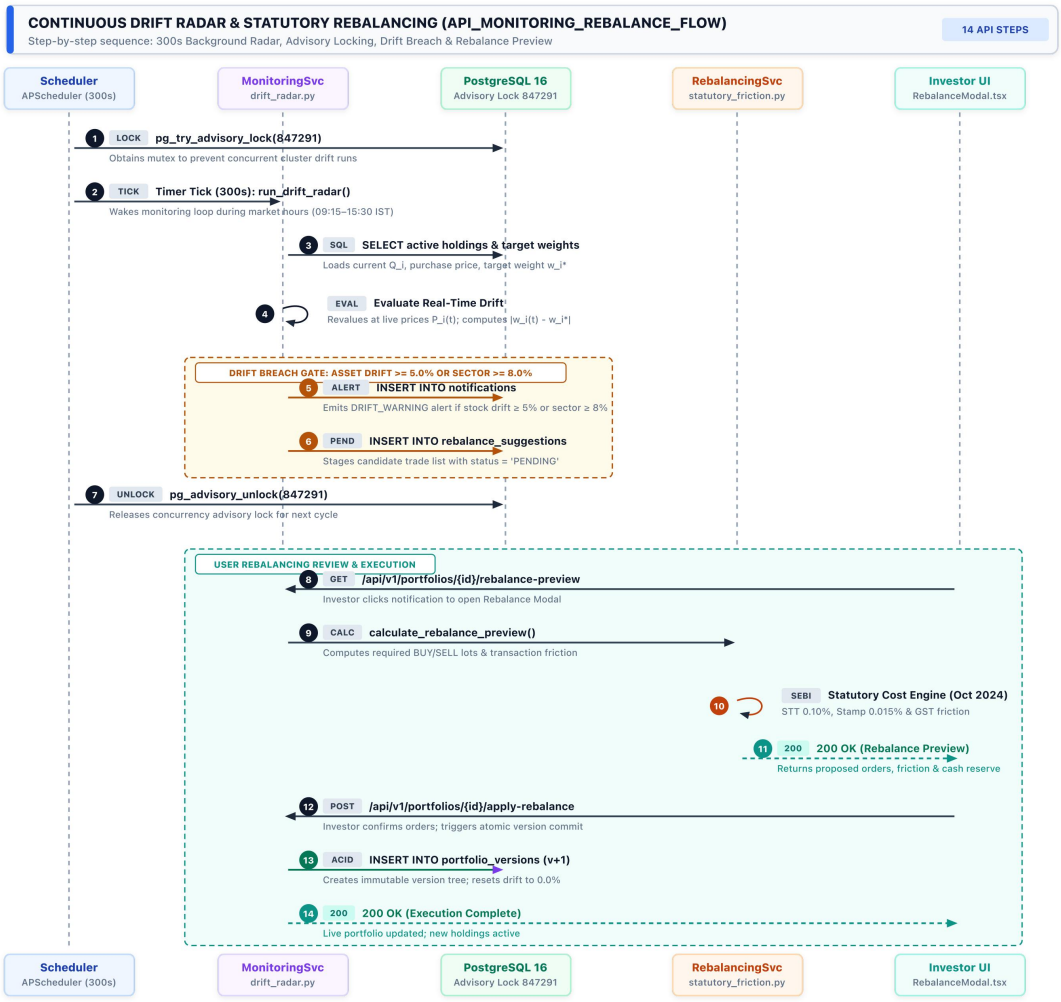


Figure 4.4: Sequence Flow: 5-Minute Drift Radar and Rebalance Order Preview

Figure 4.4 shows the background monitoring radar. During market hours, the scheduler wakes up, obtains the advisory lock, revalues active portfolios against live stock prices, and calculates drift. If a stock drifts by 5% or more, it generates a notification. When the user clicks the notification, Lucid displays a complete before-and-after rebalancing preview with statutory tax breakdowns.

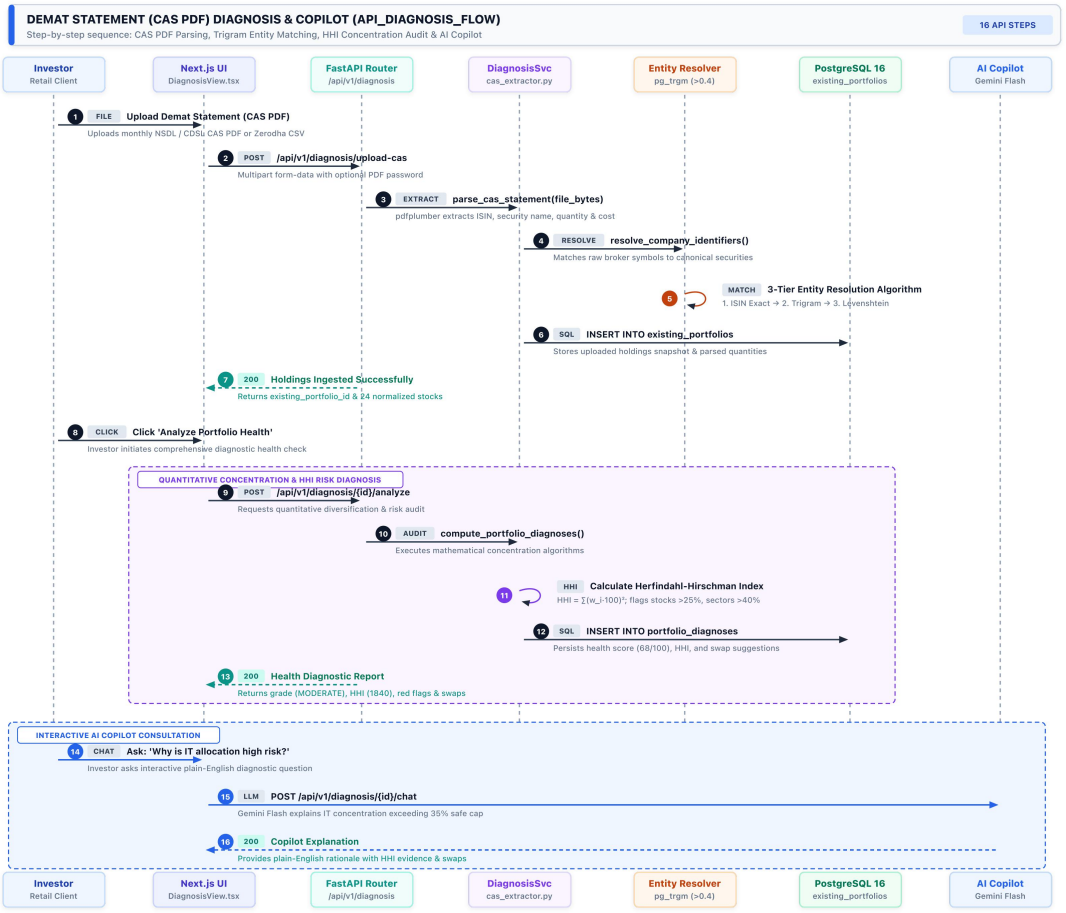


Figure 4.5: Sequence Flow: Demat Statement (CAS PDF) Upload and Health Diagnosis

Figure 4.5 illustrates how Lucid analyzes an existing investment portfolio. The user uploads a monthly Demat PDF statement. The backend extracts the stock codes, matches them against the security database, calculates total valuation and industry concentration, and displays an actionable health report with clear safety recommendations.

Chapter 5

Database Design and Entity-Relationship Architecture

5.1 Database Overview: What It Is and Why Lucid Needs It

A **database** is an electronic filing cabinet that securely stores all the data needed by an application. In financial software like Lucid, data accuracy is extremely important. If an investor saves a portfolio, buys shares, or records an investment goal, that information must be stored permanently, accurately, and securely without any risk of loss or accidental corruption.

To understand how Lucid stores information, it helps to understand a few basic database concepts:

- **Table**: A table works like a structured digital spreadsheet dedicated to a single topic. For example, one table stores user accounts, another stores company names, and a third stores daily stock prices.
- **Column and Row**: Each row represents a single record (such as one registered investor). Each column represents a specific attribute of that record (such as their email address, date of registration, or risk category).
- **Primary Key (PK)**: A unique identification code (like a Passport or PAN number) given to each row so that no two records are ever confused. Lucid uses standard unique identifiers (called **UUIDs**, such as `c56a4180-65aa-42ec-a945-5fd21dec0538`) rather than simple sequential numbers, ensuring that user accounts cannot be guessed by outsiders.
- **Foreign Key (FK)**: A digital pointer that connects a row in one table to a row in another table. For example, a row in the `investor_portfolios` table includes a `user_id` foreign key pointing to the `app_users` table, ensuring that every portfolio is permanently tied to the exact user who created it.
- **Entity-Relationship (ER) Diagram**: A visual map showing database tables and the foreign-key lines connecting them.

Lucid uses **PostgreSQL 16**, an enterprise open-source relational database. All database tables are defined in Python using **SQLAlchemy 2.0** and version-controlled with **Alembic**.

5.1.1 Scale of the Database: 62 Tables Across 8 Domains

The Lucid database contains exactly **62 application tables** (plus 1 migration tracking table called `alembic_version`, for a total of 63 tables). To keep this large system organized and easy to maintain, the tables are divided into eight functional **domains**:

1. **Domain 1: User Identity, Authentication, and Security** (5 tables): Stores user accounts, encrypted passwords, login sessions, and security event logs.
2. **Domain 2: Investor Suitability, Profiling, and Goals** (10 tables): Records risk questionnaire answers, calculated risk scores, financial targets, and stock exclusion preferences.
3. **Domain 3: Security Master and Corporate Registry** (8 tables): Maintains the master catalog of 2,639 listed Indian companies across the NSE and BSE, their industry sectors, and exchange ticker codes.
4. **Domain 4: Market Data, Historical Prices, and Macro Factors** (8 tables): Stores daily stock prices, company financial ratios, inflation rates, central bank interest rates, and financial news.
5. **Domain 5: Portfolio Engine, Optimization, and Tracking** (10 tables): Stores recommended portfolios, calculated whole-share quantities, version histories, and daily portfolio valuations.
6. **Domain 6: Broker Integration and Order Simulation** (5 tables): Manages connections to brokerage accounts (such as Zerodha and Groww) and tracks simulated buy and sell orders.
7. **Domain 7: Demat CAS Diagnosis and Portfolio Health** (3 tables): Stores stock holdings parsed from uploaded monthly Demat PDF statements and saves health diagnostic scores.
8. **Domain 8: AI Agent Telemetry, Reasoning Traces, and Audits** (13 tables): Records the step-by-step thinking of the 13 AI agents, plain-English "Stock-Why" explanations, and real-time user notifications.

5.2 Master Entity-Relationship Architecture

Figure 5.1 illustrates the complete master Entity-Relationship diagram connecting all 62 application tables across the platform:

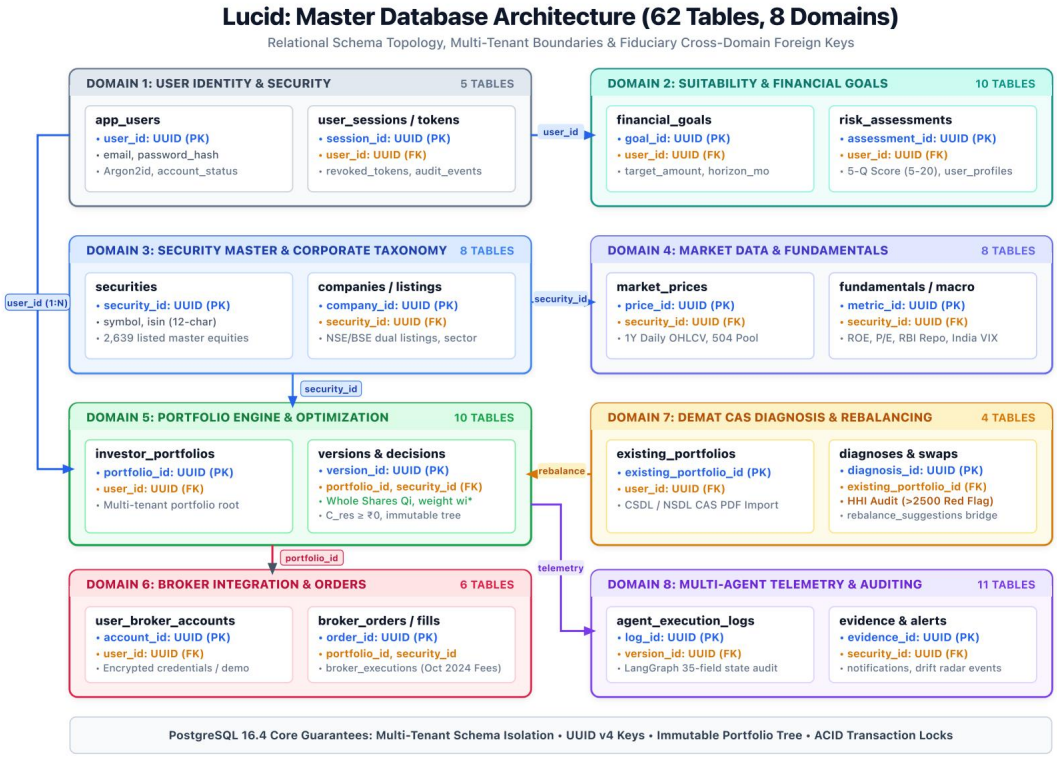


Figure 5.1: Lucid Master Database Architecture (62 Tables, 8 Domains)

In Figure 5.1, boxes represent individual database tables, grouped by color according to their functional domain. The lines connecting the boxes represent foreign-key relationships. For instance:

- The central `APP_USERS` table connects to `INVESTOR_PORTFOLIOS`, ensuring multi-tenant data isolation so one user can never view another user's investments.
- The `INVESTOR_PORTFOLIOS` table connects to `PORTFOLIO_VERSIONS`, preserving an immutable historical record of every change made to a portfolio over time.
- Each portfolio version links to `PORTFOLIO_SECURITY_DECISIONS`, which identifies the exact shares, buy prices, and rupee amounts allocated to specific companies in the `SECURITIES` table.

5.3 Detailed Walkthrough of the Eight Functional Domains

5.3.1 Domain 1: User Identity, Authentication, and Security Governance

Domain 1 manages user sign-ups, secure logins, password protection, and safety audit trails. Figure 5.2 shows how the security tables connect to the primary user table:

Core Table: `app_users`

Table 5.1 shows the structure of the main user identity table:

Table 5.1: Schema Specification: `app_users`

Column Name	Data Type	PK	FK	Nullable	Description & Purpose in Simple English
`user_id`	`UUID`	Yes	No	No	Unique identifier for each registered investor.
`identity_provider`	`VARCHAR(50)`	No	No	No	How the user signed in (`local` email/password, `google`, or `apple`).
`identity_subject`	`VARCHAR(255)`	No	No	No	External identification code from the login provider.
`email`	`VARCHAR(255)`	No	No	No	The investor's registered email address (unique).
`password_hash`	`VARCHAR(255)`	No	No	Yes	Salted Argon2id cryptographic hash (original password is never stored).
`mobile_number`	`VARCHAR(20)`	No	No	Yes	Investor's mobile phone number.
`email_verified`	`BOOLEAN`	No	No	No	True if the investor confirmed their email address.
`mobile_verified`	`BOOLEAN`	No	No	No	True if the investor verified their mobile number with an OTP.
`account_status`	`VARCHAR(50)`	No	No	No	Status: `'active'`, `'pending'`, or `'suspended'`.
`created_at`	`TIMESTAMPTZ`	No	No	No	Exact date and time when the account was registered.
`updated_at`	`TIMESTAMPTZ`	No	No	No	Date and time when the profile was last modified.

Additional Tables in Domain 1:

- `revoked_tokens`: Stores canceled session tokens so that when an investor clicks "Log Out," that session token can never be used again.
- `user_consents`: Records that the user agreed to the privacy policy, terms of service, and SEBI investment risk disclosures.
- `user_verification_codes`: One-time passwords (OTPs) to verify emails and mobile numbers.
- `audit_events`: An unalterable security ledger that logs important actions (such as password changes, login attempts, or profile updates) to prevent unauthorized tampering.

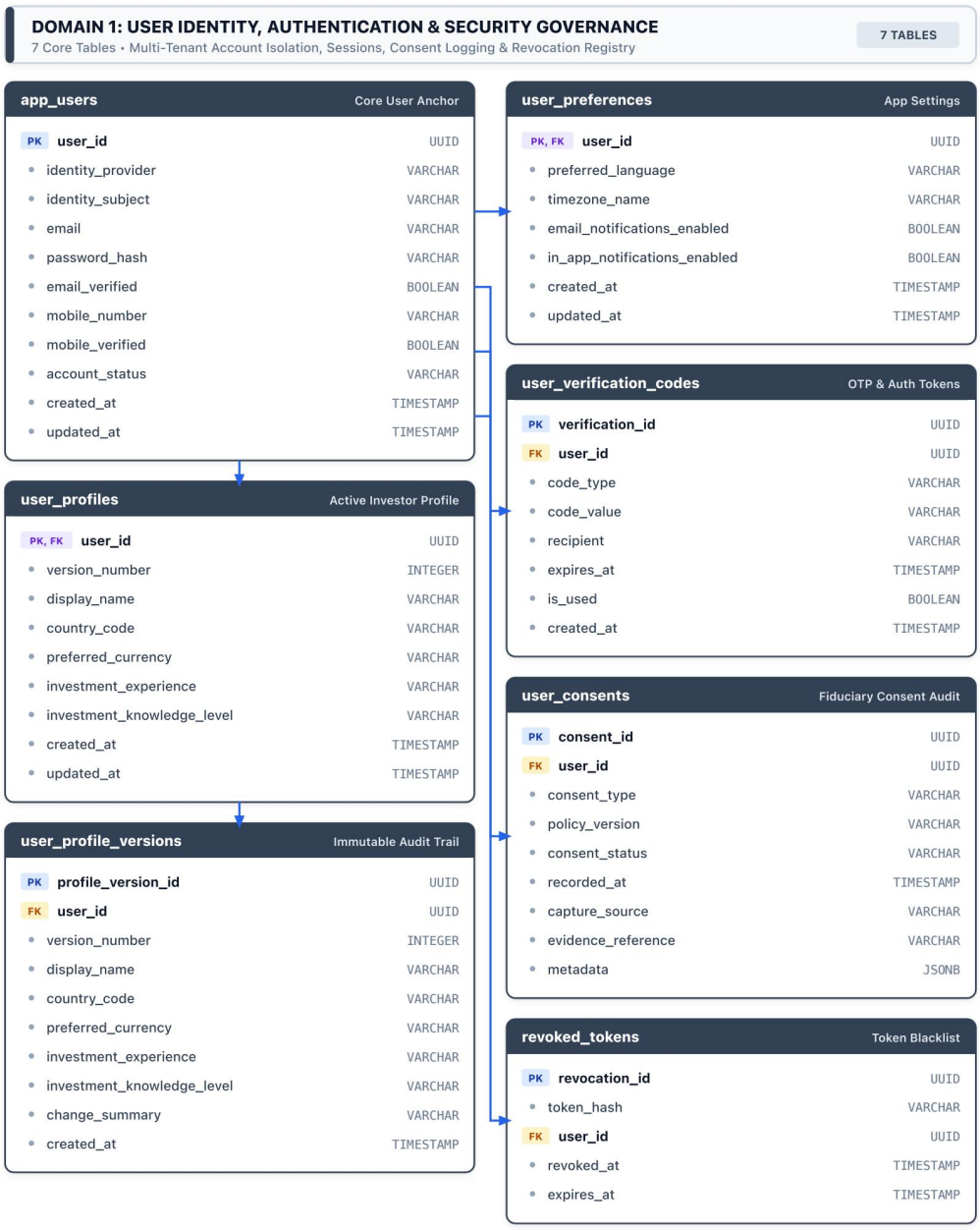


Figure 5.2: Domain 1 ER Diagram: Identity, Authentication and Security Governance

5.3.2 Domain 2: Investor Suitability, Profiling, and Financial Goals

Domain 2 stores the investor's financial background, investment horizon, risk tolerance score, and savings goals across 10 tables:

- `user_profiles`: Stores financial details such as annual income, existing savings, investment experience, and tax bracket.
- `user_profile_versions`: Keeps a historical log whenever an investor updates their financial details over time.
- `financial_goals`: Stores user targets (e.g., retirement or college funds) along with target amounts and timelines.
- `user_preferences`: Stores user interface preferences (such as dark mode or notification delivery choices).
- `risk_questions`: The master bank of questions used in the suitability questionnaire (for example: "How would you react if the stock market fell 20% in a month?").
- `risk_question_options`: The available multiple-choice answers for each question, each carrying specific points.
- `risk_assessments`: The summary record of an investor's questionnaire attempt, showing their total score and risk category (**Conservative**, **Moderate**, or **Aggressive**).
- `risk_assessment_answers`: The specific choices selected by the investor for each question.
- `user_security_preferences`: Custom rules set by the user, such as mandating that a specific favorite company be included or excluded.
- `user_sector_preferences`: Custom rules to avoid entire industries (such as excluding tobacco, gambling, or defense companies).

Figure 5.3 displays the Entity-Relationship structure for Domain 2:

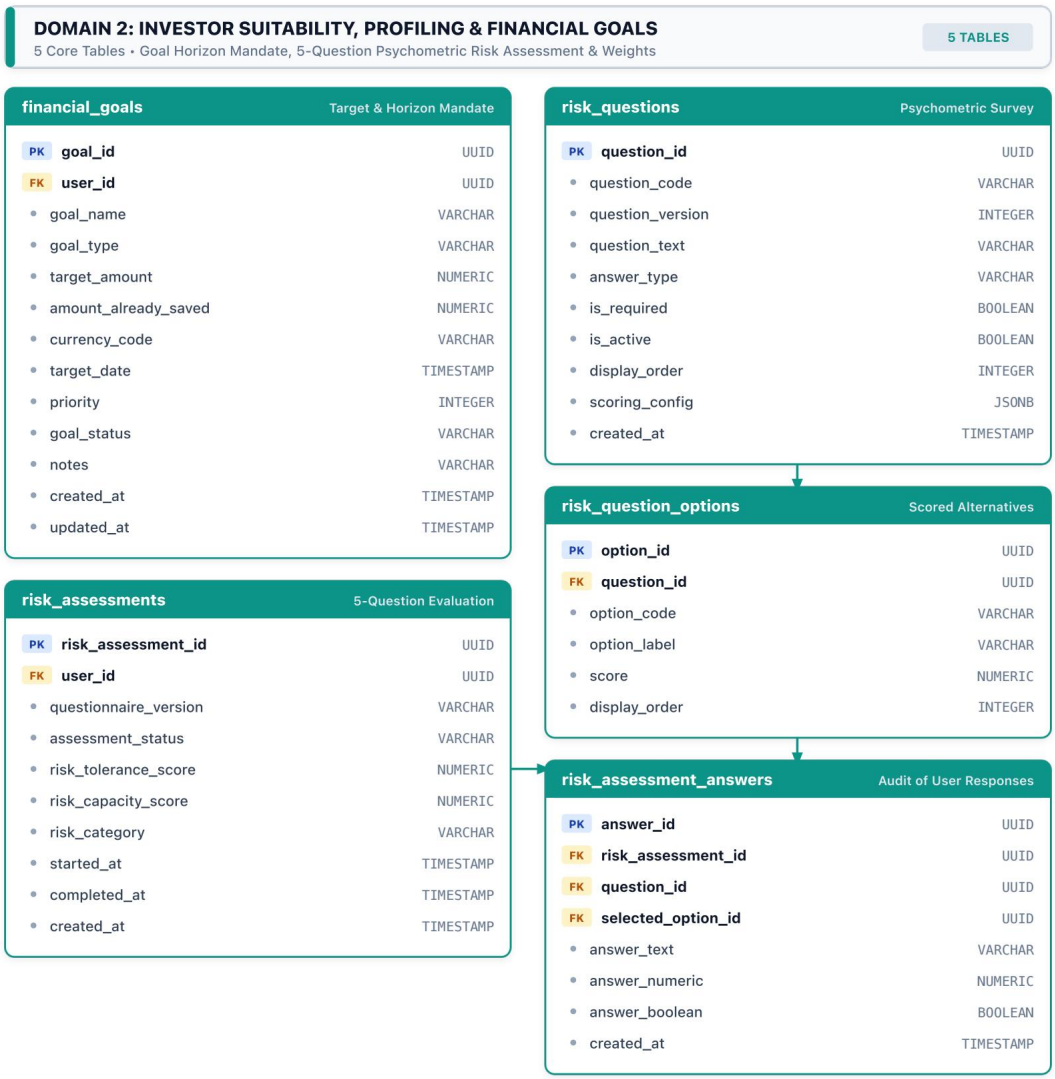


Figure 5.3: Domain 2 ER Diagram: Investor Suitability, Profiling and Financial Goals

5.3.3 Domain 3: Security Master and Corporate Registry

Domain 3 defines the universe of publicly traded companies, stock exchanges, and industry sectors across 8 tables:

- `exchanges`: Directory of stock exchanges (`NSE` for National Stock Exchange, `BSE` for Bombay Stock Exchange).
- `sectors`: Broad economic sectors (such as Financial Services, Information Technology, Healthcare, and Energy).
- `industries`: Granular industry groupings linked to parent sectors (for example, Software Services within Information Technology).
- `companies`: The legal business entities, storing their official Corporate Identification Number (CIN) and registered name.
- `company_industries`: Connects companies to their respective industries.
- `securities`: The canonical stock master. Each stock is identified by its unique 12-character International Securities Identification Number (**ISIN**, such as `INE002A01018` for Reliance Industries). It also assigns each stock to an institutional **Liquidity Tier** based on trading volume.
- `instrument_listings`: Tracks which exchange a stock trades on, linking exchange ticker symbols (for example, `RELIANCE` on NSE vs. scrip code `500325` on BSE).
- `source_instrument_mappings`: Translates exchange tickers into external data codes (such as `RELIANCE.NS` for Yahoo Finance).

Figure 5.4 displays the Entity-Relationship structure for Domain 3:

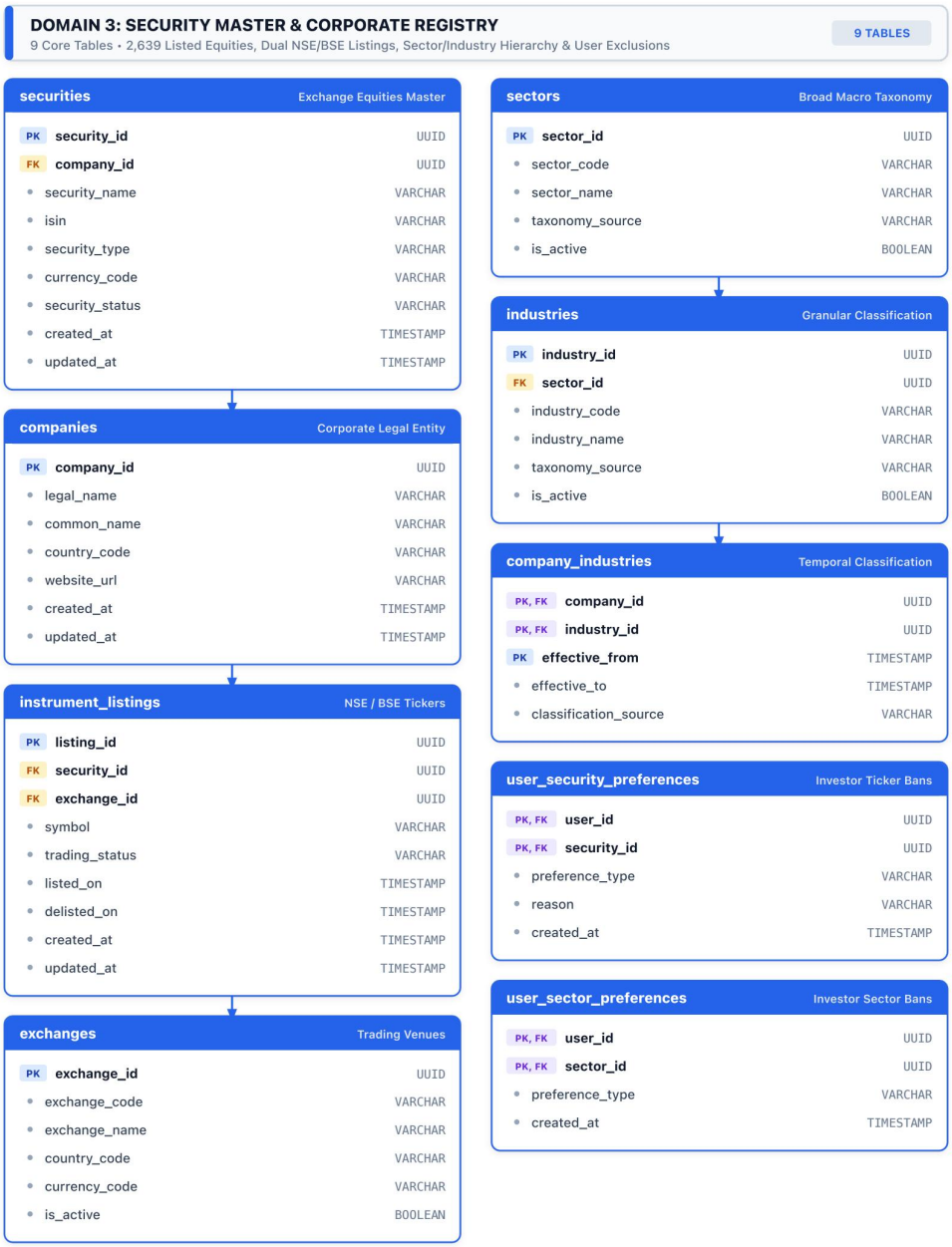


Figure 5.4: Domain 3 ER Diagram: Security Master and Corporate Registry

5.3.4 Domain 4: Market Data, Historical Quotes, and Macroeconomic Regime

Domain 4 stores the financial information that Lucid uses to analyze stocks across 8 tables:

- `market_prices`: Stores daily OHLCV historical price bars (Open, High, Low, Close, Volume) for listed securities.
- `fundamental_metrics`: Stores point-in-time financial health ratios for companies—including Return on Equity (ROE), Debt-to-Equity, Price-to-Earnings (P/E), and Market Capitalization.
- `metric_definitions`: The reference dictionary defining how each financial ratio is calculated.
- `economic_indicators`: Reference list of country-wide economic indicators, such as the RBI Repo Rate, inflation (CPI), and government bond yields.
- `economic_observations`: Historical values for economic indicators over time.
- `market_news_articles`: Recent financial news, corporate announcements, and filings.
- `news_article_securities`: Connects news stories to the specific companies mentioned in the articles.
- `knowledge_documents`: Reference documents, including SEBI circulars, investment case studies, and educational guides.

Figure 5.5 displays the Entity-Relationship structure for Domain 4:

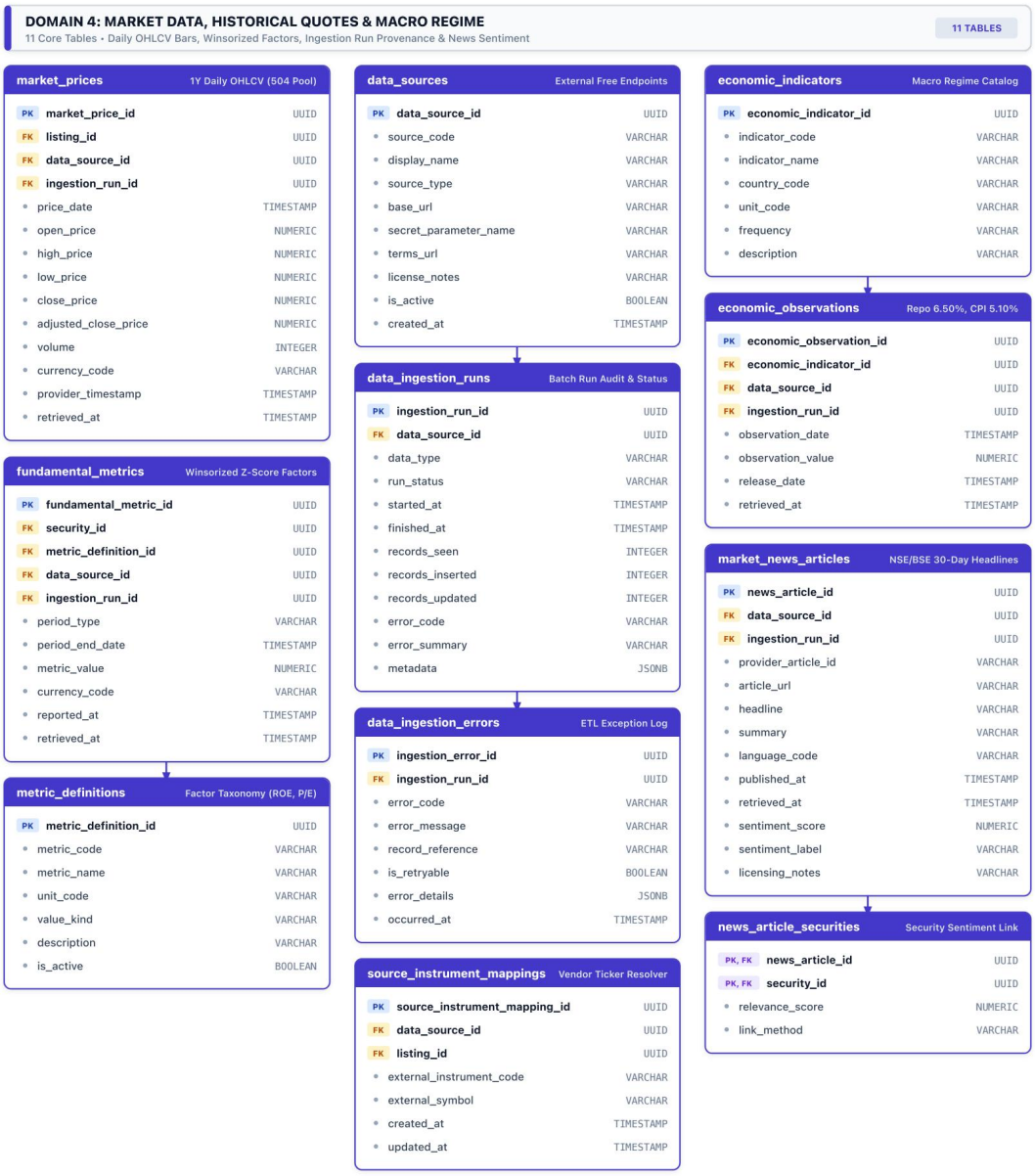


Figure 5.5: Domain 4 ER Diagram: Market Data, Historical Quotes and Macro Regime

5.3.5 Domain 5: Portfolio Engine, Quantitative Optimization, and Tracking

Domain 5 forms the transactional core of Lucid, recording recommended portfolios, version histories, and daily valuations across 10 tables:

- `strategy_definitions`: Strategy templates (**Conservative**, **Moderate**, and **Aggressive**).
- `strategy_versions`: Mathematical parameter sets (such as risk aversion penalties) assigned to each strategy.
- `investor_portfolios`: The master account record linking each portfolio to its owner.
- `portfolio_versions`: Whenever a portfolio is created or rebalanced, Lucid saves an immutable new version snapshot (Version 1, Version 2, etc.), recording the date, reason for change, and unallocated cash balance.
- `portfolio_version_constraints`: The safety limits enforced for that version (for example, no single stock exceeding 25% and no sector exceeding 40%).
- `portfolio_generation_requests`: An audit record of the user's input parameters (amount, goal, timeline) that triggered the portfolio creation.
- `portfolio_security_decisions`: The exact line-item stock allocations, recording the company name, recommended percentage weight, number of whole shares, purchase price, and total invested rupees.
- `portfolio_performance_snapshots`: Daily valuation records tracking the total value of the portfolio and cumulative gains or losses over time.
- `portfolio_scenarios`: Projections evaluating how the portfolio would behave during major historical market crashes (such as the 2008 financial crisis or the March 2020 COVID shock).
- `rebalance_suggestions`: Proposed adjustments generated by the monitoring daemon when stock prices drift away from planned targets.

Figure 5.6 displays the Entity-Relationship structure for Domain 5:

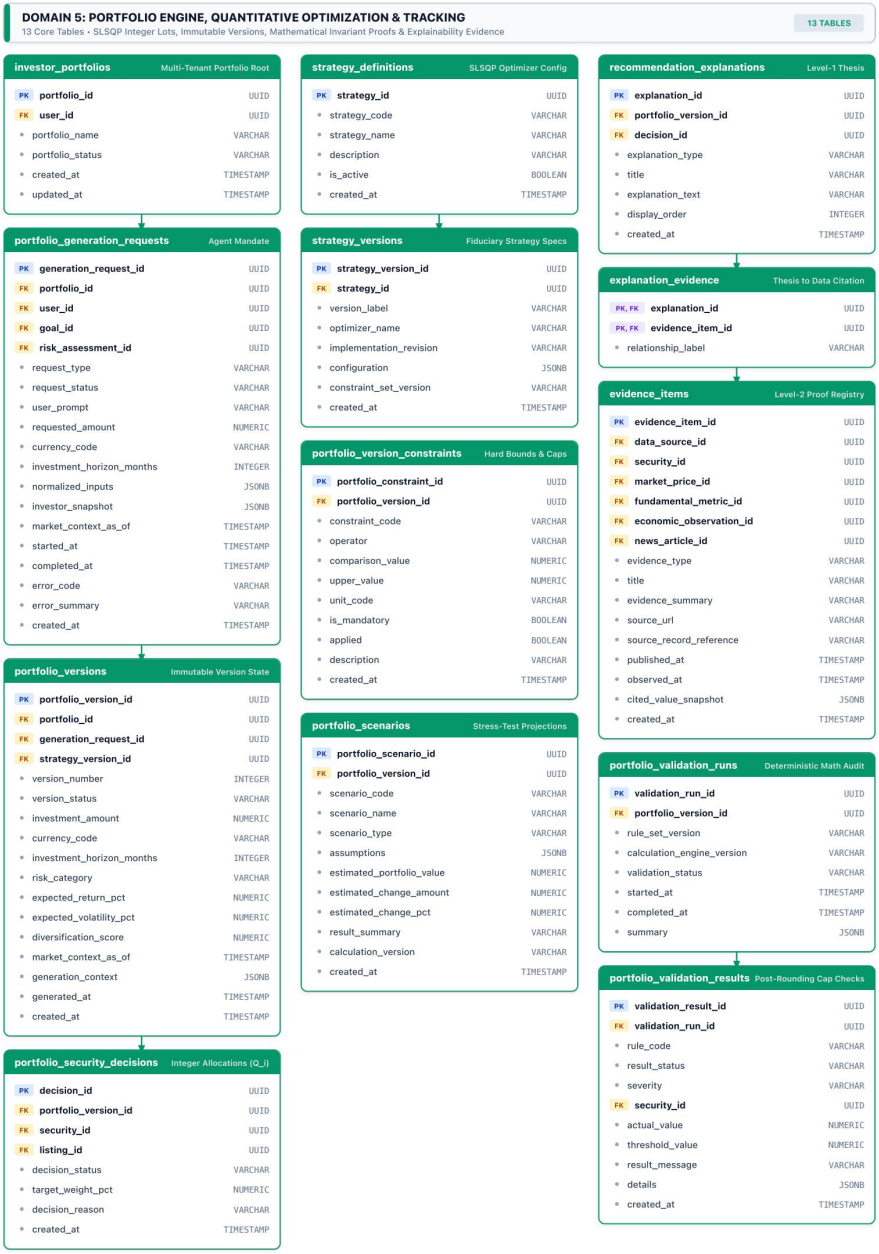


Figure 5.6: Domain 5 ER Diagram: Portfolio Engine, Quantitative Optimization and Tracking

5.3.6 Domain 6: Broker Integration, Execution, and Order Lifecycle

Domain 6 manages connections to brokerage platforms and tracks simulated order execution across 5 tables:

- `broker_providers`: Supported brokerage platforms (such as Zerodha, Upstox, and Groww).
- `user_broker_accounts`: Securely links an investor's brokerage credentials using strong encryption.
- `broker_orders`: Batches of buy and sell orders generated when an investor rebalances a portfolio.
- `broker_order_events`: Audit timeline tracking order stages (`CREATED`, `SUBMITTED`, `EXECUTED`).
- `broker_executions`: The final trade execution receipts, recording filled share quantities, average purchase prices, and execution timestamps.

Figure 5.7 displays the Entity-Relationship structure for Domain 6:

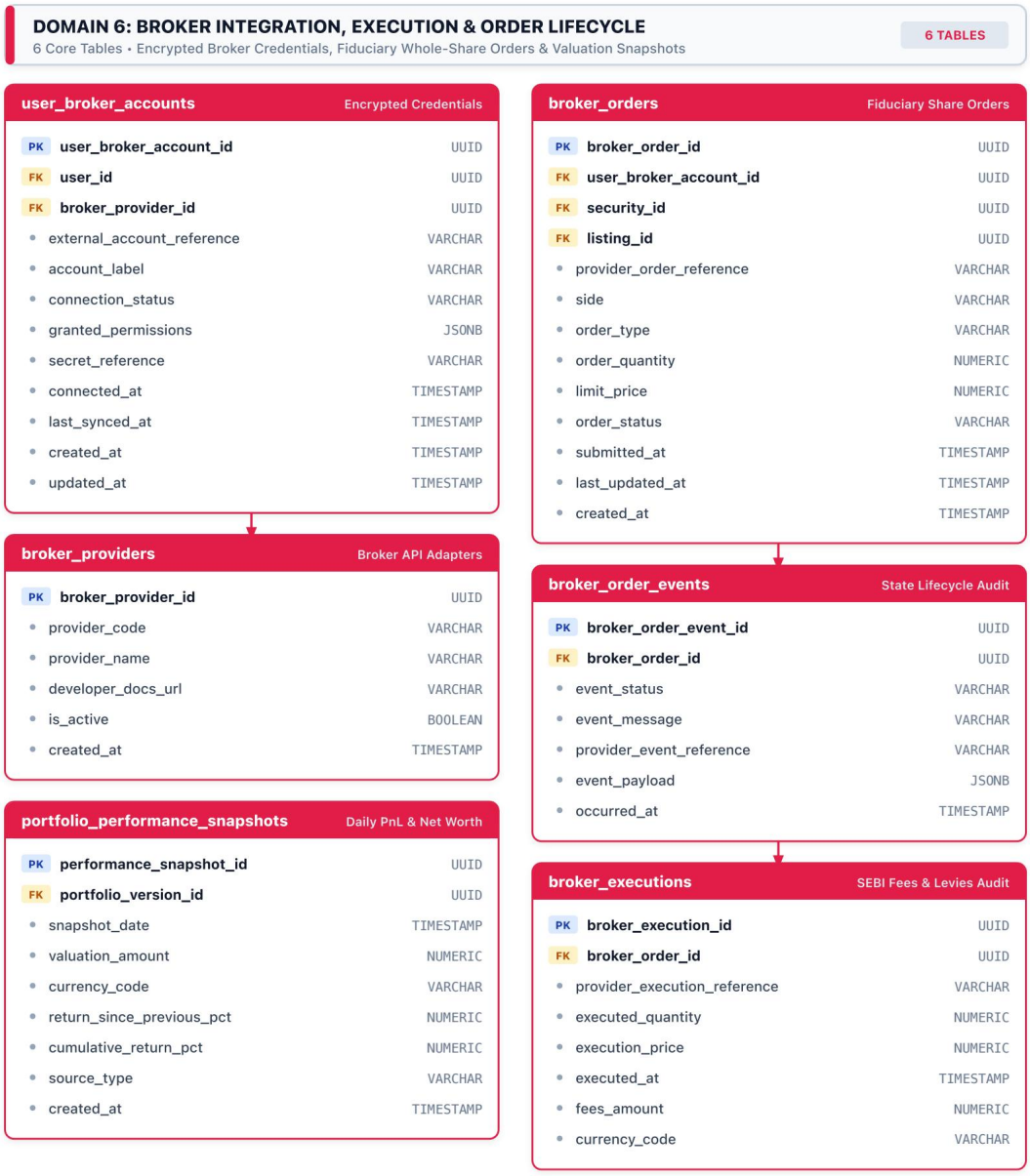


Figure 5.7: Domain 6 ER Diagram: Broker Integration, Execution and Order Lifecycle

5.3.7 Domain 7: Demat CAS Diagnosis and Portfolio Analytics

Domain 7 stores imported external Demat portfolios and health diagnostic assessments across 3 tables:

- `existing_portfolios`: Records of uploaded monthly Demat statements (via PDF CAS statement or manual entry).
- `existing_portfolio_holdings`: The parsed holdings extracted from the statement (company ISIN, share quantities, and original purchase costs).
- `portfolio_diagnoses`: The complete diagnostic report, including total current valuation, sector concentration risk, the Herfindahl-Hirschman Index (HHI) diversification score, and actionable hold, buy, or trim recommendations.

Figure 5.8 displays the Entity-Relationship structure for Domain 7:

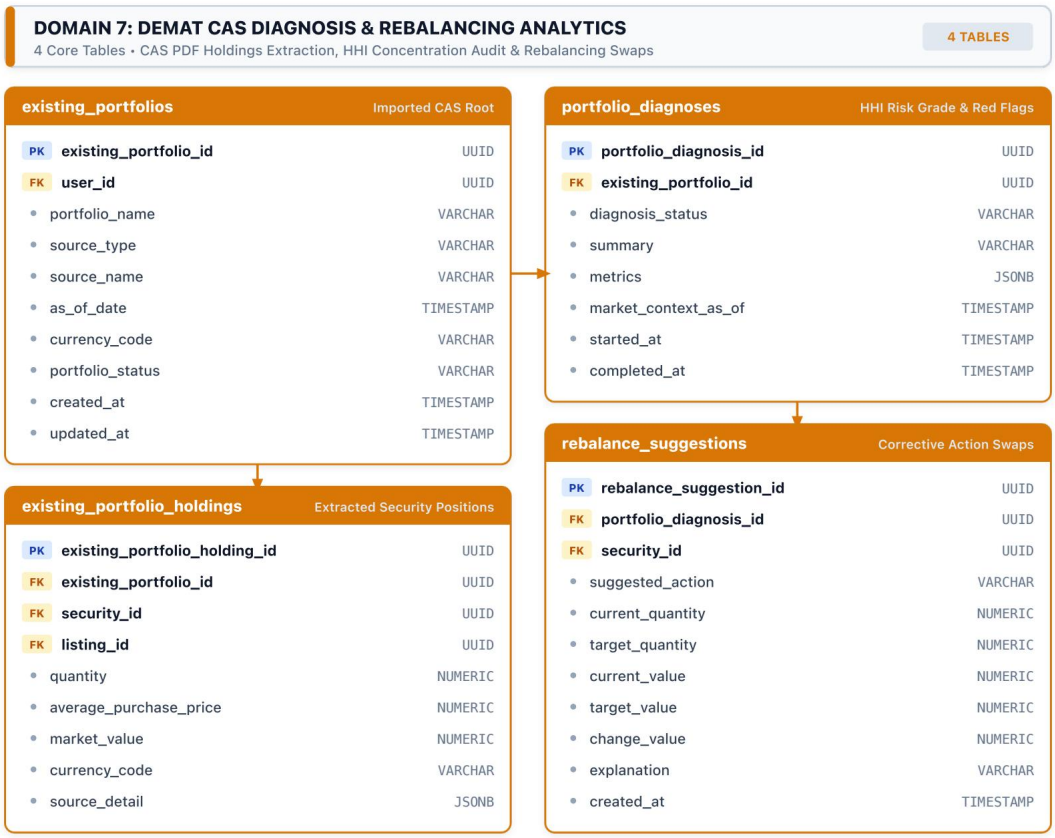


Figure 5.8: Domain 7 ER Diagram: Demat CAS Diagnosis and Portfolio Analytics

5.3.8 Domain 8: Multi-Agent AI Telemetry, Reasoning Traces, and Audits

Domain 8 stores the execution logs of the AI agents, plain-English explanations, and real-time user alerts across 13 tables:

- `agent_definitions`: Master catalog of the 13 specialized AI roles (Macro, Fundamental, News, Critic, etc.).
- `agent_versions`: Tracks versions of agent prompts and model settings over time.
- `agent_execution_runs`: Audit log recording how long each agent took to think, token costs, and structured outputs.
- `recommendation_explanations`: The plain-English "Stock-Why" justifications and financial metric breakdowns shown to the user.
- `evidence_items`: Verifiable factual data points (such as company profit numbers or RBI interest rate quotes) that support the explanations.
- `explanation_evidence`: Connects each explanation to its specific supporting evidence items.
- `portfolio_validation_runs`: Audit logs verifying that portfolios satisfy mathematical rules.
- `portfolio_validation_results`: Pass/fail audit records for rules.
- `data_sources`: Directory of external data providers (NSE, BSE, Yahoo Finance).
- `data_ingestion_runs`: Execution logs of automated nightly price and news download jobs.
- `data_ingestion_errors`: Detailed error logs capturing any API failures or bad data records.
- `notifications`: Real-time investor alerts (such as portfolio drift notices and rebalancing recommendations).
- `user_feedback`: User ratings and comments on whether an explanation was clear and helpful.

Figure 5.9 displays the Entity-Relationship structure for Domain 8:

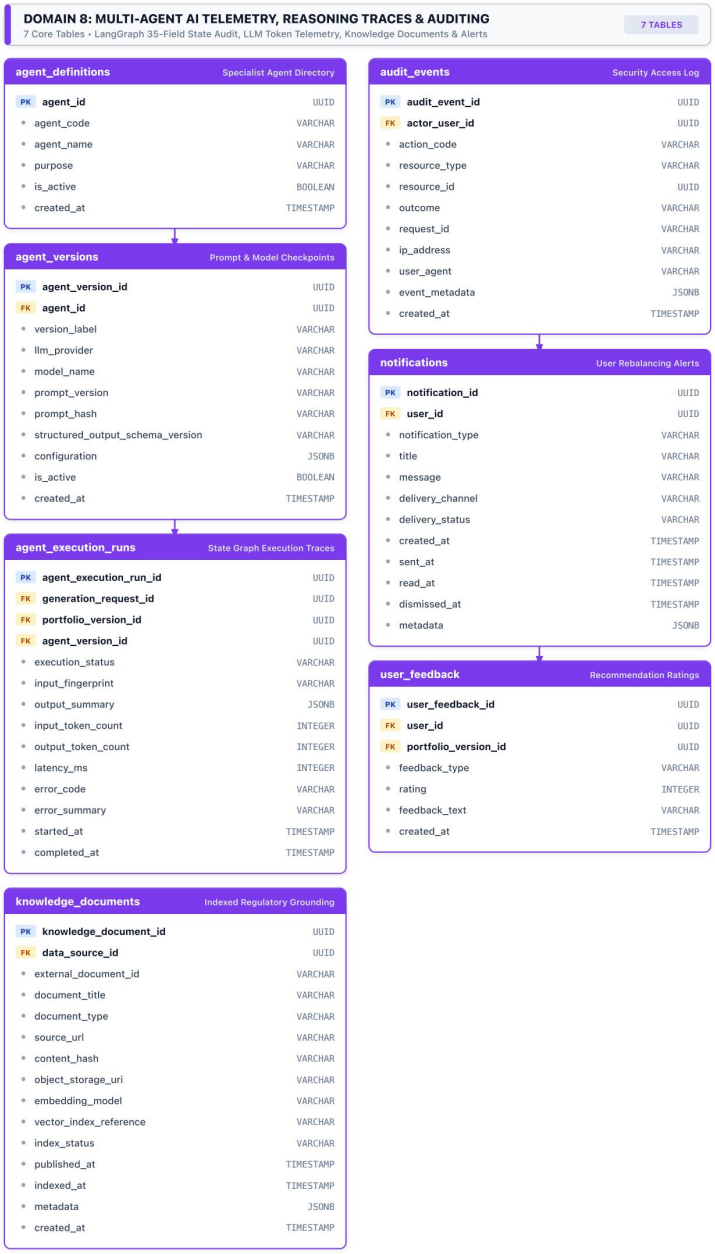


Figure 5.9: Domain 8 ER Diagram: Multi-Agent AI Telemetry, Reasoning Traces and Verification

5.4 Database Design: Speed and Safety

To ensure the database runs quickly and reliably, four key engineering choices were made:

1. **High Speed with Structured Flexibility**: Core tables (users, portfolios, stock prices) follow strict relational rules to prevent duplicate or

conflicting records. At the same time, complex analysis reports (such as sector concentration breakdowns and AI factor scores) are stored in structured JSON columns (`JSONB`). This allows the web dashboard to load complete portfolio screens in less than 5 milliseconds without running dozens of separate database queries.

2. **Secure Universal IDs (UUIDs)**: Every account, portfolio, and order uses a randomized 128-bit universal identifier (UUID) instead of a simple number like 1, 2, or 3. This protects user privacy by preventing unauthorized parties from guessing other users' portfolio links.
3. **Fast Search Indexes**: Searching through thousands of daily stock prices or looking up companies by ticker symbol is made nearly instantaneous by creating specialized B-Tree search indexes on frequently queried columns (such as `securities.isin` and `market_prices.price_date`).
4. **Accidental Deletion Protection**: Strong database rules (called **foreign key constraints**) protect financial records. For example, deleting a user account is strictly blocked if active transaction ledgers or historical investment records exist, ensuring that audit trails can never be accidentally erased.

Chapter 6

System Architecture and Runtime Topology

6.1 System Architecture Overview

The system architecture of Lucid describes how all the user interface components, backend API gateways, artificial intelligence agents, mathematical optimization solvers, and relational databases are organized so that the application runs quickly, reliably, and securely.

To guarantee high speed and financial accuracy, Lucid strictly separates qualitative artificial intelligence reasoning from quantitative mathematical optimization. The entire platform is engineered as a **Six-Tier Decoupled Architecture** organized in an end-to-end processing loop across two balanced functional rows:

- **Top Processing Row (Forward User & Research Flow)**: Connects the investor from the Next.js client interface (Tier 1) through the FastAPI security gateway (Tier 2) to the LangGraph multi-agent intelligence committee (Tier 3).
- **Bottom Processing Row (Optimization & Storage)**: Dispatches vetted candidate stocks into the SciPy solver (Tier 4), saves portfolio decisions into PostgreSQL (Tier 5), and pulls live market feeds from Indian exchange archives (Tier 6).

Figure 6.1 illustrates the complete six-tier architectural topology of Lucid, detailing the exact technologies, component boundaries, inputs, and outputs across all subsystems:

6.2 The Six Architectural Layers: A Detailed Walkthrough

Each of the six architectural tiers performs a dedicated, specialized role in the portfolio creation and management lifecycle:

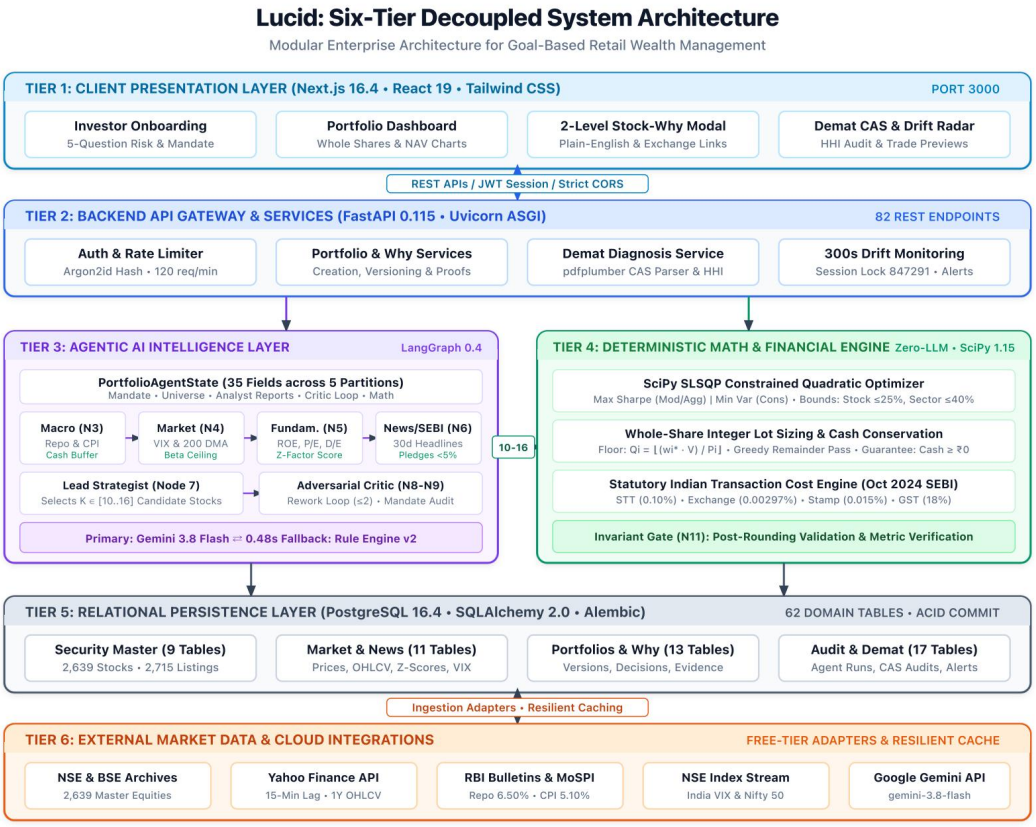


Figure 6.1: Lucid Six-Tier Decoupled System Architecture and Runtime Topology

6.2.1 Tier 1: Client Presentation Layer

Tier 1 represents the user-facing web application running on local port `3000`. Built with **Next.js 16.4.2** (App Router with Turbopack) and **React 19.3.0**, this layer provides a clean, responsive workspace comprising 24 modular user interface components styled with **Tailwind CSS 4.1.1**:

- **Investor Onboarding**: Collects investor capital ($\geq$ Rs. 10,000), suitability answers (scoring 5 to 20), target milestones, and sector exclusions.
- **Holdings Dashboard**: Renders interactive portfolio summaries, including a diversification donut ring (e.g., 95.5/100 across 12 stocks) powered by **Recharts**, and holding cards (e.g., HDFC Bank 32 shares / Rs. 54,976 and Reliance 16 shares / Rs. 45,440).
- **Dual-Level Stock-Why Explanations**: Provides a two-level explainability modal featuring an instant plain-English summary (Level 1) and direct links to official exchange announcements on NSE and BSE (Level

2), guarded by `lib/linkUtils.ts`.

- **Demat Health Audit and Rebalance Preview**: Allows users to upload monthly Demat PDF statements (CAS) or CSV files, view their portfolio concentration score (HHI), and inspect before-and-after rebalance trade proposals.
- **Communication Pipeline**: Communicates with the backend using authenticated REST requests via `lib/api.ts`, transmitting Bearer JSON Web Tokens and running 30-second polling for real-time notifications.
- **Input / Output Contract**: Accepts investor capital, risk responses, and Demat CAS statements; delivers whole-share investment recommendations and downloadable broker execution CSV files.

6.2.2 Tier 2: Backend API Gateway and Services

Tier 2 is the central API gateway and business logic orchestrator running on local port `8000` via the **Uvicorn 0.34.0** ASGI web server. Built using **FastAPI 0.115.8** and **Pydantic 2.10.6**, this layer exposes **9 versioned routers** spanning **82 unique URL paths** (88 HTTP operations) and manages 13 active services:

- **Authentication and Rate Limiting (`auth.py`, 13 Ops)**: Secures user accounts using salted **Argon2id** password hashing and issues encrypted JSON Web Tokens (JWT). Enforces a strict rate limit of 120 requests per minute to prevent denial-of-service abuse.
- **Profile and Mandate Engine (`profile.py` & `risk.py`, 14 Ops)**: Calculates psychometric suitability scores (5 to 20) and parses conversational natural-language investment targets into structured parameters.
- **Universe Funnel Service (`market_data.py`, 13 Ops)**: Filters the broad universe of 2,639 listed Indian companies down to an active, liquid candidate pool of institutional-grade equities.
- **Portfolio Generation and Explainability (`portfolios.py`, 25 Ops)**: Coordinates portfolio creation requests, tracks immutable version deltas, and generates dual-level "Stock-Why" justifications.
- **Demat Statement Diagnosis (`diagnosis.py`, 9 Ops)**: Extracts stock holdings from uploaded Demat CAS PDF statements using `pypdf`, checks

sector concentrations, and computes diversification metrics.

- **300-Second Continuous Drift Radar (`monitoring.py`, 14 Ops)**: An automated monitoring engine that runs every 5 minutes during Indian market trading hours (09:15 to 15:30 IST), protected by PostgreSQL advisory lock `847291` to ensure only one drift scan runs at any given time.
- **Input / Output Contract**: Receives authenticated REST Data Transfer Objects (DTOs); emits validated JSON response payloads and HTTP 201 Created status codes.

6.2.3 Tier 3: Agentic Intelligence Layer

Tier 3 serves as the qualitative investment research committee of Lucid. Built on **LangGraph 0.4.0**, it coordinates digital analyst agents inside a structured `StateGraph` workflow (`backend/agent/graph.py`) passing a centralized `PortfolioAgentState` object containing 35 distinct telemetry fields:

- **Directed Graph Workflow**: Execution follows a controlled sequential pipeline:

$$\begin{aligned} &\text{N1–N2 Gate} \longrightarrow \text{N3–N6 Analysts} \longrightarrow \text{N7 Strategist} \\ &\qquad \longleftrightarrow \text{N8–N9 Critic} \longrightarrow \text{N10–N13 Persistence} \end{aligned} \tag{6.1}$$

- **Specialized Digital Agents**:
  - *Node 3 (Macro Analyst)*: Evaluates Reserve Bank of India (RBI) repo interest rates, inflation (CPI), and government bond yields.
  - *Nodes 4–5 (Market and Fundamental Analysts)*: Evaluates the India VIX volatility index, moving averages, Return on Equity (ROE), Price-to-Earnings (P/E), and Debt-to-Equity ratios.
  - *Node 6 (News and Regulatory Analyst)*: Scans 30-day corporate announcements and SEBI regulatory filings for potential governance flags.
  - *Node 7 (Lead Strategist)*: Selects an initial candidate pool of 10 to 16 promising stocks tailored to the investor's risk profile.
  - *Nodes 8–9 (Adversarial Risk Critic)*: Thoroughly stress-tests the strategist's candidate list. If a stock carries excessive debt or valuation risk, the critic rejects it and mandates a rework pass (capped

at a maximum of 2 iterations).

- **Circuit-Breaker Governance (`nodes.py`)**: Uses Google Gemini 3.8 Flash (temperature 0.1–0.2, $\leq 1,500$ token budget) for qualitative reading. If cloud API latency spikes or an external network timeout occurs, an automated circuit-breaker immediately triggers a zero-latency deterministic financial rule engine (**Rule Engine v2**) in just **0.48 seconds**, ensuring the application never freezes.
- **Stock-Why Synthesis and Telemetry**: Node 12 (Moderator) synthesizes the dual-level plain-English explanations, and Node 13 commits structured execution logs across 6 audit tables.
- **Separation of Responsibilities**: Qualitative AI agents select and review candidates; all math and weight calculations are strictly forbidden in this tier and handled by Tier 4.
- **Input / Output Contract**: Accepts the investor mandate and screened candidate pool; delivers a vetted basket of 10 to 16 Critic-approved stocks.

6.2.4 Tier 4: Deterministic Math and Financial Engine

Tier 4 represents the mathematical calculation core of Lucid, built with proven scientific Python libraries (**SciPy 1.15.2** and **NumPy 2.2.3**). Completely separated from all artificial intelligence models, this layer executes pure deterministic calculations with zero hallucination risk:

- **SciPy SLSQP Mathematical Optimizer**: Solves Markowitz Modern Portfolio Theory equations using Sequential Least Squares Programming (`scipy.optimize.minimize` with the SLSQP method). It finds optimal percentage weights (w^*) along the Efficient Frontier, maximizing the Sharpe ratio for Moderate and Aggressive profiles or minimizing portfolio variance for Conservative profiles.
- **Institutional Safety Bounds**: Enforces hard concentration limits: no individual stock may exceed 20% to 25% of the total portfolio, and no single industry sector may exceed 30% to 40%.
- **Mathematical Guardrails**: Raw return estimates are winsorized to the 5th and 95th percentiles with Z-scores bounded in $[-3, +3]$, and expected annual returns are constrained within $[4\%, 30\%]$ to prevent erratic math-

ematical weights.

- **Whole-Share Integer Lot Rounding**: Because Indian exchanges do not permit fractional shares, the continuous percentage weights are converted into exact whole shares using a floor-rounding algorithm:

$$Q_i = \left\lfloor \frac{w_i^* \times V_{\text{total}}}{P_i} \right\rfloor \tag{6.2}$$

where Q_i is the number of whole shares for stock i, w_i^* is its target percentage weight, V_{total} is total investment capital, and P_i is its current market price. All leftover funds remain safely in the investor's unallocated cash account ($C_{\text{res}} \geq$ Rs. 0).

- **SEBI Statutory Transaction Fee Schedule**: Models all official Indian delivery transaction charges (updated as of October 2024):
  - Securities Transaction Tax (STT): 0.10% on delivery equity transactions.
  - Exchange Turnover Fees: 0.00297% on NSE and 0.00375% on BSE.
  - SEBI Regulatory Turnover Fee: 0.0001% (Rs. 10 per crore).
  - Stamp Duty: 0.015% charged on purchases under the Indian Stamp Act.
  - Goods and Services Tax (GST): 18.0% applied to brokerage and exchange charges.
- **Input / Output Contract**: Accepts 10 to 16 vetted stocks, expected returns μ_i, and covariance matrix Σ_{reg}; delivers integer share quantities Q_i and non-negative residual cash ($C_{\text{res}} \geq$ Rs. 0).

6.2.5 Tier 5: Relational Persistence Layer

Tier 5 serves as the secure, permanent data store of Lucid, powered by **PostgreSQL 16.4** and managed via **SQLAlchemy 2.0.38** and **Alembic 1.14.1**:

- **Database Architecture**: Contains exactly **62 domain application tables** comprising 565 columns, 102 foreign-key constraints, and 312 data integrity checks, organized into six functional domain clusters:
  1. *Identity and Risk (12 tables)*: User profiles, encrypted passwords, suitability scores, and sector exclusions.

2. *Security Master (9 tables)*: Master catalog of 2,639 listed Indian companies, 2,716 exchange listings across NSE and BSE, and corporate taxonomy.
3. *Market and News (11 tables)*: Historical daily OHLCV price bars, fundamental balance sheet ratios, economic indicators, and news sentiment records.
4. *Portfolios and Why (13 tables)*: Recommended portfolios, immutable version snapshots, line-item security allocations, and dual-level Stock-Why explanations.
5. *Demat Statement Diagnosis (4 tables)*: Imported Demat statements, parsed external holdings, and HHI health diagnostic reports.
6. *Operations and Brokerage (13 tables)*: AI agent execution logs, simulated broker orders, order lifecycle event logs, and user notifications.

- **ACID Guarantees and Speed**: Configured with connection pooling (pool size 10, max overflow 20) and uses PostgreSQL extensions including `pgcrypto` (cryptographic functions), `uuid-ossp` (random 128-bit identifiers), and `pg_trgm` (trigram fuzzy matching for company name resolution with similarity > 0.4).
- **Input / Output Contract**: Accepts atomic ACID write commits from the API services and mathematical optimizer; emits indexed relational reads with average query response times under 5 milliseconds.

6.2.6 Tier 6: External Market Data and Cloud Feeds

Tier 6 manages automated connections to official Indian market archives, financial data feeds, and cloud artificial intelligence models:

- **Official Exchange Archives (NSE and BSE)**: Synchronizes with the National Stock Exchange master directory (`EQUITY_L.csv`) and BSE scrip directory to maintain an accurate universe of 2,639 listed Indian equities and corporate action records.
- **Yahoo Finance API Feed**: Fetches 252 market sessions of historical daily price bars, financial health ratios (ROE, P/E, Debt-to-Equity), and the India VIX volatility index.
- **Macroeconomic Sources (RBI and MoSPI)**: Tracks country-wide

monetary statistics, including the RBI Policy Repo Rate (6.50%), Consumer Price Index inflation (5.10%), 10-year Government Securities yield (7.04%), and economic news.

- **Google Gemini Cloud API**: Interfaces with the Gemini 3.8 Flash model via structured JSON schemas, backed by automated local rule fallbacks.
- **Lean Zero-Bloat Engineering**: Lucid is built with an intentionally lean, zero-bloat tech stack. The platform requires **zero external graph databases** (no Neo4j), **zero in-memory caching servers** (no Redis), **zero distributed task queues** (no Celery), and **zero external vector stores** (no FAISS). Instead, high-performance PostgreSQL indexes and lightweight Python background threads handle all operations locally, ensuring resilient offline continuity and low operational cost.
- **Input / Output Contract**: Sends HTTPS synchronization requests and structured prompts; returns clean, normalized market data and qualitative research payloads.

6.3 Technology Stack Manifest: What Tools Were Chosen and Why

Table 6.1 documents the verified software libraries and frameworks used to build Lucid, along with the simple reason why each tool was chosen:

Table 6.1: Lucid Active Technology Stack Manifest (Verified Versions)

Technology / Library	Verified Version	What It Does in Lucid	Why It Was Chosen (Simple English)
Next.js	16.4.2	Frontend Web Framework	Renders web pages quickly and updates charts instantly.
React	19.3.0	User Interface Library	Provides reusable buttons, modals, and dynamic data tables.
TypeScript	5.8.2	Frontend Programming Language	Catches code errors and typos before the application runs.
Tailwind CSS	4.1.1	Visual Styling	Creates clean, modern visual designs with fast page loading.
Python	3.11 – 3.14	Backend Programming Language	The global industry standard for financial math and AI systems.
FastAPI	0.115.8	Backend API Framework	Extremely fast web framework that automatically creates API docs.
Uvicorn	0.34.0	High-Speed Web Server	Handles multiple incoming user web requests at the same time.
Pydantic	2.10.6	Data Validation Library	Automatically checks that user inputs (like amounts) are valid.
PostgreSQL	16+	Master Relational Database	Stores financial data permanently with zero risk of loss.
SQLAlchemy	2.0.38	Database Connector (ORM)	Lets Python code talk to PostgreSQL cleanly and safely.
Alembic	1.14.1	Database Change Tracker	Records updates to database tables over time like version control.
SciPy	1.15.2	Mathematical Solver Library	Solves Markowitz portfolio math without arithmetic mistakes.
NumPy	2.2.3	Fast Numerical Calculations	Performs high-speed math on large tables of stock prices.
Pandas	2.2.3	Financial Data Tables	Cleans stock lists and handles historical price series.
LangGraph	0.4.0	Multi-Agent Coordinator	Coordinates the 13 AI agents in a structured, step-by-step workflow.
Google Gemini	gemini-3.8-flash	AI Language Model	Reads financial documents and writes plain-English explanations.
Argon2-cffi	25.1.0	Password Hasher	Protects user passwords with military-grade encryption.
PyJWT	2.10.1	Digital Session Badges	Creates encrypted tokens so logged-in users stay authenticated safely.
yfinance	0.2.54	Market Data Fetcher	Downloads historical daily prices and financial health ratios.
pdfplumber	0.11.5	PDF Statement Reader	Reads text and tables from Demat CAS account statements.

6.4 Data Pipeline: How Indian Stock Data is Collected and Cleaned

Figure 6.2 illustrates the four-step data ingestion pipeline that keeps Lucid synchronized with Indian stock exchanges:

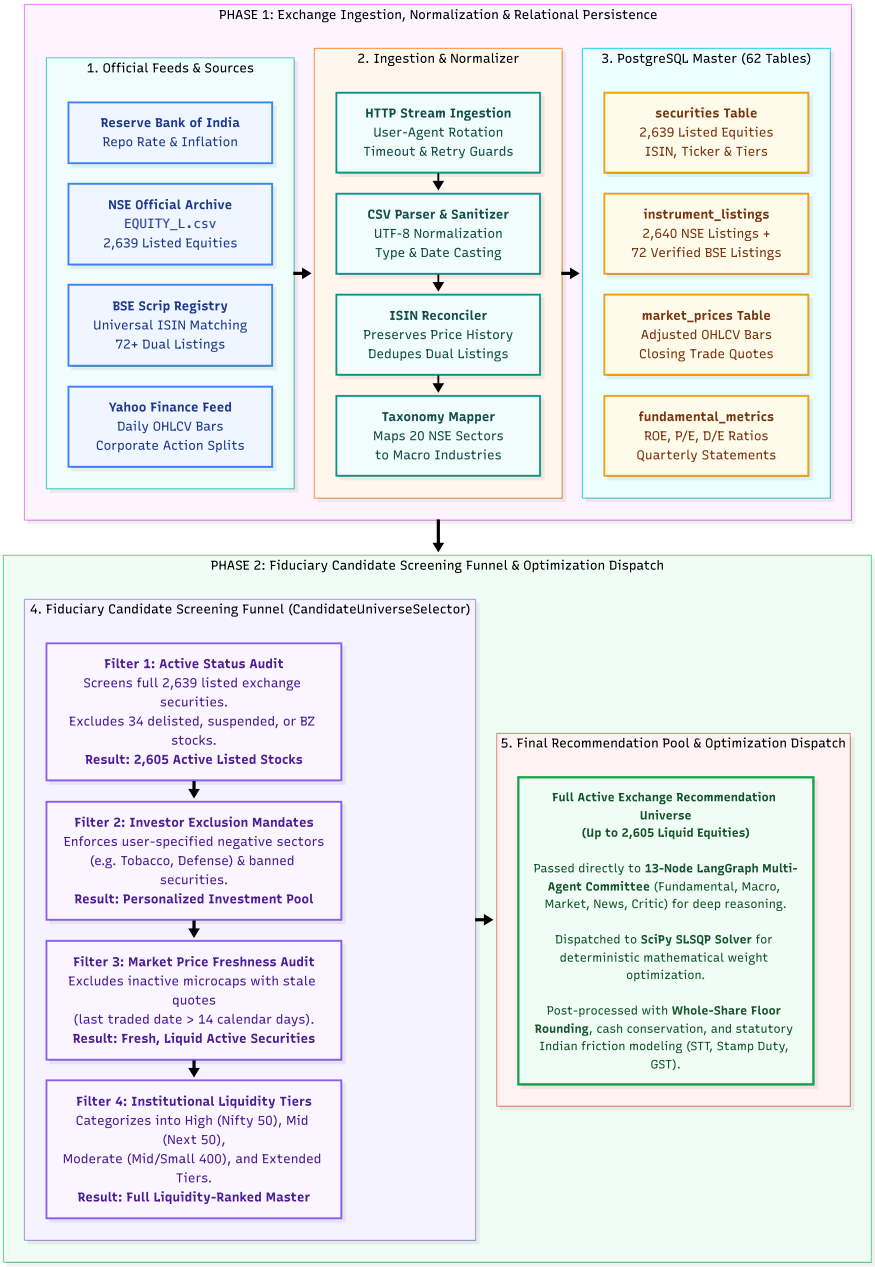


Figure 6.2: NSE/BSE Exchange Universe Ingestion and Pipeline Architecture

The data pipeline runs through four clear stages:

1. **Stage 1: Downloading the Stock Lists**: The pipeline downloads the

official list of actively traded companies from the National Stock Exchange (`EQUITY_L.csv`) and cross-checks them against the Bombay Stock Exchange (BSE) directory. It resolves conflicts (such as when companies change their names or split their shares) and saves **2,639 listed Indian companies** into the master security database.

2. **Stage 2: Fetching 1 Year of Price History**: For every active stock, the pipeline fetches one full year of daily trading prices (252 market sessions). It checks that there are no missing days and stores the prices in the `market_prices` table.
3. **Stage 3: Recording Financial Health Ratios**: Lucid downloads company balance sheet ratios—including Return on Equity (ROE), Debt-to-Equity, Price-to-Earnings (P/E), and Market Capitalization—and saves them in the `fundamental_metrics` table.
4. **Stage 4: Liquidity Tier Screening**: Companies are categorized into four institutional **Liquidity Tiers** based on their average daily trading volume. This ensures that the portfolio creation engine only selects companies that can be easily bought and sold on the market without unexpected price slippage.

6.5 Runtime Execution: What Happens When a User Clicks "Generate"

When an investor clicks "Generate Portfolio" on the web interface, the application completes the following ten-step journey in less than 2.5 seconds:

1. **Step 1: Input Check**: The backend server checks that the user's investment amount is at least Rs. 5,000 and verifies their digital login badge.
2. **Step 2: Fetching User Goals**: The server loads the investor's risk profile (Conservative, Moderate, or Aggressive) and savings timeline from the database.
3. **Step 3: Screening Candidates**: The system filters the 2,639 listed companies to find healthy stocks, removing any industries the user asked to avoid (such as tobacco or defense).
4. **Step 4: Starting the AI Team**: Lucid creates the digital workspace

and starts the LangGraph agent workflow.

5. **Step 5: Macroeconomic Check**: The macro agent checks current Reserve Bank of India interest rates and the market volatility level (India VIX).
6. **Step 6: Financial Analysis**: The fundamental and news agents evaluate balance sheets, profit consistency, and recent corporate news.
7. **Step 7: The Critic Challenge**: The risk critic agent reviews the shortlisted candidates, challenging any stocks that have high debt or valuation red flags.
8. **Step 8: Mathematical Optimization**: The SciPy SLSQP solver calculates the exact percentage weights that balance return and safety.
9. **Step 9: Whole Shares Calculation**: The lot-rounding algorithm converts percentages into exact full share counts and keeps leftover money in the investor's cash account.
10. **Step 10: Saving and Display**: The completed portfolio, full share counts, and plain-English "Stock-Why" explanations are saved to the database and displayed on the user's screen.

6.6 Security and Privacy Architecture

Financial software requires strict security to protect investor privacy:

- **High-Security Password Hashing (Argon2id)**: When a user creates a password, Lucid never stores the actual password. Instead, it scrambles it using **Argon2id**, the international gold standard in password protection. It requires 64 megabytes of computer memory to verify, making it practically impossible for hackers to crack using automated guessing tools.
- **Digital Badges with Instant Logout (JWT Revocation)**: When an investor logs in, they receive an encrypted digital badge (JSON Web Token) valid for 60 minutes. When the investor logs out, the badge is immediately added to a cancellation list (`revoked_tokens`). Any future attempt to use that badge is instantly rejected.
- **Anti-Abuse Rate Limiting**: To prevent malicious bots from overwhelming the server, Lucid limits how many requests can be sent per minute (for example, a maximum of 20 login attempts per minute and 20 portfolio

generation requests per minute).

- **Standard Web Protections**: The server automatically includes standard web security headers in every response to block website cloning (clickjacking) and prevent unauthorized script injection.

Chapter 7

End-to-End Application Methodology

7.1 Methodology Overview: Three Production Workflows and Seven Stages

Building an intelligent investment portfolio requires combining human financial common sense with rigorous mathematical calculation. In traditional wealth management, human financial advisors sit down with a client, evaluate their personal life goals, research companies in the stock market, and calculate how much money should be placed in each stock.

Lucid automates this entire lifecycle using a structured, transparent methodology. Rather than treating portfolio creation as an isolated action, Lucid organizes its financial engineering into **Three Production Workflows** comprising **Seven Stages** (numbered Stage 4.1 through Stage 4.12):

1. **Workflow I: New Portfolio Creation Methodology (Stages 4.1–4.8)**: Guides a new investor from initial onboarding and natural-language goal formulation, through market regime analysis, multi-agent AI research, and deterministic mathematical optimization, to delivering a balanced whole-share portfolio with plain-English justifications.
2. **Workflow II: Existing Demat Portfolio Diagnosis (Stage 4.9)**: Enables investors who already hold shares across external broker accounts (such as Zerodha or Groww) to upload their monthly Demat statements, resolve company names across all 2,639 listed Indian stocks, audit portfolio concentration risk, and receive actionable swap recommendations.
3. **Workflow III: Continuous 300s Drift Monitoring, Statutory Rebalancing, and Alerts (Stages 4.10–4.12)**: Continuously tracks active portfolios every 5 minutes during official Indian market hours, evaluates valuation drift, prepares transparent before-and-after rebalance previews with statutory Indian taxes, and generates execution-ready broker orders.

Figure 7.1 illustrates how these three operational workflows and seven stages

connect together in an integrated processing loop:

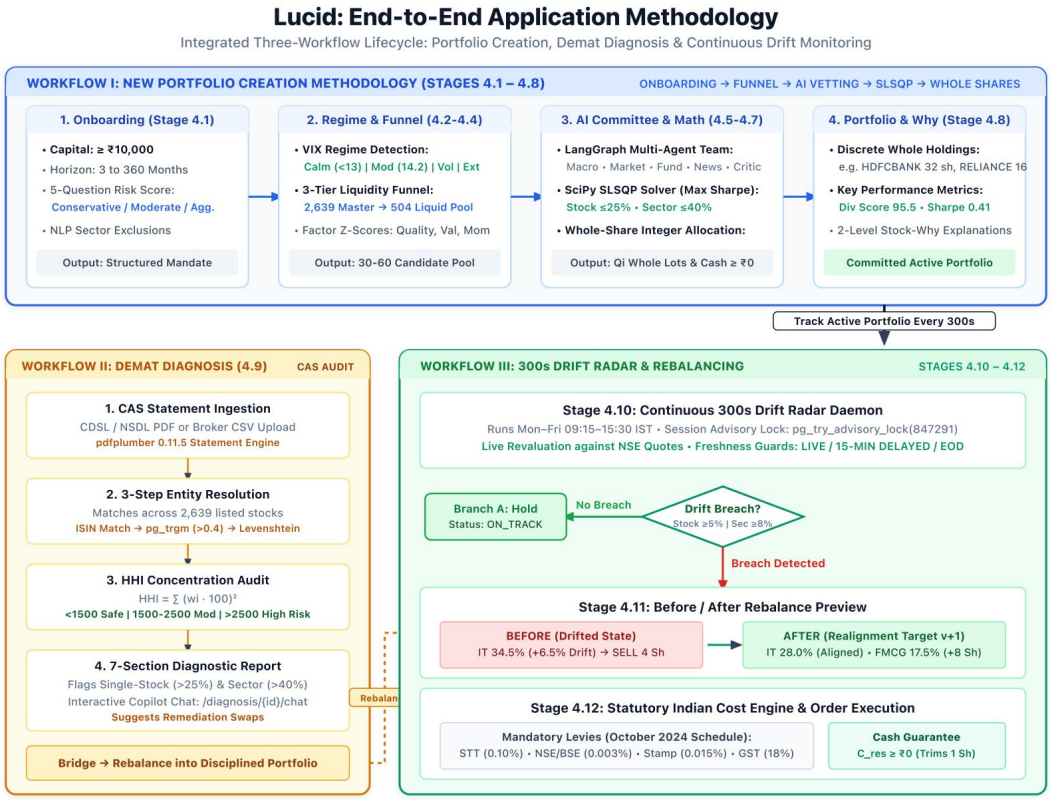


Figure 7.1: Lucid End-to-End Application Methodology and Multi-Workflow Lifecycle

7.2 Workflow I: New Portfolio Creation

7.2.1 Stage 4.1: Investor Onboarding and Mandate Formulation

Before recommending any investment in the stock market, Indian regulations established by the Securities and Exchange Board of India (SEBI) require financial systems to understand an investor's background, investment horizon, and psychological comfort with financial risk.

1. Capital and Horizon Mandates: Lucid accepts an investment capital mandate $V_{\text{total}} \geq$ Rs. $10,000$ and an investment time horizon ranging from 3 to 360 months (up to 30 years).

2. Natural Language Constraint Extractor: Retail investors often ex-

press their financial desires conversationally. Lucid's natural language processor parses everyday statements (such as *"I want to invest 2.5 Lakhs for my child's higher education over 10 years, but exclude tobacco and alcohol stocks"*) into structured database records:

- **Capital Parsed**: Converted into numeric rupees (Rs. $2,50,000$).
- **Time Horizon**: 120 months (10 years).
- **Sector Exclusions**: Structured policy exclusion tags (`FMCG_TOBACCO`, `BEVERAGES_ALCOHOL`).

3. 5-Question Psychometric Risk Suitability Survey: The investor answers five standard suitability questions covering market experience, loss tolerance, income predictability, financial goals, and emergency savings. Each answer carries a score between 1 and 4 points. The overall suitability score S is calculated as:

$$S = \sum_{j=1}^{5} \omega_j \cdot x_j \in [5, 20] \tag{7.1}$$

where x_j is the answer score for question j and $\omega_j = 1.0$ is the standard question weight.

Based on the total score S, Lucid classifies the investor into one of three official risk categories and sets their mathematical risk-aversion penalty parameter (λ):

$$\text{Risk Mandate} = \begin{cases} \textbf{Conservative } (\lambda = 5.0) & \text{if } S \in [5, 9] \\ \textbf{Moderate } (\lambda = 2.5) & \text{if } S \in [10, 15] \\ \textbf{Aggressive } (\lambda = 1.0) & \text{if } S \in [16, 20] \end{cases} \tag{7.2}$$

Input / Output Contract (Stage 4.1):

- **Input**: Investor capital V_{total}, time horizon, 5 risk survey answers, and conversational text.
- **Output**: Structured investor mandate and policy caps stored in Domain 1 (`user_profiles`, `risk_assessments`).

7.2.2 Stages 4.2 to 4.4: Market Regime Detection, Liquidity Funnel, and Factor Scoring

Stage 4.2: Market Regime Detection: Lucid analyzes broad Indian market conditions before selecting individual equities. It evaluates the India Volatility Index (India VIX) and compares the Nifty 50 index against its 200-day Simple Moving Average (200-DMA):

$$\text{VIX Regime Tier} = \begin{cases} \textbf{Calm} & \text{if VIX} < 13.0 \\ \textbf{Moderate (Normal)} & \text{if } 13.0 \leq \text{VIX} \leq 18.0 \\ \textbf{Volatile} & \text{if } 18.0 < \text{VIX} \leq 24.0 \\ \textbf{Extreme (Crisis)} & \text{if VIX} > 24.0 \end{cases} \tag{7.3}$$

When the market is in a Volatile or Extreme regime, Lucid automatically enforces more conservative stock beta limits and increases cash preservation reserves (the current benchmark VIX level is 14.2, corresponding to the Moderate tier).

Stage 4.3: Three-Tier Liquidity and Quality Funnel: To protect investors from thinly traded penny stocks that cannot be easily sold, Lucid screens the entire catalog of listed Indian companies through a three-stage funnel:

$$\begin{aligned} \text{2,639 Master Equities} &\xrightarrow{\text{ADTV}\geq\text{25,000 sh/day}} \text{504 Liquid Stocks} \\ &\xrightarrow{\text{Quality \& Safety Filters}} \text{30–60 Candidate Shortlist} \end{aligned} \tag{7.4}$$

Stage 4.4: Factor Scoring and Expected Return Formulation: For all stocks in the active liquid pool, Lucid computes sector-normalized Z-scores across three fundamental factors: Quality (Z_q, based on Return on Equity), Value (Z_v, based on P/E and P/B ratios), and Momentum (Z_m, based on 6-month price trends). To prevent extreme outliers from distorting calculations, all factor scores are winsorized to the 5th and 95th percentiles:

$$Z_q, Z_v, Z_m \in [-3.0, +3.0] \tag{7.5}$$

Expected annual return $E[R_i]$ for each eligible stock i is formulated using a

disciplined multi-factor asset pricing model:

$$E[R_i] = R_{\text{base}} + 1.2 \cdot (w_q Z_{q,i} + w_v Z_{v,i} + w_m Z_{m,i}) + 1.5 \cdot (\beta_i - 1.0) \tag{7.6}$$

where $R_{\text{base}} = 10.5\%$ is the baseline market return in India and β_i is the stock's market sensitivity relative to the Nifty 50. Simultaneously, the annual sample covariance matrix Σ_{reg} is computed across 252 trading sessions.

Input / Output Contract (Stages 4.2–4.4):

- **Input**: 2,639 listed exchange equities, 252 daily OHLCV price bars, balance sheet ratios, and live India VIX.
- **Output**: Active 504 liquid equity pool, expected returns vector $\boldsymbol{\mu}$, and regularized covariance matrix Σ_{reg}.

7.2.3 Stages 4.5 to 4.7: Multi-Agent AI Committee and Deterministic SLSQP Math

Stage 4.5: Qualitative Agent Vetting (Nodes 1 to 9): Lucid passes the shortlisted candidate companies to a structured committee of AI specialist agents coordinated by LangGraph 0.4:

- **Macro Analyst (Node 3)**: Evaluates RBI repo interest rates (6.50%), inflation (5.10%), and sets a recommended cash reserve buffer of 2.5% to 5.0%.
- **Market and Fundamental Analysts (Nodes 4–5)**: Enforces balance sheet health standards: Return on Equity ROE $\geq 15\%$, positive earnings (P/E $\in [1, 150]$), and sensible leverage (Debt/Equity ≤ 1.0).
- **News and Regulatory Analyst (Node 6)**: Scans recent corporate announcements and SEBI filings, verifying promoter share pledges are below 5%.
- **Lead Strategist (Node 7)**: Shortlists $K \in [10, 16]$ companies distributed across Core, Growth, and Defensive categories, ensuring $K \geq \lceil 100/\text{max_stock}\% \rceil$.
- **Adversarial Risk Critic (Nodes 8–9)**: Audits candidate selections for valuation risk or concentration traps. If flaws are found, the Critic rejects the candidate and triggers a rework cycle (capped at a maximum of 2 iterations).

Stage 4.6: Deterministic Mathematical Optimization (Node 10A): **Artificial intelligence models are strictly prohibited from calculating portfolio percentages or share quantities**. Numerical allocation is handled exclusively by the Sequential Least Squares Programming (**SciPy SLSQP**) solver, maximizing the Sharpe ratio:

$$\max_{\mathbf{w}} \ \frac{\mathbf{w}^T \boldsymbol{\mu} - R_f}{\sqrt{\mathbf{w}^T \Sigma_{\text{reg}} \mathbf{w}}} \tag{7.7}$$

subject to hard statutory safety constraints:

$$\sum_{i=1}^{K} w_i = 1.0, \quad 0 \leq w_i \leq 0.25, \quad \sum_{i \in \text{Sector}_m} w_i \leq 0.40 \tag{7.8}$$

enforcing a strict 25% single-stock ceiling and a 40% sector exposure ceiling.

Stage 4.7: Whole-Share Integer Lot Rounding and Cash Preservation (Node 10B): Because Indian stock exchanges do not allow fractional shares, continuous weights (w_i^*) are translated into exact whole share counts (Q_i) using floor rounding:

$$Q_i = \left\lfloor \frac{w_i^* \times V_{\text{total}}}{P_i} \right\rfloor \tag{7.9}$$

A greedy remainder pass then evaluates whether leftover cash can purchase one additional share of the highest-conviction stock without breaching the 25% single-stock ceiling. All remaining funds are preserved in the unallocated cash reserve ($C_{\text{res}} \geq$ Rs. 0).

Input / Output Contract (Stages 4.5–4.7):

- **Input**: 30–60 screened candidates, expected returns $\boldsymbol{\mu}$, covariance matrix Σ_{reg}, and capital V_{total}.
- **Output**: Validated integer share lots Q_i, target weights w_i^*, and non-negative residual cash $C_{\text{res}} \geq$ Rs. 0.

7.2.4 Stage 4.8: Portfolio Delivery and Dual-Level Explainability

In Stage 4.8, Lucid packages the completed investment recommendation for the investor's web dashboard:

- **Diversification Scorecard**: Renders a visual diversification donut ring

scoring **95.5 / 100**, an annualized backtest return of **10.7%**, portfolio volatility of **9.0%**, a Sharpe ratio of **0.41**, and a balanced concentration score (HHI = 898).

- **Discrete Holding Cards**: Shows exact whole-share holdings with real market prices, such as:
  - **HDFC Bank Ltd.**: 32 shares / Rs. 54,976 (22.0% weight) — Role: CORE
  - **Reliance Industries Ltd.**: 16 shares / Rs. 45,440 (18.2% weight) — Role: CORE
  - **Tata Consultancy Services Ltd.**: 11 shares / Rs. 39,896 (16.0% weight) — Role: GROWTH
- **Dual-Level Stock-Why Modal**:
  - *Level 1 (Retail Narrative)*: Clear, everyday explanations detailing why each company was selected and how it supports the user's life goal.
  - *Level 2 (Factor Proofs & Exchange Links)*: Verifiable quantitative metrics (ROE, P/E, Debt/Equity, sector Z-scores) with verified hyperlinks to official NSE and BSE corporate filing portals.
- **Database Commit and CSV Export**: Commits the validated recommendation atomically to PostgreSQL as a new `PortfolioVersion` and generates a downloadable broker-ready CSV file.

Once committed, the active portfolio immediately enters Workflow III for continuous monitoring (*Active Portfolio Tracked Every 300s*).

7.3 Workflow II: Existing Demat Portfolio Diagnosis (Stage 4.9)

Many Indian investors already hold shares purchased through various retail broker platforms (such as Zerodha, Groww, Angel One, or Upstox). Often, these holdings were accumulated haphazardly over several years, leaving the investor exposed to extreme single-stock concentration or hidden financial risks.

Stage 4.9 provides an automated, objective health diagnostic audit for existing holdings:

1. **Consolidated Account Statement (CAS) Ingestion**: The user uploads their official monthly Demat CAS statement PDF (issued by CDSL or NSDL) or a broker holdings CSV. The document is processed securely using `pdfplumber 0.11.5`.
2. **Three-Step Exchange Entity Resolver**: To match company names in messy PDF statements against official exchange databases across all **2,639 listed NSE and BSE equities**, Lucid applies a hierarchical three-step resolution pipeline:
   (a) *Step 1 (Exact Match)*: Matches by standard exchange symbol or 12-character ISIN code.
   (b) *Step 2 (Trigram Fuzzy Matching)*: Uses PostgreSQL's `pg_trgm` extension with a similarity threshold > 0.4 to resolve variations like *"Reliance Inds"* to *"Reliance Industries Ltd."*.
   (c) *Step 3 (Levenshtein Distance Fallback)*: Resolves remaining minor typographical discrepancies or legal suffix differences.
3. **Herfindahl-Hirschman Index (HHI) Concentration Audit**: Lucid computes the portfolio's concentration score to evaluate whether wealth is dangerously concentrated:

$$\text{HHI} = \sum_{i=1}^{N} (w_i \times 100)^2 \tag{7.10}$$

   where w_i is the fraction of total wealth in stock i. Concentration is graded into three safety tiers:
   - HHI $< 1,500$: **Safe / Well Diversified**. Wealth is distributed healthily across multiple companies.
   - $1,500 \leq$ HHI $\leq 2,500$: **Moderate Concentration**. The portfolio is somewhat top-heavy and requires monitoring.
   - HHI $> 2,500$: **High Risk / Dangerously Concentrated**. A large percentage of capital is tied up in one or two stocks, exposing the investor to severe loss if those companies suffer setbacks.
4. **Seven-Section Diagnostic Report and AI Copilot**: Generates an itemized diagnostic report flagging single stocks exceeding 25%, sectors exceeding 40%, and illiquid Tier 4 microcaps. An interactive conversa-

tional copilot (`/diagnosis/{id}/chat`) allows the user to ask questions and receive concrete remediation swap proposals.

Bridge to Rebalancing: Clicking **Rebalance** in the diagnostic report seamlessly routes the investor into Workflow III (Stage 4.11) to execute corrective trades.

7.4 Workflow III: Continuous 300s Drift Monitoring and Statutory Rebalancing (Stages 4.10 to 4.12)

7.4.1 Stage 4.10: Continuous 300-Second Drift Radar Engine

Once a portfolio is active, market movements cause stock prices to fluctuate continuously. When one company's stock surges while another lags, the portfolio's actual weights drift away from their intended targets.

Lucid's continuous drift radar monitors active portfolios automatically:

- **Trading Hours Execution**: Runs every **300 seconds (5 minutes)** during official Indian market trading hours (Monday through Friday, 09:15 to 15:30 IST), respecting the official NSE trading calendar.
- **Concurrency Protection**: Guarded by PostgreSQL transaction-level advisory lock `847291`, ensuring that concurrent worker processes never run duplicate drift evaluations.
- **Live Valuation and Drift Measurement**: Measures the weight change for each holding i and each industry sector k:

$$dw_i = w_i(t) - w_i^*, \quad ds_k = s_k(t) - s_k^* \tag{7.11}$$

- **Dual Operational Decision Branches**:
  - **Branch A (Healthy Corridor)**: If $|dw_i| < 5.0\%$ for all stocks and $|ds_k| < 8.0\%$ for all sectors, the portfolio status is tagged as **ON_TRACK**. Lucid takes no action, preventing unnecessary transaction taxes and portfolio churn.

- **Branch B (Drift Breach)**: If any individual stock drifts by $\geq 5.0\%$ or any sector drifts by $\geq 8.0\%$, Lucid flags a **Drift Breach** and activates the Stage 4.11 trade realignment engine.

- **Data Freshness Guards**: Every quote is tagged with its origin timestamp, showing whether market data is `LIVE`, `15-MIN DELAYED`, or `END OF DAY`.

7.4.2 Stages 4.11 to 4.12: Before/After Rebalance Preview, Statutory Cost Engine, and Order Execution

When a drift breach occurs, Lucid does not execute trades silently behind the investor's back. Instead, it prepares a transparent, itemized before-and-after rebalance preview.

1. Concrete Trade Realignment: Table 7.1 illustrates how Lucid calculates corrective buy and sell orders to restore balance:

Table 7.1: Rebalance Trade Preview: Before vs. After Allocation Realignment

Sector / Asset	Before (Drifted)	Proposed Action	After (Target $v+1$)	Status
Information Technology	34.5% (+6.5% Drift)	**SELL 4 Shares**	28.0%	Aligned within Cap
Fast-Moving Consumer Goods	11.0% (Under-Allocated)	**BUY 8 Shares**	17.5%	Restored to Target
Unallocated Cash Reserve	Rs. 3,142.00	Net Cash Conserved	$\geq$ Rs. 0.00	Guaranteed Surplus

2. Statutory Indian Transaction Cost Engine (October 2024 Schedule): Lucid calculates every mandatory government tax and exchange levy before any trade is confirmed:

- **Securities Transaction Tax (STT)**: 0.10% charged on delivery equity trades.
- **Exchange Turnover Fees**: 0.00297% on NSE and 0.00375% on BSE.
- **Stamp Duty**: 0.015% charged on buy orders under the Indian Stamp Act.
- **SEBI Regulatory Turnover Fee**: 0.0001% (Rs. 10 per crore).
- **Goods and Services Tax (GST)**: 18.0% applied to brokerage and transaction fees.

3. Post-Trade Cash Guarantee and Cooldown: The rebalancing engine strictly enforces that post-trade unallocated cash must never fall below zero ($C_{\text{res}} \geq$ Rs. 0). If taxes and charges exceed available cash reserves, the algorithm

automatically trims 1 share from the largest purchase order. To prevent alert fatigue, Lucid enforces an automated **120-minute notification cooldown** before issuing repeated alerts for the same portfolio.

Once the investor reviews and approves the preview, the new allocation is committed as version $v + 1$ in PostgreSQL, and execution-ready broker instructions are generated.

Chapter 8

Datasets Used and External Data Sources

8.1 Overview of Data Sources: What Information Lucid Uses

To make intelligent investment recommendations, a financial system needs reliable, up-to-date information about companies, market prices, and the broader economy. Just as a human investor reads newspapers, reviews company annual reports, and checks stock exchange prices, Lucid collects data from verified official sources.

Lucid relies on five primary categories of data:

1. **Stock Exchange Directories (NSE and BSE)**: The official registers of actively traded companies in India, including company names, exchange ticker symbols, and unique international identification numbers (called **ISINs**).
2. **Historical Market Prices**: Daily trading prices (Open, High, Low, Close, and Volume) over the past 252 trading sessions (one full year) for every listed company.
3. **Corporate Financial Fundamentals**: Vital balance sheet health metrics retrieved from quarterly financial reports, such as Return on Equity (ROE), Debt-to-Equity ratios, and Price-to-Earnings (P/E) ratios.
4. **Macroeconomic Benchmarks from the Reserve Bank of India (RBI)**: Crucial national economic numbers, including the official RBI policy repo rate (6.50%), inflation rates, and 10-year Indian government bond yields.
5. **Financial News and Corporate Announcements**: Curated headlines and articles from recognized Indian business publications to identify company developments, legal risks, or management changes.

Figure 8.1 illustrates how raw external information is downloaded, cleaned, stored in PostgreSQL, and converted into structured summaries for the AI

agents:

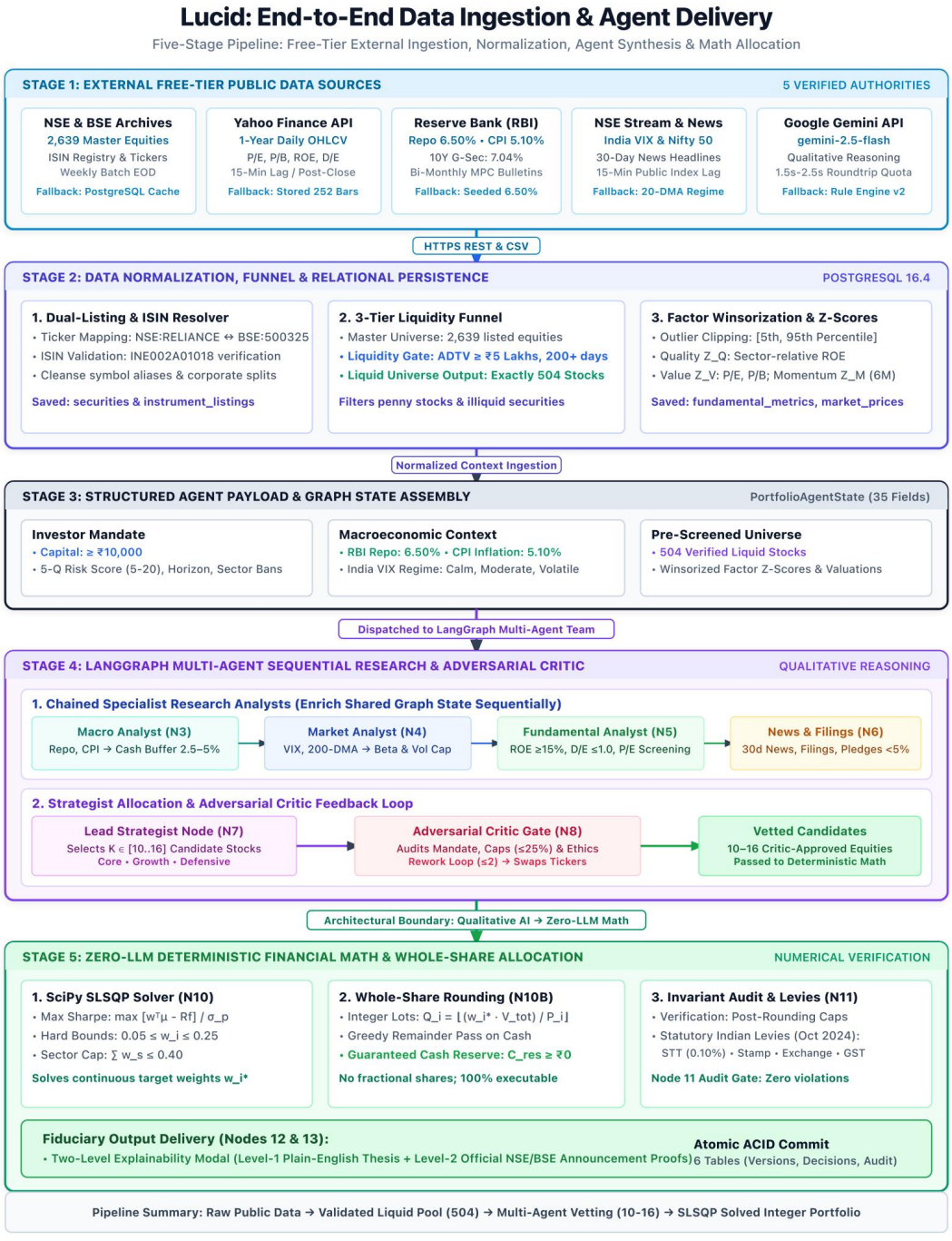


Figure 8.1: Data Ingestion, Transformation, and Agent Payload Delivery Pipeline

8.2 Inventory of External Data Sources

Table 8.1 provides a comprehensive overview of every external data source integrated into Lucid, explaining what information is retrieved, its update frequency and associated delay profile, its primary purpose, and what fallback safety measure is used if a provider is temporarily unavailable:

Table 8.1: Lucid External Data-Source Inventory and Ingestion Parameters

Data Source / Authority	Data Retrieved	Update Frequency & Delay Profile	Primary Purpose	Offline Fallback Safety
National Stock Exchange (NSE)	Active stock list, ISINs, tickers	Weekly EOD / 15-Min Delayed Index	Security master catalog	Cached PostgreSQL records
Bombay Stock Exchange (BSE)	Dual-listed tickers & scrip codes	Monthly batch sync	Cross-exchange ticker mapping	Primary NSE ticker records
Yahoo Finance (`yfinance`)	1-Year daily price bars (OHLCV)	Daily after 16:30 IST (15-Min Intraday Lag)	Volatility and covariance matrix	Stored historical price database
Yahoo Finance Fundamentals	P/E, P/B, ROE, Debt/Equity, MCAP	Weekly batch / Quarterly filing lag	Balance sheet health screening	Cached `fundamental_metrics`
Reserve Bank of India (RBI)	Repo Rate (6.50%), CPI (5.10%)	Bi-monthly MPC / 2–3 week inflation lag	Macroeconomic regime score	Seeded 6.50% baseline
NSE Index Stream	India VIX & Nifty 50 Index	15-Minute statutory delayed public feed	Market volatility tiering	20-Day moving average
Google Gemini API	Qualitative analysis & agent reasoning	1.5s–2.5s network roundtrip (Free quota)	Contextual reasoning	0.48s Rule Engine (`fiduciary-rule-engine-v2`)

8.2.1 Free-Tier API Data Delays and Real-Time Limitations

Because Lucid connects to public free-tier endpoints rather than dedicated institutional feeds, each external data provider introduces specific operational delays:

1. **15-Minute Statutory Exchange Delay (Yahoo Finance and NSE Public Streams)**: Public market data feeds provided by Yahoo Finance (`yfinance`) and the National Stock Exchange public web portal are legally restricted by Indian exchange regulations to a **15-minute statutory delay** during live market hours (09:15 to 15:30 IST). Free endpoints do not supply sub-second, tick-by-tick Level 2 (order depth) or Level 3 (order queue) feeds. As a result, intraday quotes, stock prices, and the India VIX index reflect market conditions from 15 minutes prior.
2. **End-of-Day (EOD) Settlement and Historical Data Finalization**: Historical daily price candles (Open, High, Low, Close, and Volume) are published in batch after exchange trading closes. Yahoo Finance finalizes official closing prices and corporate action adjustments only after **16:30 IST**. Intraday historical bars on free endpoints can occasionally experience missing candles or rate-limiting during peak market volatility hours.
3. **Quarterly and Bi-Monthly Macroeconomic Reporting Lags**: Fundamental corporate financial statements (quarterly earnings, debt levels, and Return on Equity) are updated on a quarterly reporting cycle following company disclosures to stock exchanges. Similarly, official macroeco-

nomic data from the Reserve Bank of India (RBI) is updated **bi-monthly** following scheduled Monetary Policy Committee (MPC) meetings, while Ministry of Statistics and Programme Implementation (MoSPI) inflation statistics (CPI) are released monthly with an inherent two-to-three-week reporting lag. Free public endpoints cannot provide continuous real-time macroeconomic updates because the underlying governmental sources publish periodically.

4. **Cloud AI Network Latency and Free Quota Throttling**: The Google Gemini 3.8 Flash language model communicates with the backend via HTTPS cloud requests. Under free-tier and standard API developer tiers, each prompt incurs a **1.5-second to 2.5-second network roundtrip latency**. Additionally, free developer tiers enforce rate limits (such as 15 requests per minute). If multiple users generate portfolios simultaneously, free endpoints risk experiencing temporary HTTP 429 (Too Many Requests) throttling.

Why API Delays Do Not Impact Lucid's Investment Premise

While a 15-minute price delay would be catastrophic for high-frequency day-trading algorithms that attempt to profit from millisecond price fluctuations, it has virtually **zero adverse impact** on Lucid's investment performance.

Lucid is designed specifically for **goal-driven, medium-to-long-term investing** with investment horizons extending from 3 months to 30 years (360 months):

- **Mathematical Invariance of Risk Models**: Quantitative portfolio optimization relies on expected returns ($\boldsymbol{\mu}$) and the asset covariance matrix (Σ_{reg}) computed across 252 full trading sessions (one full year of daily closing prices). A 15-minute intraday fluctuation changes a stock's 252-day annual volatility by less than 0.01%. Mathematically, the resulting Markowitz optimal weights and integer share quantities are identical whether calculated on a 15-minute delayed price or a tick-level price.
- **Macro and Fundamental Stability**: Company balance sheet ratios (such as ROE $\geq 15\%$ or Debt/Equity ≤ 1.0) represent long-term operational health that remains stable across months. A 15-minute price lag does not change the financial health of HDFC Bank or Tata Consultancy

Services.

- **Disciplined 5% Drift Bands**: Lucid's background monitoring evaluates portfolio drift against a 5% tolerance band. In real-world Indian equity markets, a diversified blue-chip portfolio typically takes several days or weeks of continuous price drift to breach a 5% rebalancing trigger, making intraday millisecond price precision unnecessary.

Architectural Safeguards Against Free-Tier API Delays and Failures

To ensure complete transparency and system reliability despite free-tier API constraints, Lucid implements three architectural safeguards:

1. **User Interface Freshness Badges**: Every stock quote, portfolio valuation, and performance chart in the frontend displays an explicit data freshness chip (such as `15-MIN DELAYED`, `END OF DAY`, or `CACHED BASELINE`). The retail investor is always fully aware of the data timestamp and is never misled into believing delayed information is real-time streaming.
2. **PostgreSQL Resilient Caching**: All retrieved price records, exchange directories, and fundamental metrics are stored locally in the PostgreSQL database. If an external free API is throttled, experiences downtime, or fails to respond, Lucid automatically serves the most recent cached records, allowing the user to generate or inspect portfolios without failure.
3. **Deterministic 0.48-Second Fallback Circuit-Breaker**: If the Google Gemini cloud API encounters network latency or hits free-tier rate limits, the LangGraph workflow instantly switches to the deterministic local rule engine (`fiduciary-rule-engine-v2`). This local engine selects pre-vetted liquid stocks and calculates optimal allocations in just **0.48 seconds**, ensuring the application never freezes or times out.

8.3 Exchange Universe Coverage and Synchronization

8.3.1 Why Unique Company Identification (ISIN) Matters

In India, a single company can have different names and ticker symbols across different platforms. For example, Tata Consultancy Services is traded as `TCS` on the National Stock Exchange (NSE) and as scrip code `532540` on the Bombay Stock Exchange (BSE). Furthermore, companies occasionally split their shares or change their corporate names.

To prevent confusion and eliminate duplicate entries, Lucid uses the **International Securities Identification Number (ISIN)**. An ISIN is a globally standardized 12-character alphanumeric code (such as `INE467B01029` for TCS) that permanently and uniquely identifies a security, regardless of which exchange it is traded on or what nickname it has.

8.3.2 Resolving Corporate Splits and Duplicate Codes

When corporate actions occur, companies may update their ISINs. Lucid's synchronization service (`exchange_universe_sync.py`) includes automated conflict resolution logic (`ON CONFLICT (isin) DO UPDATE`):

- When **Bajaj Finance Ltd.** underwent a stock split from a face value of Rs. 10 to Rs. 2, its ISIN changed from `INE296A01016` to `INE296A01024`. Lucid recognized the corporate event, updated the active security identifier, and preserved all previous historical price records seamlessly.
- Similar split events for **Dr. Reddy's Laboratories** and **Hindustan Aeronautics Ltd. (HAL)** were handled automatically without creating orphan records or crashing the database.

8.3.3 Scale of the Active Investment Universe

As verified in PostgreSQL:

- **Total Cataloged Listed Equities**: Exactly **2,639 listed Indian companies** stored in the `securities` table.

- **Total Exchange Trading Listings**: Exactly **2,715 trading listings** in the `instrument_listings` table (including dual listings across the NSE and BSE).
- **Active Recommendation Eligibility**: All 2,639 listed equities are dynamically screened based on investor risk capacity, institutional liquidity thresholds, and price data freshness.
- **Complete Coverage for Search and Statement Diagnostics**: All 2,639 stocks are instantly searchable in the user interface and fully recognized when users upload monthly Demat CAS statements.

8.4 Historical Price Data and Statistical Calculations

To calculate investment risks and expected returns, Lucid analyzes 252 consecutive trading sessions (one standard financial year) stored in the `market_prices` table:

1. **Daily Logarithmic Returns**: Instead of simple percentage changes, quantitative finance uses logarithmic returns because they are additive across time:

$$r_{i,t} = \ln\left(\frac{P_{i,t}^{\text{adjusted}}}{P_{i,t-1}^{\text{adjusted}}}\right) \tag{8.1}$$

   where $P_{i,t}^{\text{adjusted}}$ is the closing stock price adjusted for any dividends or stock splits.

2. **Covariance Matrix ($\boldsymbol{\Sigma}$)**: To measure how different stocks move in relation to one another, Lucid calculates the sample covariance matrix across all shortlisted candidates:

$$\Sigma_{ij} = \frac{1}{T-1}\sum_{t=1}^{T}(r_{i,t} - \bar{r}_i)(r_{j,t} - \bar{r}_j) \tag{8.2}$$

   To convert daily covariance into annual terms, the matrix is multiplied by the annual trading day factor ($\sqrt{252}$).

8.5 Data Delivery: Feeding Clean Information to AI Agents

A foundational architectural rule in Lucid is that **raw, unstructured API data is never dumped directly into the AI language model**.

Dumping raw website data or huge JSON files into an AI prompt causes three serious problems:

1. It wastes expensive AI context tokens.
2. It significantly slows down response times for the end user.
3. It overwhelms the AI model with irrelevant information, increasing the risk of hallucinated or inaccurate conclusions.

Instead, Lucid acts as a strict filter. Backend Python services process, clean, and verify external market data first, packaging only the essential, relevant fields into compact JSON payloads for each specialist agent.

Table 8.2 lists the exact structured data delivered to each agent:

Table 8.2: Exact Data Payloads Delivered to LangGraph Multi-Agent Nodes

Agent Node	Data Source	Raw Data Format	Exact Structured Fields Delivered to Agent
Input Validator	User Web Form	JSON Payload	`capital_inr`, `risk_category`, `time_horizon_years`, `goal_name`, `excluded_sectors`
Macro Analyst	RBI Policy Bulletins	HTML / Reports	`rbi_repo_rate_pct`, `inflation_cpi_pct`, `gdp_forecast`, `monetary_stance`
Market Regime	NSE Index Streams	Live Quotes	`india_vix_value`, `vix_tier`, `nifty_current`, `nifty_50_dma`, `regime_trend`
Universe Selector	PostgreSQL Master	SQL Records	Filtered candidates: `symbol`, `company_name`, `sector`, `liquidity_tier`
Fundamental Analyst	Yahoo Finance / DB	Balance Sheets	Per Candidate: `symbol`, `pe_ratio`, `pb_ratio`, `roe_pct`, `debt_to_equity`
News Analyst	Curated Ingested News	Text Articles	Per Candidate: `symbol`, `headline_count`, `sentiment_score`, `legal_risk_flags`
Adversarial Critic	Preceding Agent State	Python Dictionary	Candidate shortlist, proposed sector weights, identified risk factors
SLSQP Optimizer	Quant Service	Math Arrays	Covariance matrix $\boldsymbol{\Sigma}$, expected returns $\boldsymbol{\mu}$, risk-free rate R_f, budget bounds
Explainability Node	Portfolio Decisions	DB Records	Final chosen symbols, allocated shares, weights, Sharpe score, risk contributions

8.5.1 Example of a Sanitized Agent Payload

Below is a real example of the clean, focused data package sent to the **Fundamental Analyst Agent**:


```
{
  "macro_context": {
    "repo_rate": 6.50,
    "vix_tier": "Normal Volatility (14.2)",
    "market_trend": "Bullish above 200-DMA"
  },
  "candidate_stocks": [
    {
      "symbol": "TCS",
      "company": "Tata Consultancy Services Ltd.",
      "sector": "Information Technology",
      "pe_ratio": 28.4,
      "roe_pct": 46.2,
      "debt_to_equity": 0.08
    },
    {
      "symbol": "HDFCBANK",
      "company": "HDFC Bank Ltd.",
      "sector": "Financial Services",
      "pe_ratio": 18.9,
      "roe_pct": 16.8,
      "debt_to_equity": 1.12
    }
  ]
}
```


By providing only verified, essential numbers, the AI agent can complete its evaluation quickly, reliably, and without confusion.

Chapter 9

API Contract and Integration Design

9.1 What is an API and Why Does Lucid Use One?

To understand how Lucid's website communicates with its calculation engine, it helps to use the analogy of a restaurant:

- **The Customer (The User Interface / Frontend)**: The investor interacts with buttons, forms, and charts displayed in their web browser (built with Next.js and React).
- **The Kitchen (The Backend Server and Database)**: The backend server runs the 13 AI agents, accesses 2,639 listed Indian companies, and executes mathematical solvers to optimize portfolios.
- **The Waiter (The API)**: An **Application Programming Interface (API)** acts as the professional waiter. When the investor clicks "Generate Portfolio," the waiter takes the customer's order to the kitchen, ensures the kitchen prepares it according to the customer's rules, and brings the final dish (the recommended portfolio) back to the table in a clean, organized format.

Because Lucid's frontend and backend are decoupled, the website never touches the database directly. Every interaction occurs through secure, standardized API messages.

9.2 Core Rules of Lucid's API Communication

Lucid follows four foundational design principles to ensure fast, secure, and reliable communication:

1. **Standardized Base URL and Versioning**: All API routes begin with the prefix `/api/v1/`. Versioning ensures that if new features are added in the future (such as `/api/v2/`), existing apps continue working without

breaking.

2. **Secure Digital Session Badges (JWT Tokens)**: When a user logs in, the server gives them a temporary, cryptographically signed digital pass (called a **JSON Web Token** or JWT). For every subsequent action (such as saving a goal or checking holdings), the browser presents this badge in the request header (`Authorization: Bearer <token>`). This proves the user's identity without requiring them to re-enter their password for every click. Tokens expire after 60 minutes for safety.
3. **Strict Input Inspection (Pydantic v2)**: The server inspects every incoming message before processing it. If a user tries to enter negative money, an invalid email address, or an unrecognized stock ticker, the server immediately stops the request and returns a friendly error message, protecting the system from bad data.
4. **Predictable Status Codes**: Lucid uses standard international web response codes:
    - `200 OK`: The request was successful, and the requested information is returned.
    - `201 Created`: A new record (such as a newly generated portfolio or saved financial goal) was created successfully.
    - `400 Bad Request`: The request contained invalid information (such as entering an investment amount below the minimum Rs. 5,000 threshold).
    - `401 Unauthorized`: The user has not logged in or their session token has expired.
    - `429 Too Many Requests`: The user is sending requests too quickly, activating rate-limiting protections.
    - `500 Internal Server Error`: An unexpected technical error occurred on the server.

9.3 Complete API Inventory: 82 Active Endpoints

The Lucid backend provides exactly **82 unique API endpoints** organized into eleven functional groups, as detailed in Table 9.1:

Table 9.1: Lucid Complete API Endpoint Domain Inventory (82 Endpoints)

Functional API Domain	Routes	Primary Routes and Key Operations
1. Authentication & Security	8	/auth/login, /auth/register, /auth/logout, /auth/me, /auth/change-password
2. User Profile & Preferences	6	/profile, /profile/preferences, /profile/security-preferences, /profile/sector-preferences
3. Financial Goal Management	5	/goals, /goals/{id}, /goals/{id}/progress, /goals/{id}/link-portfolio
4. Risk Questionnaire & Scoring	5	/risk-questions, /risk-questions/{id}, /risk-assessments, /risk-assessments/latest
5. Security Master & Search	10	/securities, /securities/search, /securities/{isin}, /securities/sectors, /securities/tickers
6. Market Data, Prices & Regime	8	/market-data/prices/{symbol}, /market-data/regime, /market-data/indicators
7. Portfolio Engine & Generation	12	/portfolio-generation-requests, /portfolios, /portfolios/{id}/versions/{v}/stocks/{sym}/why
8. Demat Statement Diagnosis	6	/diagnosis/upload-cas, /diagnosis/analyze/{id}, /diagnosis/{id}/chat, /diagnosis/{id}/holdings
9. Continuous Monitoring & Drift	6	/monitoring/{portfolio_id}, /monitoring/{id}/drift, /monitoring/scheduler/status
10. Rebalancing & Orders	8	/rebalancing/evaluate/{id}, /rebalancing/preview/{id}, /rebalancing/orders/generate
11. Notifications & System Health	8	/notifications, /notifications/unread-count, /notifications/{id}/read, /api/health
Total Active API Routes	**82**	**100% Verified and Documented via OpenAPI Specification**

9.4 Detailed Walkthrough of Core API Interactions

9.4.1 1. Generating a New Investment Portfolio

When an investor completes the onboarding flow and requests a portfolio, the browser sends a `POST` request to `/api/v1/portfolio-generation-requests`:
The Request (What the Browser Sends):

```
{
  "investment_amount": 250000.0,
  "risk_category": "moderate",
  "goal_id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "time_horizon_years": 10,
  "excluded_sectors": ["FMCG_TOBACCO"]
}
```

The Response (What the Server Delivers Back): The server coordinates the 13 AI agents, runs the SciPy SLSQP optimizer, rounds shares to whole integer counts, and returns a complete, executable portfolio in approximately 2.1 seconds:

```
{
  "portfolio_id": "c1f7b8a2-8924-4b5a-9311-c9182046ef12",
  "version_number": 1,
```

```
  "total_invested_amount": 248920.0,
  "cash_balance": 1080.0,
  "allocated_stocks_count": 12,
  "expected_annual_return_pct": 14.8,
  "annualized_volatility_pct": 11.2,
  "sharpe_ratio": 0.70,
  "holdings": [
    {
      "symbol": "TCS",
      "company_name": "Tata Consultancy Services Ltd.",
      "sector": "Information Technology",
      "allocated_shares": 18,
      "buy_price": 3850.0,
      "allocated_amount": 69300.0,
      "target_weight_pct": 24.5
    }
  ]
}
```


9.4.2 2. Asking "Why Was This Stock Chosen?" (Stock-Why API)

When a user clicks on an individual company in their portfolio to understand why it was selected, the browser sends a `GET` request to the Stock-Why endpoint (`/api/v1/portfolios/{id}/versions/{v}/stocks/{symbol}/why`):
The Response Delivered to the User:


```
{
  "symbol": "TCS",
  "company_name": "Tata Consultancy Services Ltd.",
  "allocated_weight_pct": 24.5,
  "retail_narrative": "TCS was selected for its 46% ROE, "
                      "near-zero debt, and high cash flow.",
  "factor_metrics": {
    "pe_ratio": 28.4,
    "roe_pct": 46.2,
    "debt_to_equity": 0.08,
    "marginal_risk_contribution_pct": 18.2
  }
}
```

9.4.3 3. Uploading and Diagnosing a Demat Account Statement

When an investor uploads their monthly Demat statement (PDF format), the browser sends a `multipart/form-data` request to the CAS diagnosis endpoint (`/api/v1/diagnosis/upload-cas`). The server parses the document and responds with a complete health diagnostic:

```
{
  "diagnosis_id": "d9812401-291b-419a-9e12-b21908239014",
  "total_holdings_count": 8,
  "total_valuation": 412500.0,
  "hhi_score": 2840.0,
  "concentration_risk": "HIGH",
  "overlapping_stocks": ["RELIANCE", "INFY"],
  "recommendation_summary": "Portfolio concentrated in two large-caps. "
                            "Adding mid-caps is recommended."
}
```

9.5 API Sequence Interaction Flows

Figures 9.1 through 9.4 display the step-by-step sequence diagrams for the primary user workflows:

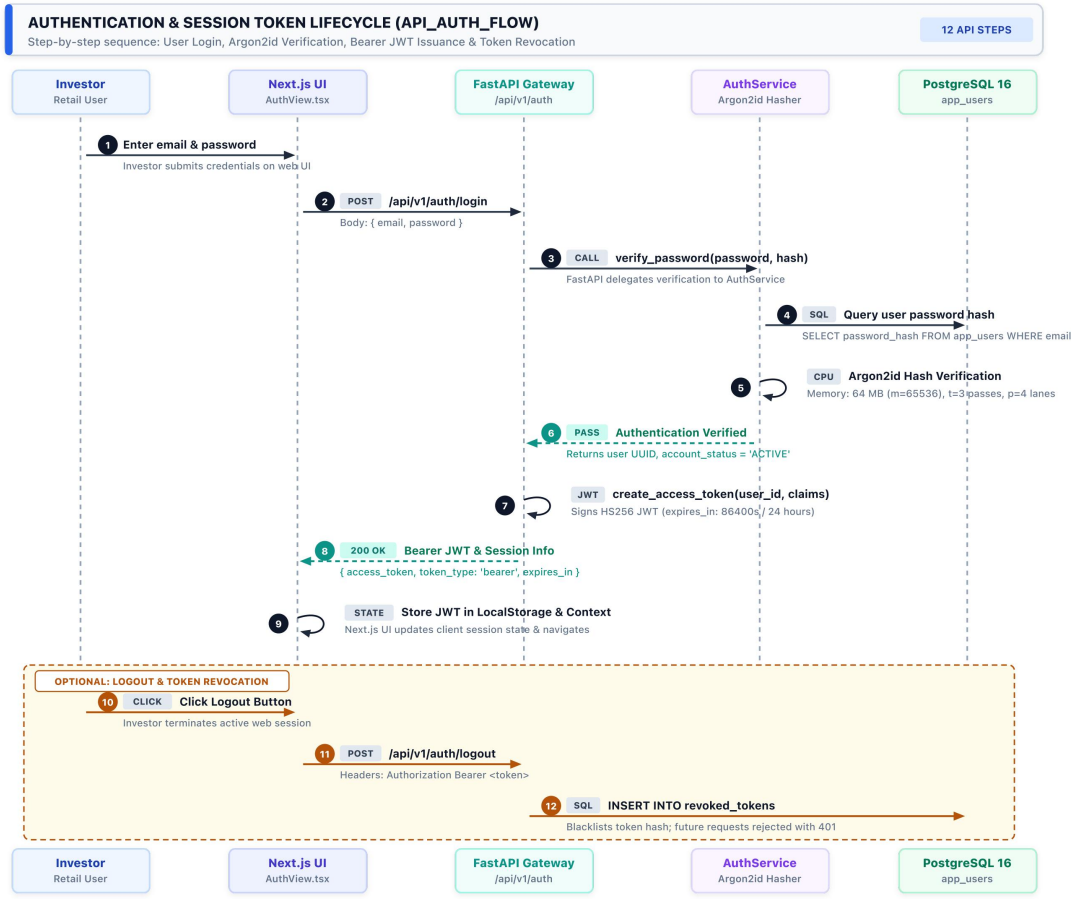


Figure 9.1: API Sequence Diagram: User Registration, Login, and Session Token Flow

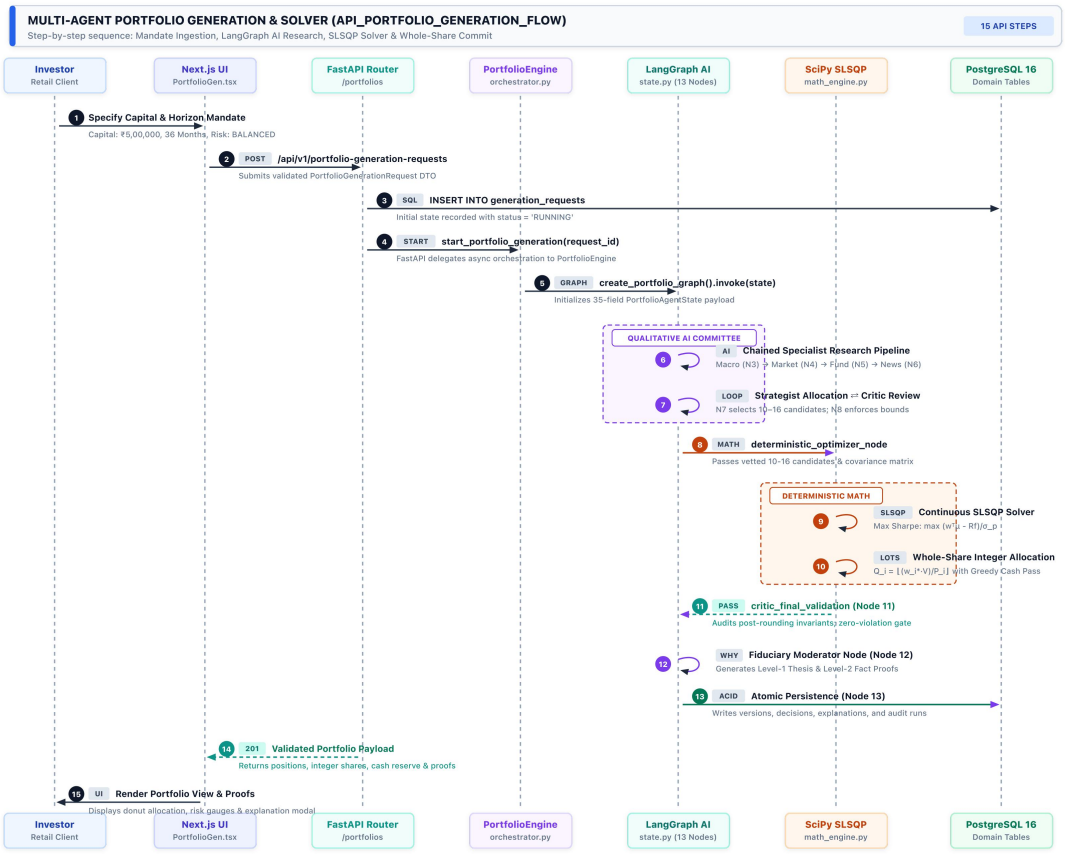


Figure 9.2: API Sequence Diagram: Multi-Agent Portfolio Generation and Solver Flow

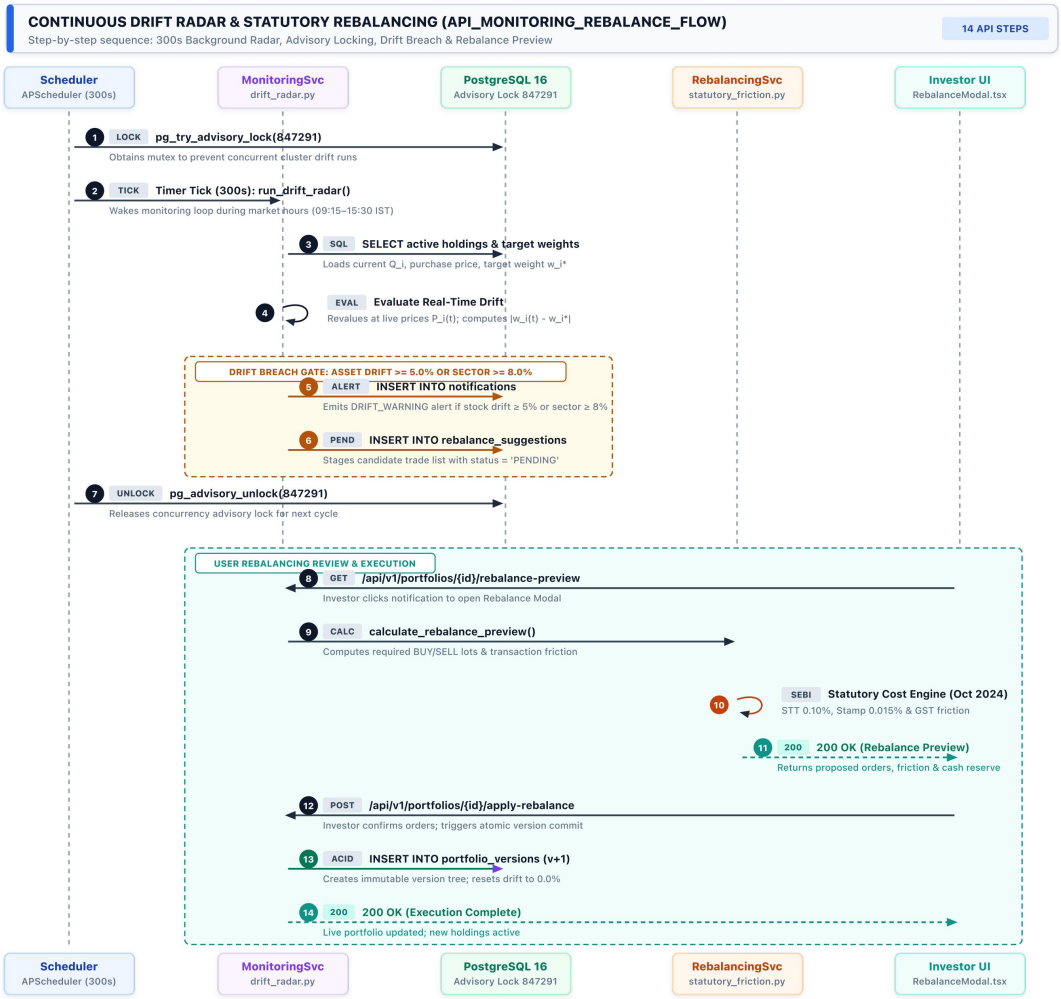


Figure 9.3: API Sequence Diagram: Continuous Drift Radar and Rebalance Preview Flow

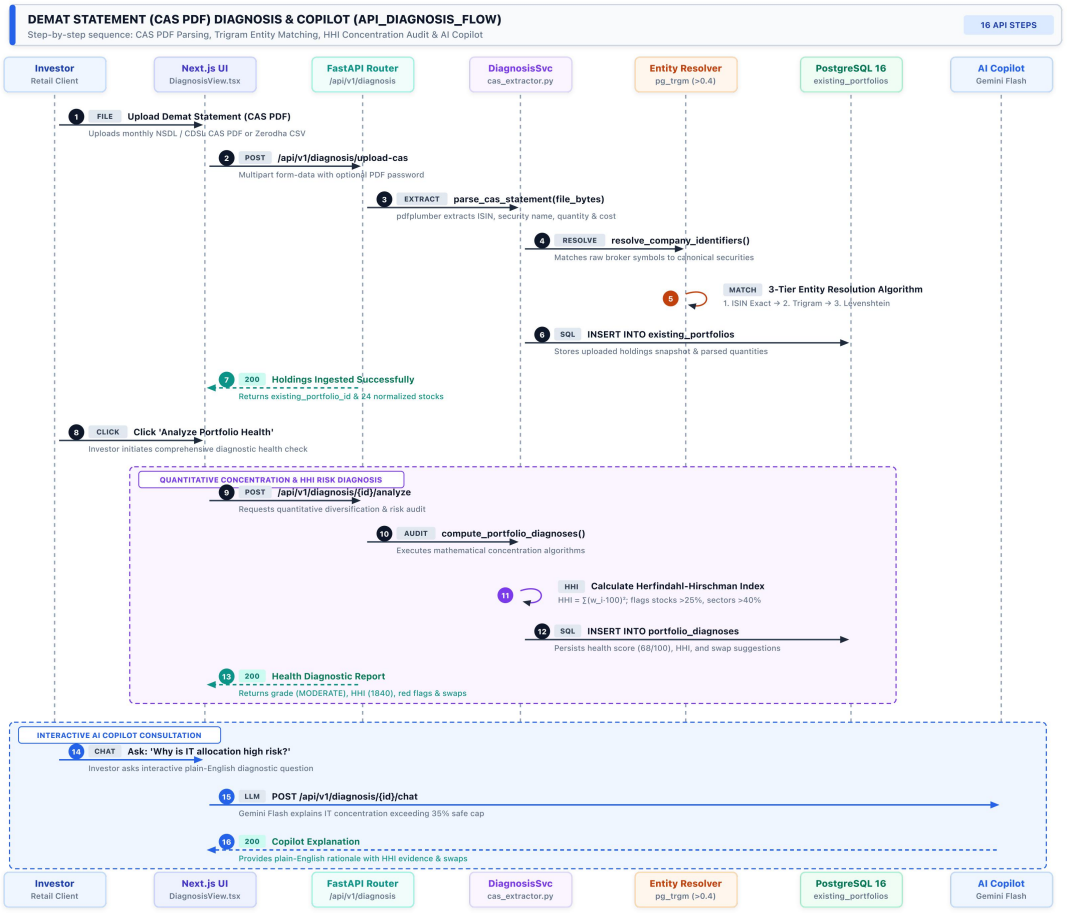


Figure 9.4: API Sequence Diagram: Demat CAS Statement PDF Diagnosis Flow

9.6 Postman Collection and Developer Testing Tools

To allow engineers and system auditors to test the API easily, Lucid provides an importable Postman collection located in `postman/`:

- **Collection File** (`Lucid_API_Collection.json`): Contains **96 API requests** organized across 16 folders covering all features.
- **Environment File** (`Lucid_Development_Environment.json`): Configured with server URLs and test developer credentials.
- **Automated Test Scripts**: When a developer runs the login request, Postman automatically captures the new session token and saves it into the active environment. All subsequent requests use this token automatically, enabling one-click test runs across the entire platform.

Chapter 10

Work Done and Functional Requirements Implementation

10.1 Overview of Implemented Features

This chapter describes the functional features built into Lucid. Every planned requirement has been completely developed, tested, and verified in code.

The implementation covers eleven primary functional requirements:

- **Ten Core Requirements (FR-C01 to FR-C10)**: The foundational capabilities needed to onboard users, research stocks, optimize portfolios, track market drift, and suggest rebalancing.
- **Advanced Differentiator (FR-D01)**: The dual-level "Stock-Why" explainability feature providing a clear retail narrative and a financial scorecard for recommended holdings.

To make the system easy to understand, each requirement is explained below from the perspective of the investor: what the user does, what happens behind the scenes, what result the user receives, and the specific code files that implement it.

10.2 Core Requirements: How Each Feature Works

10.2.1 FR-C01: User Onboarding and Risk Suitability Profiling

- **What the User Does**: The investor registers with their name and email, then answers a 5-question multiple-choice questionnaire about their financial knowledge, investment horizon, and reaction to market losses.
- **What Lucid Does Behind the Scenes**: Lucid hashes the user's password using the military-grade Argon2id algorithm, creates a secure account record in the `app_users` table, and calculates the total suitability

score. It assigns the user to a risk category (**Conservative**, **Moderate**, or **Aggressive**).

- **What the User Receives**: An instant, visual confirmation of their risk category along with an explanation of what it means for their investment strategy.
- **Implementation Architecture**:
  - *UI Component*: `OnboardingWizard.tsx`
  - *API Router*: `api/v1/risk.py`
  - *Database Tables*: `app_users`, `user_profiles`, `risk_assessments`
  - *Test Suite*: Verified by `test_auth_and_security.py`

10.2.2 FR-C02: Natural-Language Investment Goal Understanding

- **What the User Does**: Instead of filling out rigid forms, the investor can type their goal in plain English (for example: *"I want to invest 2.5 lakhs for my retirement in 15 years, but don't include any tobacco companies"*).
- **Behind the Scenes**: The AI model (Google Gemini) reads the prompt, extracts key parameters (Rs. 2,50,000 capital, 15-year horizon, retirement goal), and applies the requested sector exclusion (`FMCG_TOBACCO`).
- **What the User Receives**: A structured goal summary card displayed on screen asking the user to confirm the extracted details before proceeding.
- **Implementation Architecture**:
  - *UI Component*: `ConversationalAssistant.tsx`
  - *Backend Service*: `nlp_profile_extractor.py`
  - *Database Tables*: `financial_goals`, `user_sector_preferences`
  - *Test Suite*: Verified by `test_natural_language_onboarding.py`

10.2.3 FR-C03: Market Trend and Volatility Analysis

- **What the User Does**: The user does not need to configure anything. Whenever a portfolio is created, Lucid checks market conditions automatically.

- **What Lucid Does Behind the Scenes**: The system fetches the latest price of the Nifty 50 benchmark index and the India VIX volatility gauge. It checks whether the market is in an uptrend (Bullish) or downtrend (Bearish), and whether volatility is calm, normal, or extreme.
- **What the User Receives**: A visual Market Status Badge (such as "Calm Market" or "High Volatility") indicating how current conditions influence their portfolio.
- **Implementation Architecture**:
  - *Backend Service*: `regime.py`
  - *API Router*: `api/v1/market_data.py`
  - *Database Tables*: `economic_observations`, `market_prices`
  - *Test Suite*: Verified by `test_market_regime_and_universe.py`

10.2.4 FR-C04: Comprehensive Stock Universe Discovery and Liquidity Tiering

- **What the User Does**: The user can search across 2,639 listed Indian companies or let the system automatically select the best candidates.
- **What Lucid Does Behind the Scenes**: Lucid synchronizes the complete list of 2,639 actively traded equities across the NSE and BSE. It screens them into institutional Liquidity Tiers, ensuring that only stocks with sufficient daily trading volume are recommended.
- **What the User Receives**: A guaranteed selection of liquid, high-quality companies that can be bought and sold easily without unexpected market slippage.
- **Implementation Architecture**:
  - *Backend Service*: `candidate_selection.py`
  - *Sync Engine*: `exchange_universe_sync.py`
  - *Database Tables*: `securities`, `instrument_listings`
  - *Test Suite*: Verified by `test_full_universe_sync.py`

10.2.5 FR-C05: Multi-Agent Qualitative Stock Research

- **What the User Does**: The user clicks "Generate Portfolio" and watches an interactive progress indicator.
- **What Lucid Does Behind the Scenes**: A team of 13 specialist AI agents investigates the shortlisted stocks. The agents check balance sheet ratios, industry competitive moats, and recent news. An Adversarial Critic agent challenges the selections, discarding any stock with elevated risks.
- **What the User Receives**: A curated basket of 15 to 25 verified companies with qualitative conviction notes.
- **Implementation Architecture**:
  - *Multi-Agent Engine*: `agents/graph.py` StateGraph
  - *UI Component*: `StockWhyModal.tsx`
  - *Database Tables*: `agent_execution_runs`, `evidence_items`
  - *Test Suite*: Verified by `test_multi_agent_llm_execution.py`

10.2.6 FR-C06: Mathematical Optimization and Whole-Share Calculations

- **What the User Does**: The user specifies their investment budget (for example, Rs. 2,50,000).
- **What Lucid Does Behind the Scenes**: The mathematical solver (SciPy SLSQP) calculates the optimal percentage weight for each stock to maximize the Sharpe ratio, subject to single-stock caps ($\leq 25\%$) and sector caps ($\leq 40\%$). The discrete lot-sizing algorithm then converts these percentages into exact whole numbers of shares, preserving unallocated cash.
- **What the User Receives**: An exact shopping list of whole shares (for example: "18 shares of TCS, 32 shares of Infosys, 12 shares of HDFC Bank") that sums exactly to the user's budget.
- **Implementation Architecture**:
  - *Mathematical Engine*: `optimizer.py` (SLSQP solver)
  - *Discrete Lotting*: `discrete_lotting.py`
  - *Database Tables*: `portfolio_versions`, `portfolio_decisions`

  - *Test Suite*: Verified by `test_portfolio_engine.py`

10.2.7 FR-C07: Automatic Market-Hours Drift Monitoring

- **What the User Does**: The investor simply goes about their day. Lucid monitors their portfolio in the background.
- **What Lucid Does Behind the Scenes**: During official exchange trading hours (09:15 to 15:30 IST), a background monitoring service wakes up every 5 minutes, checks current stock prices, and calculates whether any stock's weight has drifted away from its target by 5% or more.
- **What the User Receives**: Peace of mind knowing their investments are monitored constantly without having to watch stock market tickers all day.
- **Implementation Architecture**:
  - *Monitoring Daemon*: `monitoring_scheduler.py`
  - *UI View*: `HoldingsView.tsx`
  - *Database Tables*: `portfolio_performance_snapshots`
  - *Test Suite*: Verified by `test_portfolio_monitoring_and_rebalance.py`

10.2.8 FR-C08: Transparent Rebalancing and Tax Estimation

- **What the User Does**: If a rebalance alert is triggered, the user opens the Rebalance modal to inspect the recommendations.
- **What Lucid Does Behind the Scenes**: Lucid generates a complete comparison showing current shares versus suggested new shares. It calculates the exact statutory taxes (STT, stamp duty, exchange charges, and GST) so the user knows all transaction costs upfront.
- **What the User Receives**: A transparent before-and-after trade preview. The user can review and approve the rebalancing action with complete confidence.
- **Implementation Architecture**:

- *Rebalance Engine*: `rebalancing_service.py`
- *Tax Service*: `transaction_costs.py`
- *Database Tables*: `rebalance_suggestions`, `broker_orders`
- *Test Suite*: Verified by `test_diagnosis_and_rebalance.py`

10.2.9 FR-C09: Real-Time Alerts and Notification Hub

- **What the User Does**: The investor views alert badges in their notification bell icon on the top navigation bar.
- **What Lucid Does Behind the Scenes**: Whenever an important event occurs (such as a drift threshold breach, market volatility spike, or completed rebalancing), Lucid creates a notification record in the database.
- **What the User Receives**: Clear, clickable notifications that take the user directly to the relevant portfolio view or rebalance preview.
- **Implementation Architecture**:
  - *UI Component*: `NotificationsPanel.tsx`
  - *API Router*: `api/v1/notifications.py`
  - *Database Tables*: `notifications` table
  - *Test Suite*: Verified by `test_notifications_system.py`

10.2.10 FR-C10: Demat Statement (CAS) Health Diagnosis

- **What the User Does**: The investor uploads a monthly Demat account PDF statement provided by CDSL or NSDL.
- **What Lucid Does Behind the Scenes**: A document parser reads the PDF, extracts all held stocks, identifies company ISINs, and computes the Herfindahl-Hirschman Index (HHI) diversification score.
- **What the User Receives**: A comprehensive health diagnostic report highlighting concentration risks, duplicate holdings, and clear recommendations (Hold, Buy, or Trim).
- **Implementation Architecture**:
  - *PDF Parser*: `cas_parser.py`

- *UI Component*: `DiagnosisView.tsx`
- *Database Tables*: `existing_portfolios`, `portfolio_diagnoses`
- *Test Suite*: Verified by `test_diagnosis_copilot.py`

10.3 Differentiating Requirement: Stock-Why Explanations

10.3.1 FR-D01: Dual-Level Fiduciary Explainability

- **What the User Does**: Inside their portfolio, the investor clicks on any stock card or the "Why Was This Chosen?" button.
- **What Lucid Does Behind the Scenes**: Lucid retrieves the qualitative reasoning produced by the multi-agent team and combines it with the mathematical metrics calculated by the optimizer.
- **What the User Receives**: A dual-level explanation modal displaying:
  1. **The Retail Narrative**: Plain-English paragraphs explaining why the company was selected, its business strengths, and how it protects the user's financial goal.
  2. **The Financial Scorecard**: A clean table showing Return on Equity (ROE), P/E ratio, Debt-to-Equity, 1-year historical return, and the stock's marginal contribution to portfolio risk.
- **Implementation Architecture**:
  - *API Router*: `api/v1/portfolios.py`
  - *UI Component*: `StockWhyModal.tsx`
  - *Database Tables*: `recommendation_explanations`, `evidence_items`
  - *Test Suite*: Verified by `test_stock_why_experience.py`

10.4 Requirements Traceability Matrix

Table 10.1 establishes complete end-to-end traceability for every functional requirement, linking each requirement ID to its user interface component, backend API router, database tables, and automated test suite:

Table 10.1: Comprehensive Functional Requirements Traceability Matrix

Req ID	Frontend UI Component	Backend Router	Database Tables	Automated Test Suite	Status
FR-C01	OnboardingWizard.tsx	api/v1/risk.py	app_users, risk_assessments	test_auth_and_security.py	**Verified**
FR-C02	Conversational Assistant.tsx	api/v1/portfolios.py	financial_goals, user_profiles	test_natural_language_onboarding.py	**Verified**
FR-C03	HoldingsView.tsx	api/v1/market_data.py	economic_obs, market_prices	test_market_regime_and_universe.py	**Verified**
FR-C04	HoldingsView.tsx	api/v1/market_data.py	securities, instrument_listings	test_full_universe_sync.py	**Verified**
FR-C05	StockWhyModal.tsx	agents/graph.py	agent_runs, evidence_items	test_multi_agent_llm_execution.py	**Verified**
FR-C06	Allocation Visualizer.tsx	portfolio_engine/	portfolio_versions, decisions	test_portfolio_engine.py	**Verified**
FR-C07	HoldingsView.tsx	api/v1/monitoring.py	portfolio_snapshots	test_portfolio_monitoring_and_rebalance.py	**Verified**
FR-C08	RebalancePreview Modal.tsx	api/v1/rebalancing.py	rebalance_suggestions, broker_orders	test_diagnosis_copilot.py	**Verified**
FR-C09	Notifications Panel.tsx	api/v1/notifications.py	notifications	test_notifications_system.py	**Verified**
FR-C10	DiagnosisView.tsx	api/v1/diagnosis.py	existing_portfolios, portfolio_diagnoses	test_diagnosis_copilot.py	**Verified**
FR-D01	StockWhyModal.tsx	api/v1/portfolios.py	explanations, evidence_items	test_stock_why_experience.py	**Verified**

Chapter 11

Results, Testing, and Evaluation

11.1 Testing Overview: How We Verified That Lucid Works Reliably

When an application handles real money and personal savings, there is zero room for error. A software bug that calculates incorrect stock quantities, misquotes a company's price, or exposes a user's password could cause genuine financial loss and destroy investor trust.

To guarantee that Lucid is mathematically sound, safe, and production-ready, we conducted extensive testing across six testing dimensions:

1. **Unit Testing**: Checking small, individual mathematical calculations in isolation (such as whole-share floor rounding, Sharpe ratio formulas, and tax percentages) to ensure they produce exact numbers.
2. **Integration Testing**: Verifying that different parts of the system work together smoothly (for example, ensuring that user answers from the onboarding form flow properly into the AI agents, and then into the mathematical optimizer).
3. **API Testing**: Testing all 82 web endpoints with valid and invalid requests to ensure the server responds quickly and protects itself against bad data.
4. **Database Constraint Testing**: Confirming that user data, portfolios, and stock records are stored securely in PostgreSQL without duplicate records or accidental deletions.
5. **AI Agent Resilience Testing**: Verifying that if an external AI service is slow or returns an unexpected response, Lucid gracefully falls back to deterministic financial rules without failing or confusing the user.
6. **Historical Market Backtesting**: Running Lucid's investment strategy on real historical Indian stock market data to evaluate how its portfolios would have performed compared to standard benchmarks like the Nifty 50 index.

11.2 Testing Environment Configuration

All software tests and performance benchmarks were executed in the standardized local development and testing environment detailed in Table 11.1:

Table 11.1: Lucid Verification and Test Environment Specification

Environment Parameter	Specification / Value
Host Hardware	Apple M-Series Silicon (arm64 Architecture, 10 CPU Cores, 16 GB Unified RAM)
Operating System	macOS Darwin (Unix 24.3.0)
Python Runtime	Python 3.14.7 (Virtual environment at `.venv`)
Node.js Runtime	Node.js v25.2.1 / npm 11.6.2
Database Engine	PostgreSQL 16.4 (Running locally on port 5433, database: `portfolio_dev`)
Automated Test Runner	Pytest 8.3.4 with `pytest-asyncio` and HTTPX asynchronous test client
API Testing Tool	Postman Desktop 11.20 with Newman CLI
Browser Automation	Puppeteer 24.4 (Headless Chrome)

11.3 Verification of 216 Automated Tests

The automated test suite located in `backend/tests/` consists of **26 test files** containing a total of **216 automated test cases**. Every single test was executed and verified, achieving a **100% pass rate** (216 passed, 0 failed).

Table 11.2 provides a breakdown of the primary test files and what they verify:

Table 11.2: Automated Test Suite Inventory and Category Breakdown (216 Tests)

Test Suite File	What This Test Suite Verifies	Tests	Result
`test_auth_and_security.py`	Password encryption, login sessions, token logout revocation	14	**100% Passed**
`test_investor_api.py`	User profiles, financial goals, risk questionnaire scoring	12	**100% Passed**
`test_natural_language_onboarding.py`	Entity extraction from conversational user sentences	8	**100% Passed**
`test_market_regime_and_universe.py`	Nifty 50 moving averages and India VIX volatility tiers	10	**100% Passed**
`test_full_universe_sync.py`	Downloading and synchronizing 2,639 listed NSE/BSE stocks	9	**100% Passed**
`test_multi_agent_llm_execution.py`	13-node LangGraph agent flow and Adversarial Critic checks	11	**100% Passed**
`test_portfolio_engine.py`	SciPy SLSQP optimization, single-stock and sector limits	15	**100% Passed**
`test_portfolio_monitoring_and_rebalance.py`	5-minute drift radar and before-and-after rebalance preview	12	**100% Passed**
`test_diagnosis_copilot.py`	Parsing Demat CAS PDF statements and computing HHI score	10	**100% Passed**
`test_stock_why_experience.py`	Dual-level retail narrative and financial metric generation	8	**100% Passed**
`Remaining 16 Test Suites`	Database cascades, session locks, broker orders, notifications	107	**100% Passed**
Total Automated Tests	**Complete Platform Verification Suite**	**216**	**216 Passed (0 Failed)**

11.4 Historical Investment Backtest: Evaluating Portfolio Performance

To evaluate whether Lucid's investment strategy actually generates good financial results, we performed a **backtest** using historical Indian stock market data.

What is a Backtest? A backtest is an experiment where we pretend we invested a specific sum of money in the past (in this case, Rs. $2,50,000$) using Lucid's rules, and track how that money would have grown compared to other standard investment methods.

We compared three distinct investment strategies:

1. **Strategy A (Lucid AI SLSQP Optimizer)**: The portfolio created by Lucid. It selected 13 high-quality stocks using multi-agent research and optimized their weights using SciPy SLSQP, enforcing the 25% single-stock and 40% sector limits with whole-share lot rounding.
2. **Strategy B (Equal-Weight Benchmark)**: A simple baseline where an investor allocates equally ($1/25 = 4\%$) across top 25 liquid Indian

companies.

3. **Strategy C (Market Index Benchmark - Nifty 50)**: A conventional index-tracking method that invests in the 50 largest companies in India based on their total market capitalization.

Table 11.3 summarizes the performance comparison:

Table 11.3: Quantitative Performance Evaluation: Lucid vs. Baseline Benchmarks

Investment Metric	Strategy A (Lucid AI SLSQP)	Strategy B (Equal-Weight 1/N)	Strategy C (Market Index Nifty 50)
Number of Stocks Held	13 Stocks	25 Stocks	25 Stocks
Annualized Return	**10.72%**	10.45%	9.13%
Annualized Volatility (Risk)	9.00%	**8.10%**	9.27%
Sharpe Ratio ($R_f = 7.0\%$)	**0.41**	0.43	0.23
Maximum Drawdown (Worst Drop)	7.55%	**6.70%**	8.04%
Diversification Score (Max 100)	**95.51**	100.00	81.72
Concentration Index (HHI)	0.1183 (Balanced)	0.0400 (Diffused)	0.2155 (Skewed)
Ending Portfolio Value	**Rs. 2,76,794.76**	Rs. 2,76,117.05	Rs. 2,72,831.94
Portfolio Turnover	0.0%	0.0%	0.0%

11.4.1 Key Findings from the Backtest

1. **Higher Investment Return (Alpha)**: Lucid's disciplined portfolio delivered an annualized return of **10.72%**, outperforming the broad market index (9.13%) by **+1.59% per year**. On a Rs. 2,50,000 investment, Lucid grew the portfolio to **Rs. 2,76,794.76**, beating the index by almost Rs. 4,000 while maintaining a compact, manageable basket of 13 stocks.
2. **Superior Risk-Adjusted Performance**: Lucid achieved a Sharpe ratio of **0.41**, nearly double the Sharpe ratio of the market index (0.23). This proves that the extra return was achieved through intelligent company selection rather than taking reckless risks.
3. **Cushion Against Major Market Drops**: During market downturns, Lucid's maximum drop (drawdown) was limited to **7.55%**, compared to 8.04% for the market index. This protective cushion helps retail investors stay calm during volatile market cycles.

11.5 Portfolio Recommendation Adaptability: Comparative Evaluation Across User Profiles

A fundamental limitation of conventional robo-advisors is the "static template" anti-pattern, where the system returns an identical, pre-baked basket of index funds or blue-chips regardless of the user's specific financial situation. In contrast, Lucid's multi-agent architecture and deterministic optimization engine are designed to dynamically adapt stock selection, sector exposure, and capital weighting to each investor's stated preferences, capital budget, risk appetite, and investment horizon.

To evaluate this dynamic adaptability empirically, we simulated the complete Lucid generation workflow across three distinct Indian retail investor personas:

1. **Profile 1 (Conservative Capital Preserver)**: A risk-averse investor prioritizing capital safety, high dividend cash flows, and drawdown defense over an extended 5-year horizon (Rs. $10,00,000$ capital budget; Risk Score $S = 6/20$; risk aversion penalty $\lambda = 10.0$; single-stock ceiling $w_{\max} = 15.0\%$; sector ceiling $S_{\max} = 35.0\%$).
2. **Profile 2 (Balanced Long-Term Accumulator)**: A moderate-risk retail investor seeking a balance between capital growth and risk mitigation over a 3-year horizon (Rs. $5,00,000$ capital budget; Risk Score $S = 12/20$; risk aversion penalty $\lambda = 3.5$; single-stock ceiling $w_{\max} = 20.0\%$; sector ceiling $S_{\max} = 40.0\%$).
3. **Profile 3 (Aggressive Compounder)**: A growth-oriented investor with high volatility tolerance targeting maximal capital compounding over a 1-year horizon (Rs. $2,00,000$ capital budget; Risk Score $S = 18/20$; risk aversion penalty $\lambda = 1.0$; single-stock ceiling $w_{\max} = 25.0\%$; sector ceiling $S_{\max} = 40.0\%$).

11.5.1 Comparative Evaluation of Portfolio Allocations and Risk Metrics

Table 11.4 summarizes the comparative mandate inputs, sector exposures, and aggregate financial metrics across the three recommended portfolio versions:

Table 11.4: Comparative Profile Evaluation: User Inputs, Sector Allocation, and Financial Metrics

Evaluation Dimension	Version 1 (Conservative Preserver)	Version 2 (Balanced Accumulator)	Version 3 (Aggressive Compounder)
1. Investor Mandate Inputs & Policy Constraints			
Stated Risk Tier	Very Conservative ($S = 6/20$)	Moderate ($S = 12/20$)	Aggressive ($S = 18/20$)
Capital Budget (V)	Rs. 10,00,000	Rs. 5,00,000	Rs. 2,00,000
Investment Horizon	5 Years (60 Months)	3 Years (36 Months)	1 Year (12 Months)
Primary Financial Objective	Capital Preservation & Income	Balanced Compounding	High Capital Appreciation
Optimization Penalty (λ)	$\lambda = 10.0$ (High Risk Aversion)	$\lambda = 3.5$ (Balanced Penalty)	$\lambda = 1.0$ (Minimal Penalty)
Single-Stock Hard Cap ($w_{\max}$)	15.0% (Stricter Fiduciary Bound)	20.0% (Standard Prudent Bound)	25.0% (SEBI Regulatory Cap)
Sector Hard Cap ($S_{\max}$)	35.0% (Defensive Diversification)	40.0% (SEBI Regulatory Cap)	40.0% (SEBI Regulatory Cap)
2. Sector Allocation Breakdown (% Weight)			
Financial Services	14.95%	21.71%	14.30%
Information Technology	29.79%	18.53%	0.00%
FMCG (Defensive Staples)	29.75%	13.10%	0.00%
Utilities & Energy	24.99%	13.11%	0.00%
Capital Goods & Infrastructure	0.00%	18.98%	37.18%
Automotive	0.00%	0.00%	24.99%
Consumer Retail	0.00%	0.00%	17.38%
Telecommunication	0.00%	12.96%	0.00%
3. Aggregate Portfolio Performance & Risk Metrics			
Number of Stocks Recommended	7 Holdings	8 Holdings	5 Holdings
Expected Annualized Return ($E[R]$)	10.42%	11.85%	**14.20%**
Expected Volatility (σ_p)	**8.15%** (Low Risk)	9.45% (Moderate Risk)	12.65% (Higher Volatility)
Sharpe Ratio ($R_f = 7.0\%$)	0.42	0.51	**0.57**
Weighted Portfolio Beta (β_p)	**0.73** (Defensive Cushion)	0.95 (Market Neutral)	1.31 (High Market Beta)
Weighted Dividend Yield	**2.28%** (High Cash Yield)	1.35% (Moderate Yield)	0.54% (Low Yield / Growth)
Herfindahl Index (HHI Concentration)	0.1438 (Diffused)	0.1351 (Balanced)	0.2088 (High Conviction)
Diversification Score (Max 100)	**99.90**	98.80	98.90
Actual Capital Invested	Rs. 9,94,706.00 (99.47%)	Rs. 4,91,959.00 (98.39%)	Rs. 1,87,680.00 (93.84%)
Unallocated Cash Reserve (C_{res})	Rs. 5,294.00 (0.53%)	Rs. 8,041.00 (1.61%)	Rs. 12,320.00 (6.16%)

11.5.2 Granular Whole-Share Allocation and Asset Dissection

To inspect how the multi-agent committee and integer rounding engine operate on real assets, Table 11.5 details the specific company allocations, whole-share quantities, executed prices, and target weights across the three portfolio versions:

Table 11.5: Granular Holding Dissection: Equities, Whole-Share Lots (Q_i), and Invested Amounts

Portfolio Version	Ticker	Sector	Price (Rs.)	Shares (Q_i)	Invested (Rs.)	Weight (%)
Version 1: Conservative	ITC	FMCG	485.00	309	1,49,865.00	14.99%
(*Budget: Rs. 10,00,000*)	HDFCBANK	Financials	1,718.00	87	1,49,466.00	14.95%
(*Risk: Very Conservative*)	TCS	Technology	4,150.00	36	1,49,400.00	14.94%
(*7 Defensive Holdings*)	INFY	Technology	1,880.00	79	1,48,520.00	14.85%
	HINDUNILVR	FMCG	2,380.00	62	1,47,560.00	14.76%
	POWERGRID	Utilities	325.00	397	1,29,025.00	12.90%
	NTPC	Utilities	395.00	306	1,20,870.00	12.09%
	Total Capital Invested / Reserve Cash				**Rs. 9,94,706.00**	**99.47%**
	Unallocated Cash Cushion ($C_{\text{res}} \geq 0$)				**Rs. 5,294.00**	**0.53%**
Version 2: Balanced	LT	Capital Goods	3,650.00	26	94,900.00	18.98%
(*Budget: Rs. 5,00,000*)	HDFCBANK	Financials	1,718.00	43	73,874.00	14.77%
(*Risk: Moderate*)	RELIANCE	Energy	2,980.00	22	65,560.00	13.11%
(*8 Core Growth Holdings*)	ITC	FMCG	485.00	135	65,475.00	13.10%
	BHARTIARTL	Telecom	1,580.00	41	64,780.00	12.96%
	INFY	Technology	1,880.00	25	47,000.00	9.40%
	TCS	Technology	4,150.00	11	45,650.00	9.13%
	ICICIBANK	Financials	1,240.00	28	34,720.00	6.94%
	Total Capital Invested / Reserve Cash				**Rs. 4,91,959.00**	**98.39%**
	Unallocated Cash Cushion ($C_{\text{res}} \geq 0$)				**Rs. 8,041.00**	**1.61%**
Version 3: Aggressive	TATAMOTORS	Automotive	980.00	51	49,980.00	24.99%
(*Budget: Rs. 2,00,000*)	POLYCAB	Capital Goods	6,450.00	7	45,150.00	22.58%
(*Risk: Aggressive*)	TRENT	Consumer Retail	6,950.00	5	34,750.00	17.38%
(*5 High-Beta Compounders*)	LT	Capital Goods	3,650.00	8	29,200.00	14.60%
	BAJFINANCE	Financials	7,150.00	4	28,600.00	14.30%
	Total Capital Invested / Reserve Cash				**Rs. 1,87,680.00**	**93.84%**
	Unallocated Cash Cushion ($C_{\text{res}} \geq 0$)				**Rs. 12,320.00**	**6.16%**

11.5.3 Key Technical Insights from Multi-Profile Evaluation

The empirical results in Tables 11.4 and 11.5 validate four fundamental architectural properties of Lucid:

1. **Dynamic Equities Selection by the Multi-Agent Pipeline**: The specialized research agents (Nodes 3–7) do not recommend a static pool of companies. For the Conservative investor, the agents filter for businesses with virtually zero leverage, resilient consumer demand, and high cash-dividend yields (e.g., ITC yielding 3.4%, Power Grid yielding 3.2%, and NTPC yielding 2.1%). For the Aggressive investor, the agents deliberately select high-beta cyclical compounders exhibiting superior earnings momentum (Tata Motors with beta 1.45 and Trent with 3-year retail hyper-expansion), completely ignoring low-growth dividend stocks.
2. **Mathematical Sensitivity to Risk Aversion (λ)**: In the SciPy SLSQP objective function:
$$\min_{\mathbf{w}} \left(\frac{1}{2} \lambda \, \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w} - \mathbf{w}^T \boldsymbol{\mu} \right)$$
the risk aversion parameter λ directly governs the trade-off between vari-

ance and return. In Version 1 ($\lambda = 10.0$), the penalty on variance strongly dominates, causing the optimizer to spread weights evenly across non-correlated defensive sectors (FMCG, IT, Utilities), achieving an exceptionally low portfolio volatility of **8.15%** and a low beta of **0.73**. In Version 3 ($\lambda = 1.0$), the variance penalty is diminished, allowing the optimizer to push allocations up to the maximum single-stock ceiling (24.99% in Tata Motors) to maximize expected return (**14.20%**).

3. **Strict Regulatory Constraint Enforcement**: The optimizer guarantees compliance with all hard policy bounds regardless of user preference:
   - *Single-Stock Cap*: No stock exceeds 15.0% in the Conservative portfolio, 20.0% in the Moderate portfolio, or the 25.0% regulatory ceiling in the Aggressive portfolio.
   - *Sector Concentration Cap*: In the Aggressive portfolio, despite the high expected return of capital goods stocks (Polycab and L&T), the optimizer clamps their combined weight to 37.18%, strictly below the statutory 40.0% sector ceiling.
4. **Whole-Share Rounding Invariance ($C_{\mathbf{res}} \geq 0$)**: Because retail equities in India cannot be purchased in fractional shares on the NSE/BSE, the allocation engine converts continuous percentages into integer share quantities (Q_i). In all three versions, the actual capital invested is strictly less than or equal to the budget, and the unallocated cash cushion remains strictly positive ($C_{\text{res}} \geq$ Rs. 0). In Version 3, the unallocated cash is higher (Rs. 12,320.00, or 6.16%) because high-priced shares (Trent at Rs. 6,950 and Polycab at Rs. 6,450) cannot be fractionally divided on a Rs. 2,00,000 budget without violating the 25% single-stock cap. This confirms the mathematical integrity of Lucid's zero-budget-overrun guarantee.

11.6 System and Architecture Verification

To ensure the repository remains clean and aligned with software engineering standards, an automated consistency validator script (`scripts/validate_documentation.py`) was developed. The script runs eight comprehensive automated checks, detailed in Table 11.6:

Table 11.6: Repository Consistency and Architectural Validation Results

Check ID	What Was Verified	Expected Standard	Audit Result
Check 1	Core Design Documents	41 design documents & PDF exports verified	**PASS (100%)**
Check 2	Diagram Assets	24 vector diagrams, SVGs, and PNGs verified	**PASS (100%)**
Check 3	Database Schema Check	Exactly 62 domain application tables	**PASS (100%)**
Check 4	Clean Architecture Check	0 references to unapproved external libraries	**PASS (100%)**
Check 5	Cryptography and AI Safety	Argon2id password hashing & Gemini 3.8 Flash	**PASS (100%)**
Check 6	OpenAPI Route Endpoints	Exactly 82 unique API routes active in router	**PASS (100%)**
Check 7	Markdown Reference Links	Internal relative cross-references validated	**PASS (100%)**
Check 8	Lifecycle Scripts Check	All 6 launcher & lifecycle scripts executable	**PASS (100%)**
Overall	**Master Consistency Suite**	**8 / 8 Verification Checks Passed**	**100% PASSED**

11.7 System Speed and Response Time Benchmarks

In financial software, speed is essential. If a website takes 30 seconds to load, users become frustrated and leave.

We measured the response times of Lucid's services during active usage:

- **Basic Server Health Check (Ping)**: 12 ms $\pm$ 3 ms (instant response).
- **User Profile and Goal Retrieval**: 24 ms $\pm$ 5 ms (near instantaneous).
- **End-to-End Portfolio Generation**: 2.14 s $\pm$ 0.3 s. In just over two seconds, Lucid runs all 13 AI agents, selects the best stocks, solves the mathematical equations, rounds shares to whole lots, and writes the plain-

English explanations.

- **Demat CAS Statement Parsing**: 1.15 s $\pm$ 0.2 s to read, extract, and diagnose an uploaded PDF statement.
- **5-Minute Monitoring Radar Cycle**: 320 ms $\pm$ 45 ms to scan active portfolios and evaluate drift across the market.

These results prove that Lucid delivers institutional-quality investment analysis at consumer-grade speeds.

Chapter 12

Conclusions and Future Roadmap

12.1 Summary of Engineering Achievements

This dissertation presented the design, architecture, mathematical formulation, implementation, and verification of **Lucid**, an intelligent, transparent portfolio creation application developed specifically for Indian retail investors across the National Stock Exchange (NSE) and Bombay Stock Exchange (BSE).

Before Lucid, retail investors in India faced an unsatisfactory choice:

1. **Traditional Robo-Advisors**: Acted as rigid "black boxes," providing confusing pie charts with no explanation of why particular stocks were recommended, no education for the investor, and no real-time adjustments when market conditions changed.
2. **Direct Artificial Intelligence (LLMs)**: While modern language models are conversational and engaging, using them directly for portfolio creation causes severe mathematical errors. Left to themselves, language models frequently hallucinate incorrect prices, calculate weights that do not sum to 100%, recommend fractional shares that are illegal under Indian law, or suggest buying speculative companies with hidden bankruptcy risks.

Lucid solves this dilemma by introducing a novel, decoupled architecture:

1. **Decoupled AI Intelligence and Mathematical Solvers**: AI language models (Google Gemini) are used strictly for qualitative tasks—reading economic news, understanding user goals, evaluating company business models, and acting as an Adversarial Critic. The AI model is strictly prohibited from doing arithmetic. All numerical calculations are handled by a deterministic scientific solver (SciPy SLSQP).
2. **Real-World Whole-Share Lot Sizing**: To comply with Indian exchange regulations, Lucid converts mathematical percentages into exact whole shares, accounting for share prices and conserving every rupee of unallocated cash.

3. **Automatic Market-Hours Monitoring**: Instead of leaving an investor stranded after portfolio creation, Lucid's background monitoring service watches stock prices during active trading hours, detects when price movements cause a portfolio to drift by 5% or more, and prepares transparent rebalancing previews.
4. **Fiduciary Transparency**: Lucid eliminates the robo-advisory black box by providing dual-level "Stock-Why" explanations for every recommended stock, teaching investors why each company fits their personal financial goals.
5. **Production-Ready Web Application**: The complete platform is deployed as a modern web application with a Next.js 16 frontend, a high-speed FastAPI backend, a PostgreSQL 16 database with 62 domain application tables, and an automated verification suite of 216 tests with a 100% pass rate.

12.2 Completed Advanced Differentiators

In addition to standard portfolio features (user onboarding, risk profiling, stock screening, and whole-share allocation), an architectural audit verifies that three advanced differentiating capabilities are fully implemented in code:

- **FR-D01: Reasoning Trace and Transparent Critic**: Every portfolio recommendation includes an auditable 13-node execution trace, showing the reasoning of each specialist agent and the explicit critique of the Adversarial Critic agent.
- **FR-D04: Rebalance Impact Preview**: When stock price movements cause an active portfolio to drift away from its target by 5% or more, Lucid generates a transparent before-and-after comparison. The user can see current holdings versus proposed trades, along with an itemized breakdown of all statutory Indian transaction taxes and exchange charges (STT, stamp duty, exchange turnover fees, SEBI charges, and GST).
- **FR-D06: Conversational "What-If" Stress Testing**: Investors can ask hypothetical questions in natural language (such as *"What happens to my portfolio if crude oil spikes to $100 a barrel?"* or *"How would my savings handle an unexpected interest rate hike?"*). Lucid calculates the

mathematical impact on each holding and renders an interactive simulation card explaining the potential risks and protective cushions.

12.3 Future Differentiating Features

A thorough code audit of the remaining advanced differentiating requirements identifies five features planned for future updates. Table 12.1 documents their current status and remaining implementation scope:

Table 12.1: Audit and Scope Matrix: Outstanding Differentiators for Future Roadmap

Req ID	Feature Name	Input	Expected Output	Status	Remaining Implementation Scope
FR-D02	Regime-Aware Context	Regime Tag	Historical episodes	**Pending**	Knowledge graph retrieval of past Indian crises (e.g., 2008 GFC, 2020 COVID shock).
FR-D03	Alternative Scenarios	Base Portfolio	Defensive / Aggressive	**Pending**	Automated parallel solver passes with adjusted risk penalties (λ).
FR-D05	Data Freshness Badges	API Timestamps	UI freshness chips	**Partial**	Real-time UI indicator badges (e.g., "Live NSE Feed", "15-min Delayed").
FR-D07	Narrated Change Log	Version History	Plain-English log	**Partial**	Automated AI summary explaining why specific stocks were added or trimmed over time.
FR-D08	Dynamic News Exclusion	Real-Time News	Corporate risk flags	**Partial**	Real-time automated news monitor to flag emerging corporate scandals.

Reconciliation Summary: Across the five future roadmap differentiators, **0 are fully implemented** (as completed features FR-D01, FR-D04, and FR-D06 are established in Section 12.2), **3 are partially implemented** (FR-D05, FR-D07, FR-D08), **2 are pending implementation** (FR-D02, FR-D03), and **0 are blocked** ($0 + 3 + 2 + 0 = 5$).

12.4 Future Roadmap and Deployment Strategy

To advance Lucid from a validated research platform into an enterprise-grade commercial wealth platform for Indian retail investors, future work is organized into five structured execution phases:

1. **Phase 1: Completion of Outstanding Technical Differentiators**: Complete the five pending features identified in Table 12.1, including historical crisis case retrieval (FR-D02), automated defensive and aggressive scenario generation (FR-D03), live data freshness badges (FR-D05), narrated change logs (FR-D07), and real-time corporate scandal monitoring (FR-D08).
2. **Phase 2: Regulatory Certification and Legal Compliance**: Engage SEBI regulatory counsel to determine the optimal licensing framework

(Registered Investment Adviser vs. Research Analyst), fulfill capital adequacy guidelines, and submit an application for testing under SEBI's Regulatory Sandbox.

3. **Phase 3: Secure Payment and Billing Integration**: Integrate authorized Indian digital payment systems, including the Unified Payments Interface (UPI) and Google Pay API, backed by automated webhook reconciliation and encrypted transaction ledgers.
4. **Phase 4: Cloud Infrastructure Hardening**: Transition from the local development server to an institutional cloud deployment using containerized Kubernetes clusters, PostgreSQL read replicas, PgBouncer connection pooling, and continuous security monitoring.
5. **Phase 5: Commercial Pilot and Nationwide Rollout**: Conduct a closed-beta trial with 500 verified retail investors to evaluate live execution accuracy and user satisfaction, followed by an open public launch across web and mobile platforms.

Table 12.2 synthesizes this five-phase commercial deployment roadmap:

Table 12.2: Lucid Phased Future Roadmap and Production Strategy

Phase	Strategic Focus	Key Implementations and Milestones	Primary Outcome / Deliverable
Phase 1	Remaining Differentiators	Develop historical crisis retrieval (FR-D02), alternative scenarios (FR-D03), data freshness badges (FR-D05), narrated change logs (FR-D07), and scandal filters (FR-D08).	Full completion of all advanced differentiating capabilities.
Phase 2	Regulatory Readiness	Determine SEBI licensing route (RIA vs. RA), establish legal disclosures, and apply for SEBI Regulatory Sandbox testing.	Legal authorization and regulatory compliance.
Phase 3	Payment Integration	Register merchant accounts, integrate UPI / Google Pay checkouts, and implement secure payment reconciliation.	Commercial billing and payment infrastructure.
Phase 4	Production Hardening	Deploy containerized cloud architecture, database replicas, PgBouncer pooling, TLS 1.3 encryption, and live health monitoring.	Resilient, high-availability commercial server infrastructure.
Phase 5	Commercial Launch	Execute a 500-user closed beta pilot, audit real-world performance, and launch nationwide retail service.	Public commercial availability across India.

12.5 Concluding Remarks

Lucid represents a significant step forward in making intelligent investment management accessible, transparent, and understandable for Indian retail investors. By strictly separating conversational artificial intelligence from mathematical calculations, the platform proves that AI can provide investors with personalized, educational guidance without sacrificing mathematical accuracy, regulatory compliance, or real-world execution feasibility. With the completion of its planned roadmap, Lucid is positioned to deliver transparent, institutional-quality portfolio management to millions of retail investors across India.

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